Technical Appendix — Commons Bonds

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Contents

§1. Foundations — Formal Definitions and Primitives

Definition 1.1 (Commons Resource)

A resource unit R ∈ 𝒞 (where 𝒞 (script C) denotes the commons-territory set; distinguished typographically from C used as cost variable C_i) is drawn from a shared stock Q with regeneration rate $\rho \geq 0$. The framework prices extraction for all values of ρ.

- When $\rho = 0$ (strict non-renewability), extraction is irreversible and foreclosure is permanent.
- When $\rho > 0$, extraction below the regeneration rate leaves Q stable and drives the foreclosure component toward zero; extraction above the regeneration rate drives Q downward and the foreclosure component upward, approaching the non-renewable case as $\rho \to 0$ or as stock depletion approaches ecological thresholds.

Definition 1.2 (Substitutability Function)

$S(t \mid t_0)$: $[t_0, \infty) \to [0, S\_max]$ where $S\_max \leq 1$, is the probability that a functionally adequate substitute for resource R exists at time t, given extraction at time t_0. S satisfies:

1. $S(t_0) \geq 0$ (at extraction, substitutability is at or near zero)
2. S is non-decreasing (substitutes, once developed, persist)
3. $\lim_{t\to\infty} S(t) = S\_max \leq 1$ (full substitution may or may not be achievable)

For computational tractability: $S(t) = S\_max \cdot (1 - e^{-\lambda(t - t_0)})$, where $\lambda > 0$ is the substitution rate estimated empirically per resource class. S(t) may be influenced by extraction timing — earlier extraction can slow innovation by reducing economic pressure to develop alternatives.

Definition 1.3 (Stock-Dependent Utility)

U(R, t, Q(t)) represents generation t's utility from resource R being available in situ, conditional on remaining stock Q(t). U is:

1. Non-negative: $U(R, t, Q) \geq 0$
2. Increasing in scarcity: $\partial U/\partial Q < 0$ (utility per remaining unit increases as stock depletes)
3. Bounded below by market price: $U(R, t, Q(t)) \geq P(t)$ for all $t \geq t_0$

U(R, t, Q(t)) includes direct economic value, option value, and ecological function value. The ecological component is small at abundant stock levels and rises nonlinearly as Q approaches ecological thresholds, capturing keystone effects and ecosystem cascades.

Notational convention (Q(t) = remaining in-situ stock at time t). The framework adopts Q(t) for remaining in-situ stock, with units consistent with the resource class (e.g., barrels of oil; tonnes of rare-earth ore; cubic meters of timber). The choice diverges from several adjacent conventions in the resource-economics + industrial-organization literatures, and the divergence is marked here for first-encounter disambiguation:

- *Hotelling 1931* uses q for extraction flow (rate of withdrawal); the framework's Q is the integrated stock state, not the flow.
- *Pindyck 1978 Journal of Political Economy* uses X(t) for stock; symbol choice differs but type-equivalent to framework's Q.
- *Dasgupta & Heal 1979 Economic Theory and Exhaustible Resources* uses S(t) for stock; symbol collides with framework's S(t) = substitutability function (Definition 1.2). The framework's choice of Q resolves the collision.
- *Mussa & Rosen 1978 Journal of Economic Theory* uses Q for quality-index in industrial-organization; the framework's Q is stock state, not quality dimension. Readers approaching the framework from IO conventions should read Q as quantity-stock, not quality.

The internal mathematical apparatus enforces the framework's reading: $\partial U/\partial Q < 0$ (utility per remaining unit increases as stock depletes; would be ill-typed under Hotelling-flow or Mussa-Rosen-quality readings) and Q-critical-threshold coupling (Q approaches ecological thresholds; would be ill-typed under flow reading). Adjacent literature: Slade & Thille 2009 *Journal of Economic Literature* on natural-resource cost-function notation; Spence 1976 *Review of Economic Studies* + Tirole 1988 *The Theory of Industrial Organization* on Q-as-quality contrast.

Definition 1.4 (Externality Tail)

$E(R, t)$ represents the externality flow at time t resulting from extraction of resource R at time t_0 (per Definition 1.3, $Q(t)$ = remaining in-situ stock at time t; E is independent of Q in this formulation). This includes ongoing environmental damage, health costs, community disruption, and ecological degradation that persist beyond the extraction event. $E(R, t) \geq 0$ for all t, and typically exhibits a long tail: $E(R, t) > 0$ for many generations after extraction ceases.

Notational disambiguation ("tail" — temporal-persistence sense, not statistics-distribution-tail). "Externality Tail" in this framework names the temporal-persistence dimension of an externality: damage that persists for many generations after extraction stops. The "tail" is in *time*, not in *probability distribution*. $E(R, t)$ is a deterministic flow function, not a probability density's tail region. Readers approaching the framework from finance / actuarial / industrial-organization conventions may pattern-match "tail" to extreme-value-theory or fat-tailed distribution conventions (Mandelbrot 1963 *Journal of Business* on fat-tailed price distributions; Taleb 2007 *The Black Swan*; Embrechts, Klüppelberg & Mikosch 1997 *Modelling Extremal Events*; Anderson 2006 *The Long Tail*); the framework's usage is distinct. The "runs on its own clock" rhetorical anchor (§7 methodological framing) reinforces the temporal-persistence reading.

Definition 1.5 (Social Discount Function)

$D(t, t_0): T \times T \to (0,1]$ is a declining discount factor following Weitzman (2001). The instantaneous discount rate $r(t)$ satisfies $r(t) = r_0 \cdot g(t)$, where r_0 is the initial discount rate and $g(t)$ is a declining function reflecting uncertainty about appropriate long-horizon discounting. The Weitzman result: under uncertainty about the true discount rate, the certainty-equivalent rate declines toward the lowest plausible rate. This is a conservative choice that biases the RCV estimate downward relative to a fixed high discount rate.

Definition 1.6 (Residual Commons Value: Core Formula)

The RCV of resource unit R, extracted at t_0, is the present value of all foregone intergenerational utility plus all ongoing externalities (per Definition 1.3, $Q(t)$ = remaining in-situ stock at time t):

$$RCV(R, t_0) = \int_{t_0}^{\infty} \{ [1 - S(t \mid t_0)] \cdot U(R, t, Q(t)) + E(R, t) \} \cdot D(t, t_0)\, dt$$

Acronym disambiguation (RCV first-introduction discipline). The framework uses RCV exclusively for Residual Commons Value as defined here. Three real-world acronym collisions are noted for first-encounter disambiguation:

- *Ranked-Choice Voting* — civic-engagement / electoral-policy usage; rising salience through 2024+ (Maine 2018; NYC 2021; Alaska 2022; multiple state initiatives). Academic anchor: Reilly 2001 *Democracy and Diversity: Political Engineering in the Asia-Pacific* (Oxford UP).
- *Replacement Cost Value* — insurance-policy usage (vernacular in U.S. homeowner / renter / auto policies; distinguished from Actual Cash Value). Reference: Insurance Information Institute *Replacement Cost vs Actual Cash Value* consumer guide; Casualty Actuarial Society methodological texts.

- *Risk Capital Value* — banking-regulatory usage (Basel III + FDIC capital adequacy frameworks). Reference: Basel Committee on Banking Supervision 2017 *Basel III: Finalising Post-Crisis Reforms* (Bank for International Settlements).

None of these usages fits the integrand-and-integral form above, so internal mathematical apparatus disambiguates after first formula encounter. To close pre-equation reading-friction, the framework's discipline is to introduce the term as "Residual Commons Value (RCV)" on first appearance per chapter or major section before reusing the acronym.

The integrand has two components:

- Foreclosure component: $[1 - S(t \mid t_0)] \cdot U(R, t, Q(t))$ — the expected utility loss when no substitute exists, weighted by the probability that no substitute exists at time t.
- Externality component: $E(R, t)$ — the ongoing damage costs regardless of substitution status. A substitute for coal does not remove carbon from the atmosphere; the externality tail runs independently.

Definition 1.7 (Accountability Bond)

B represents the total set of mechanisms that force extractors to bear costs: reclamation bonds, severance taxes, legal liability, regulatory requirements, community compensation agreements. Cost Severance is formally defined as:

$CS = RCV - B$

Cost severance occurs whenever $CS > 0$, indicating that the true social cost of extraction exceeds the mechanisms forcing extractors to bear those costs. Under current extraction regimes, B reflects only identified, legally-enforceable costs. RCV includes intergenerational foreclosure costs that no existing B mechanism prices, ensuring $CS > 0$ structurally for all non-renewable extraction under present institutions.

Forward-pricing shorthand vs. full bidirectional form. The form above — `CS = RCV − B` — is the high-level forward-pricing shorthand operative throughout this volume's chapter prose and the empirical work in §11. The canonical bidirectional articulation, `total CS = (CSD − B₁) + (RCV − B₂)` where B decomposes into B_1 (Restitution Bond, backward-paired with CSD = Cost Severance Damages already realized) and B_2 (Foreclosure Bond, forward-paired with RCV), is given in §5 (decomposition) and §5.5 (bidirectional application of the framework apparatus). The two forms coincide when $(CSD - B_1) \approx 0$ — either because backward damages have been restituted, or in pure forward-looking analysis bracketing backward damages. This volume's empirical scope operates in the latter mode by design.

Definition 1.8 (Commons Inversion Test: Methodological Protocol)

Per Theorem 10.3 (Abundance Masking) at §10, the abundance-masking phenomenon is established analytically from resource-economics cost-function primitives. CIT operationalizes detection of the phenomenon: stripping away dimensions of abundance and observing which cost components transition from $C \approx 0$ to $C \to$ large is the empirical signature of Theorem 10.3's analytic prediction. CIT's validity as a discovery methodology follows from Theorem 10.3's analytic establishment of the abundance-masking phenomenon.

To identify hidden cost components in any extraction transaction, systematically strip away dimensions of abundance and observe which costs become visible at each dimension. Formally: for a cost component C_i, commons category A_j, and scarcity threshold τ_j, the cost is latent when $A_j \gg \tau_j$ and becomes visible as $A_j \to \tau_j$. The test proceeds by hypothetically removing each commons category in sequence:

1. Habitability commons (abundant state) → reveals oxygen processing, atmospheric regulation, and radiation shielding costs
2. Spatial commons (abundant state) → reveals spatial displacement costs: distance from home, transport, community immobility
3. Temporal commons (abundant state) → reveals acceleration costs: compressing multi-century availability into decades
4. Institutional commons (abundant state) → reveals governance costs: enforcement, accountability, honest accounting infrastructure
5. Kindred commons (abundant state) → reveals knowledge/cultural costs: transmission, continuity, community expertise below critical thresholds
6. Ecosystem commons (abundant state) → reveals ecosystem service costs: pollination, hydrological cycles, soil formation, waste processing
7. Political commons (abundant state) → reveals governance capture costs: lobbying-investment, regulatory capture, suppression of scientific findings, institutional corrosion
8. Cohesion commons (abundant state) → reveals civic-trust erosion costs: stranger-coordination collapse, commons-governance failures, civic-institution defunding, platform-mediation displacing civic trust
9. Epistemic commons (abundant state) → reveals epistemic-asymmetry costs: information suppression, AI-model opacity, algorithmic-accountability gaps, education-availability asymmetry
10. Autonomy commons (abundant state) → reveals coercion-shield costs: employment-relationship power asymmetry, consent-normalization, behavioral-architecture manipulation, addiction-induced choice-restriction

The test is complete when no further commons category can be identified whose removal reveals new costs.

Discipline note (commons categories as illustrative, not ontological). The ten categories above are organizational scaffolding over commons that CIT has surfaced across the cases this volume examines. They are presented as illustrative examples, not a closed ontological taxonomy — the framework's job is methodology (CIT + Four Gates); a political-philosophical tradition supplies the candidate commons. Different traditions may surface different categories or refine these; the methodology operates on whatever candidate commons a tradition treats as load-bearing.

§1.10 — Per-category commons profiles

Each profile names what the commons category prices, an abundant-state ↔ scarcity-grounded-cost reading, and the academic lineages that the framework engages on first introduction. Detailed per-category case clusters, distinctness analyses, and pre-drafting items are research-supporting material; what follows is the academic-reviewer summary.

Habitability commons. Life-support substrate (atmosphere; climate; radiation shielding; water cycle; soil-fertility cycle). Under abundance, atmosphere processes oxygen and regulates climate without explicit pricing; under scarcity, costs of oxygen processing, climate-regulation services, and shielding services become legible. Lineage: ecological-economics (Daly; Costanza et al. 1997 · 2014); planetary-boundaries (Rockström et al. 2009; Steffen et al. 2015); Stern Review 2007 climate-economics.

Spatial commons. Geographic / place-access (mobility, distance-to-services, neighborhood substrate, place-attachment). Under abundance, distance is small or transport is low-cost; under scarcity, costs of community immobility, place-loss, and spatial displacement become legible. Lineage: cultural geography (Tuan 1977 *Space and Place*); urban sociology (Jacobs 1961 *The Death and Life of Great American Cities*); environmental justice (Bullard 1990 *Dumping in Dixie*).

Temporal commons. Multi-generational / multi-decadal availability of resources and conditions. Under abundance, multi-century availability is unconstrained; under scarcity, acceleration costs (compressing multi-century availability into decades; accumulating temporal-deferred costs) become legible. Lineage: Hartwick rule 1977; Brundtland Commission 1987 *Our Common Future*; intergenerational equity (Brown Weiss 1989 *In Fairness to Future Generations*).

Institutional commons. Governance + accountability infrastructure (regulatory capacity, enforcement capability, honest accounting, oversight institutions). Under abundance, regulators have capacity and accounting is honest; under scarcity, costs of regulatory capture, enforcement gaps, and accountability hollowing become legible. Lineage: institutional economics (North 1990; Acemoglu & Robinson 2012 *Why Nations Fail*); regulatory-capture theory (Stigler 1971); state-capacity literature (Mann 1984; Tilly 1990).

Kindred commons. Identity-deep specific bonds (family, kinship, intimate-community ties, language community, indigenous-people bonds). Under abundance, identity-deep relationships persist intact across generations; under scarcity, costs of family severance, language loss, community-belonging loss, and intergenerational-trauma transmission become legible. Lineage: kinship-as-chosen-and-given (Weston 1991 *Families We Choose*); decolonial theory (Coulthard 2014 *Red Skin, White Masks*; Tuck & Yang 2012 "Decolonization is not a metaphor"); intergenerational-trauma scholarship.

Ecosystem commons. Biospheric services (pollination, hydrological cycles, soil formation, waste processing, biodiversity). Under abundance, ecosystem services are uncosted because abundant; under scarcity, costs of ecosystem-service erosion become legible. Lineage: Daily 1997 *Nature's Services*; Costanza et al. 1997 *Nature* and Costanza et al. 2014 update; planetary boundaries (Rockström et al. 2009; Steffen et al. 2015).

Political commons. Democratic-process infrastructure (representation, distributional contestation, electoral integrity). Under abundance, distributional contestation operates and political pricing of extractor costs is enforced; under scarcity, costs of political capture (lobbying-investment in suppression of honest accounting, regulatory enforcement, and democratic pricing) become legible. Lineage: Stigler 1971 capture theory; Olson 1965 *The Logic of Collective Action*; Acemoglu & Robinson 2012 on inclusive vs extractive institutions.

Cohesion commons. Civic-trust / collective-action infrastructure / non-kin coordination capacity. Under abundance, stranger-coordination is low-friction and civic infrastructure is intact; under scarcity, costs of civic-trust erosion (gig-economy atomization, civic-institution defunding, platform-mediation displacing neighborhood trust, commons-governance failures) become legible. Distinct from Kindred (which prices specific identity-deep bonds): the canonical non-co-occurring case is Kindred-abundance with Cohesion-scarcity (e.g., dense family ties + atomized civic life). Lineage: Putnam 2000 *Bowling Alone*; Coleman 1990

Foundations of Social Theory; Ostrom 1990 *Governing the Commons*; Tönnies 1887 *Gemeinschaft und Gesellschaft*; Klinenberg 2018 *Palaces for the People* on social infrastructure.

Epistemic commons. Access to evidence, knowledge, knowledge-structures (information availability, evidentiary access, education capacity, knowledge-evaluation infrastructure). Under abundance, all relevant facts and evidence are equally accessible and education-baseline is universal; under scarcity, costs of epistemic-asymmetry-as-shield (information suppression, AI-model opacity, algorithmic-accountability gaps, education-availability asymmetry) become legible. Forms a Kantian consent-pair with Autonomy: genuine consent requires both knowledge and non-coercion. Lineage: Akerlof 1970 information asymmetry (Nobel 2001); Fricker 2007 *Epistemic Injustice*; Hess & Ostrom 2007 *Understanding Knowledge as a Commons*; Habermas 1981 *Theory of Communicative Action*; AI-ethics literature (Crawford 2021 *Atlas of AI*; Benjamin 2019 *Race After Technology*; O'Neil 2016 *Weapons of Math Destruction*).

Autonomy commons. Non-coerced-choice capacity (workplace, consumer, civic, medical, cultural). Under abundance, choice is genuinely uncompelled and exit-from-arrangement is low-cost; under scarcity, costs of coercion-shield (employment-relationship power asymmetry, consent-normalization for 120-hour weeks, behavioral-architecture manipulation, addiction-induced choice-restriction, cultural-assimilation coercion) become legible. Forms a Kantian consent-pair with Epistemic. Sen's capability approach operates at the Epistemic ↔ Autonomy cross-reference: capability is developed through education (Epistemic) and exercised through non-coerced choice (Autonomy). Lineage: Kant 1785 *Groundwork of the Metaphysics of Morals*; Mill 1859 *On Liberty*; Sen 1999 *Development as Freedom*; Nussbaum 2011 *Creating Capabilities*; Anderson 2017 *Private Government*.

Scope-of-applicability boundary at the extreme-scarcity end. At the extreme-scarcity end of the Autonomy commons (forced labor; slavery; coerced extraction generally), the framework's analytical apparatus reaches a boundary. The framework's primary case-class operates through information-asymmetry vectors (cost-bearers proceed because they do not know what they will bear — NFL concussions; cigarette industry concealment; opioid Appalachia; Libby vermiculite). Coerced extraction is structurally distinct: cost-bearers know what they will bear and are forced to bear it nonetheless. The direct pricing of the *coercion vector itself* — what should be assigned to the coercive dimension as cost — remains methodologically unresolved in the reparations-economics field (Darity 2026). The framework's methodology extends to coerced-extraction cases through the *legacy-effects pricing pathway* — longevity gaps, wealth gaps, cultural-knowledge transmission disruption, intergenerational trauma — computed using Method 1 + Method 2 + Method 3 triangulation applied backward (see §5.5 below). The named coercion-variable is acknowledged here at first introduction; the direct pricing of coercion-as-cost-dimension is outside this volume's scope.

Heterogeneous-stakeholder commons scope. A second scope-of-applicability boundary the framework names explicitly is the heterogeneous-stakeholder commons. Where the cost-bearing community drawing from a shared commons is structurally divided along racial, class, ethnic, or analogous group lines, and carries conflicting use-preferences for the commons itself, the Ostrom-tradition apparatus does not produce a solution — Ostrom's eight design principles (Ostrom 1990) presume substantially-aligned stakeholders capable of negotiating governance institutions, and structurally divided stakeholders fall outside that scope. This framework's apparatus reaches such commons because the Commons Inversion Test operates on the commons-extraction relationship rather than on the stakeholder-coordination relationship; the cost severance the framework prices is measurable whether the cost-bearing community is internally aligned or divided. This break-point connects the framework to the analytical apparatus of *stratification economics* (Darity, Hamilton, and collaborators), which analyzes structural-extraction operating through hierarchical group-divisions of race, class, and ethnicity (Darity 2026). The framework's commons-governance extension applies stratification-economics analysis of intergroup relationships to commons that are simultaneously contested across stakeholder-group lines and being extracted from by structural outsiders.

Scope: complementarity with Public Choice / rent-seeking analysis. The framework's cost-severance accounting operates at the architecture level — identifying which parties absorb which costs given an institutional architecture's distributive properties. A complementary analytical question — who shaped the institutional architecture, by what political-economic process, with what rent-seeking expenditure — is the domain of Public Choice analysis (Buchanan & Tullock 1962; Tullock 1967; Mueller 2003 and successors). The framework is compatible with Public Choice analysis layered on top of it, and depends on it for cases where the architecture's origins are part of the substantive analytical work. The framework does not attempt to replicate Public Choice's actor-identification methodology; the framework's apparatus is calibrated for cost-bearing-party identification + magnitude, not for political-coalition-of-architecture-shapers analysis. Applicability conditions: *sharpest application of Public Choice complementarity* — cases where extractor populations and cost-bearer populations are heterogeneous (corporate operator vs downstream community; industry lobbying vs unaffiliated household; financial-services concentration vs diffuse depositor base); *blurred application* — cases where extractor and cost-bearer populations overlap (consumer-good supply chains where consumers occupy both roles; intergenerational dynamics where temporal-distance creates an actor-identification challenge that present-tense Public Choice analysis is not optimized for); *out of scope of this Technical Appendix* — rent-seeking-actor-identification methodology; the Appendix does not develop this apparatus and refers readers to the Public Choice literature for it. Readers in the Public Choice tradition should read the framework's cost-bearing-party identification as a foundation their apparatus can apply to, not as an alternative to their apparatus.

Definition 1.9 (Existentially Critical Resource / existential substitutability gap Classification)

A resource R is existentially critical if its substitutability $S_{max} \le S_{threshold}$ for any application class that may be necessary for species-level viability or major civilizational threshold events (including but not limited to: off-world settlement, advanced energy generation, high-density computing, precision manufacturing). The Civilizational Substitutability Gap (existential substitutability gap) for resource R is defined as:

existential substitutability gap(R) = $S_{max}(\text{industrial}) - S_{max}(\text{existential})$

where $S_{max}(\text{industrial})$ is the maximum achievable substitutability across current commercial applications and $S_{max}(\text{existential})$ is the maximum achievable substitutability across foreseeable civilizational-threshold applications. High existential substitutability gap resources warrant strategic in-situ preservation because the insurance cost (foregone extraction revenue from designated reserves) is typically a small fraction of projected extraction value.

§2. Two-Instrument Architecture (CSD ↔ B1; RCV ↔ B2)

§2.1 Architecture overview

The framework operates two distinct accounting instruments — one **backward-looking** (already-extracted commons), one **forward-looking** (potential / claim-bearing commons). Both are governed by the same central equation `CS = RCV − B` (Cost Severance equals Residual Commons Value minus Bond posting). The two instruments differ in which side of the equation is being measured.

CSD — Cost Severance Damages (backward-looking):

- Question answered: *What was severed?*
- Measures CS that has already accumulated through past extraction.
- Anchors restitution claims (B_1 Restitution Bond — see §4 below).
- Empirical input: realized extraction history; realized commons-degradation; realized cost-bearer claims.
- Discount-rate posture: low / zero / negative (already-realized harm; reparations-economics traditions per Darity & Mullen 2020).

RCV — Residual Commons Value (forward-looking):

- Question answered: *What's left, and what would severing it more cost?*
- Measures commons-value that the extraction relationship has NOT yet captured / consumed / extinguished.
- Anchors foreclosure-prevention claims (B_2 Foreclosure Bond — see §4 below).
- Empirical input: replacement cost (Method 1 — see §3.3) + revealed preference (Method 2 — §3.4) + scarcity-adjusted option value (Method 3 — §3.5).
- Discount-rate posture: option-value-bearing; scarcity-adjusted (commons-finite Hotelling-like rents — see §4 Hotelling Identity).

Why two instruments + one equation: the framework's structural identity is *commons-governance extended to extraction* . Past extraction (already-realized commons-loss) is governance-relevant for restitution. Pending extraction (un-realized commons-claim) is governance-relevant for foreclosure prevention. Same equation reads each direction; same Bond aggregate ($B = B_1 + B_2$) discharges both.

§2.2 CSD–RCV correlation hypothesis

An initial hypothesis was: contexts with high CSD (large past extraction) tend to have low RCV (commons largely consumed); contexts with low CSD (early-stage extraction) tend to have high RCV (commons largely intact).

Empirical finding: two-case calibration (Norway + McDowell — see §11.5, §11.6) found the inverse-correlation hypothesis NOT SUPPORTED. Reframed hypothesis: *extraction-history-driven co-elevation of CSD + RCV (positive correlation as severity-signature, not independence finding).* Both Norway (moderate CSD + high uncaptured RCV) and McDowell (very high CSD + very high uncaptured RCV) carry uncaptured RCV; McDowell more on both axes. The two instruments remain structurally independent (orthogonal measurements) but co-driven by extraction history.

§3. RCV Quantification — Three Ways of Counting

§3.1 — Core Formula (Term-by-Term Translation)

$RCV(R, t_0) = \int_{t_0}^{\infty} \{ [1 - S(t \mid t_0)] \cdot U(R, t, Q(t)) + E(R, t) \} \cdot D(t, t_0)\, dt$

Plain-language translation, term by term:

- $[1 - S(t \mid t_0)]$ is the probability that no adequate substitute exists at time t. When substitutes are fully available, this term approaches zero and the foreclosure cost vanishes. When substitutes are unavailable, this term approaches 1 and every unit of U is a genuine cost.
- $U(R, t, Q(t))$ is the utility future generation t derives from having R available in situ. This includes economic value, option value, and ecological function value. It is bounded below by the current market price and rises as Q declines.
- $E(R, t)$ is the externality cost burden at time t from extraction at t_0. This term is independent of substitution: even if a substitute for coal is found, the carbon already in the atmosphere is still there. The externality tail runs on its own clock.
- $D(t, t_0)$ is the social discount factor — the weight we assign to the welfare of generation t. Following Weitzman (2001), this declines toward zero as $t \to \infty$ but does so slowly enough that the integral converges (see Theorem 10.4).

§3.2 — The Intergenerational Pricing Gap (IPG)

The IPG is the ratio of RCV to current market price P:

$IPG(R, t_0) = RCV(R, t_0) / P(t_0)$

IPG > 1 indicates underpricing. IPG = 33 means the market price covers approximately 3% of the true intergenerational cost. All empirical cases tested (§11) yield IPG ≫ 1, consistent with Theorem 10.1b (Structural Cost Severance).

Sensitivity analysis and numerical validation results (§11) inform the IPG estimates reported below.

§3.3 — Method 1: Replacement Cost

Question: What would it cost to engineer a substitute for the commons that extraction is consuming?

Formal: $RCV_{M1} = \sum_i$ (replacement-cost per unit commons-i × commons-i remaining quantity)

Worked-example anchor (Habitability commons): direct-air-capture (DAC) cost per ton CO_2 at engineering scale. Three-horizon range across documented engineering scenarios:

- **Conservative (current operational; first-of-a-kind):** \$600–\$1,000/ton CO_2 (Climeworks Orca, Mammoth)

- **Mid-range (at-scale commercial within next decade):** \$300–\$600/ton CO_2 (Mammoth full-operation target; Carbon Engineering Stratos)
- **Optimistic (2050 nth-of-a-kind):** \$100–\$300/ton CO_2 (IEA, IPCC AR6, US National Academies)

SCC-vs-DAC distinction: Method 1 uses DAC (substitution-side; engineering reversal cost). Social Cost of Carbon (currently US EPA \$190/ton CO_2 central estimate; Rennert et al. 2022 *Nature*) is the damage-side number. Both appear in the Ch 6 convergence-table walkthrough but answer different questions.

Strengths: Most directly priced; smallest epistemic-ambition step. **Limits:** Anchors floor only — replacement is rarely full functional substitute. Underprices irreplaceable commons.

§3.4 — Method 2: Revealed Preference (Norway-Style)

Question: What does a polity that explicitly chose NOT to extract reveal about commons-value?

Formal: RCV_{M2} = (foregone-extraction value at observed restraint) + (preserved commons-quality observed) − (substitute-revenue path)

Worked-example anchor: Norway sovereign-wealth-fund + petroleum-conservation policy. Per §11.5 numerical execution: GPFG ~\$1.9T at ~75% capture rate of net petroleum wealth implies ~\$2.5T total petroleum wealth ÷ 50B BOE cumulative production ≈ **\$50/BOE Method 2 anchor** .

Strengths: Empirically grounded in actual policy choices (not hypothetical). **Limits:** Norway operates as the canonical existing B2-exemplar; under the framework's epistemic-humility discipline, M2 is a LOWER BOUND on true RCV (Norway captured rent but did not capture the full atmospheric externality; non-Norwegian populations affected by Norwegian oil's climate externalities are not in the GPFG). M2 is a revealed-preference-anchored MIDDLE-WITHIN-TRIANGULATION but LOWER-BOUND ON TRUE RCV.

§3.5 — Method 3: Scarcity-Adjusted Option Value

Question: What is the option value of preserving the commons in irreversibility-aware scarcity-aware terms?

Formal specification:

```
RCV_M3 = V_option × scarcity_multiplier(σ) × irreversibility_premium(α)

  V_option                      = standard real-options option value (Dixit-Pindyck 1994)
  σ                             = scarcity parameter (commons-stock / sustainable-flow ratio)
  α                             = irreversibility parameter (probability commons cannot be
restored
                                  at any finite cost once extracted; α ∈ [0, 1])

  scarcity_multiplier(σ)        = 1 + log(1 + σ) × Hotelling_anchor
  irreversibility_premium(α)    = 1 / (1 − α)^β     where β > 0 calibrates risk-posture
                                                       (β = 1 default; β = 2 precaution-regime)
```

Functional-form motivation (scarcity_multiplier). The log(1 + σ) form is selected for four converging reasons:

- *Hotelling rent-path conjugate*. Under Hotelling 1931 rent-path dynamics, optimal extraction prices grow exponentially at the long-run interest rate; the logarithmic form is the conjugate that linearizes this exponential under integration. The Hotelling_anchor coefficient maps σ-induced multiplier growth onto the documented Hotelling-rate range (~5%/yr proxy across documented σ regimes).
- *Welfare-economics log-utility lineage*. Atkinson 1970 *Journal of Economic Theory* ("On the Measurement of Inequality") + Cobb & Douglas 1928 *American Economic Review* establish the logarithmic form as standard in welfare-economics utility-aggregation under risk + inequality contexts. The framework adopts a parallel discipline: scarcity-induced cost amplification is logarithmic in stock-depletion ratio, consistent with bounded-concavity preferences in present-value comparison.
- *Bounded concavity + numerical tractability*. Slow-growing logarithmically; practically bounded across documented σ range; structurally unbounded as σ → ∞ (the divergence reflects the framework's commitment to finite-but-large-cost regimes near critical-stock thresholds, not a numerical pathology). Alternative asymptotic-bounded forms (e.g., σ/(1 + σ) bounded; arctan(σ) bounded) are admissible under sensitivity-robustness scoping; the cross-case α-dominance regime is preserved across all admissible concave forms (see Method 3 sensitivity analysis §11.8 for full numerical verification).
- *Adjacent intergenerational-equity lineage*. Solow 1956 *Quarterly Journal of Economics* + Bergson 1938 *Quarterly Journal of Economics* on long-horizon utility aggregation under intergenerational discounting establish the broader concave-aggregation discipline that the logarithmic form sits within.

Sensitivity finding (cross-case dominance map):

Regime	σ	α	Dominant parameter	Example case
Commons-extinction-class	any	0.9–1.0	**α**	McDowell coal; biodiversity collapse; atmospheric tipping points
Scarce-but-restorable	100+	0.1–0.5	**σ**	Aquifer overdraft; old-growth timber
Abundant-restorable	0.1–10	0–0.1	**V_option**	Asteroid iron; cosmic-scale resources
Hybrid (most real cases)	10–100	0.5–0.9	mixed; **α** → **σ** → **V_option**	Norway oil; Deepwater Horizon; opioid Appalachia

Framework debate-narrowing implication: Earth-bound civilization-scale extractions cluster in α-dominance regime. The real debate about RCV for those cases is a debate about **irreversibility** — not about discount rates, scarcity, or option-value. Tractably empirical; aligned with Asymmetric Regret Rule (§8.2); aligned with climate-tipping-point + biodiversity-collapse + cultural-loss literatures.

§3.6 — Triangulation logic

The three methods price the same commons-claim from three independent directions: M1 prices *substitution* ; M2 prices *revealed restraint* ; M3 prices *option-bearing scarcity* . **Triangulation rule:** RCV estimate is the convergence range across M1, M2, M3 with documented divergences. Where the three methods agree within a tight range, RCV estimate is well-supported. Where they diverge widely, the divergence itself is empirically informative — surfacing what feature of the commons each method captures.

Reporting discipline: every RCV estimate published under the framework reports all three methods individually + the convergence range + identified divergence sources. No single-method RCV claim has standing.

Worked-example results across calibration cases:

Case	M1 (Replacement Cost)	M2 (Revealed Preference)	M3 (Option Value, β=1)	Implied CS
Norway oil	\$300–650/BOE	\$50/BOE	\$70–1,000/BOE; mid \$280	~\$250/BOE; ~\$12.5T cumulative
Appalachian coal (McDowell)	\$310–\$1,800/ton	\$8–\$88/ton	\$420–\$13,100/ton; mid \$2,500	~\$2,400/ton; IPG ~50× (2025\$)

§4. Hotelling Identity (extension of Hotelling 1931)

Attribution discipline: Hotelling 1931 (*Journal of Political Economy* 39(2): 137–175) is the foundational source for the rent rule on exhaustible resources; that work is **Hotelling's** and is cited per standard academic convention. The framework's contribution is the *identity articulation* RCV − Hotelling rent = CS per unit, which pairs the framework's commons-side measurement (RCV, defined within this Tech Appendix) with Hotelling's extractor-side rent rule. Both contributions are clearly attributed below.

§4.1 Hotelling 1931 — the part that's Hotelling's

Hotelling (1931, "The Economics of Exhaustible Resources," *Journal of Political Economy*) established that for an exhaustible resource extracted competitively under a discount rate r, the resource rent (price minus marginal extraction cost) must rise at rate r. Otherwise the owner would prefer to extract earlier (if rent rises faster) or later (if slower). This is the **Hotelling rent rule** — a foundational result in resource economics. Hotelling's framing assumes: resource is exhaustible (finite stock); owner is private (extracts to maximize NPV of rent); cost is private extraction-cost (not commons-loss-cost).

§4.2 What the Commons Bonds framework adds (the "Identity" extension)

The Commons Bonds framework extends Hotelling 1931 with the following identity:

```
Hotelling Identity:     RCV per unit − Hotelling rent per unit = CS per unit
```

In words: *the residual commons value per unit of extraction, minus the standard Hotelling rent per unit, equals the cost severance per unit* . **What this surfaces:** when extraction prices the standard Hotelling rent (i.e., honors private resource-economics rent-rules), the *excess* residual commons value beyond that rent is precisely the cost-severance — the commons-extraction cost the extractor is not paying. The Hotelling rent prices private exhaustibility; the framework prices commons-extraction. The difference is the asymmetry the framework names.

Worked-example anchor: Norwegian oil. Norway prices Hotelling rent appropriately (sovereign-fund accumulation; depletion-rate discipline). The Hotelling Identity asks: even with proper Hotelling-rent pricing, what's the residual commons value Norway preserves through its restraint? Answer: the difference between the Norwegian-fund's accumulation and the full RCV (per Method 2 revealed preference). That difference is the commons-loss Norway is NOT incurring — and thus also reveals what other extraction contexts ARE incurring when they price Hotelling rent only.

§4.3 Citation discipline

Anywhere the framework references the Hotelling Identity, the canonical attribution form is:

> The Hotelling Identity (RCV − Hotelling rent = CS, where Hotelling rent is the standard rent rule from Hotelling 1931) extends Hotelling's exhaustible-resource economics to commons-extraction contexts. Hotelling's contribution: the rent rule for private exhaustible resources. The Commons Bonds framework's contribution: the identity extension showing that commons-extraction cost is precisely the residual value beyond standard Hotelling rent.

§5. Accountability Bond Decomposition ($B = B_1 + B_2$; Restitution + Foreclosure Bonds)

The framework's aggregate Accountability Bond B decomposes into two structurally-independent sub-instruments: a backward-looking instrument (B_1, Restitution Bond) that closes the realized-harm gap, and a forward-looking instrument (B_2, Foreclosure Bond) that closes the prospective-foreclosure gap. The decomposition refines the framework's central equation CS = RCV − B by partitioning total cost severance into two temporally-distinct accountability obligations:

```
total CS = (CSD − B₁) + (RCV − B₂)
```

where CSD = Cost Severance Damages (already-realized commons-loss, integrated over historical extraction) and RCV = Residual Commons Value (forward-looking commons-loss claim, integrated over future foreclosure horizon). B_1 pairs structurally with CSD; B_2 pairs structurally with RCV.

§5.1 Sub-instrument formal definitions

§5.1.1 — B_1 Restitution Bond

Definition. B_1 — Restitution Bond is the backward-looking sub-instrument in the framework's two-instrument decomposition of Accountability Bond ($B = B_1 + B_2$). It denotes the cluster of accountability mechanisms that close the Cost Severance Damages (CSD) gap by restoring — in monetary, legal, structural, or rights-based form — what was taken from people via realized cost severance. In the decomposed equation total CS = (CSD − B_1) + (RCV − B_2), B_1 pairs structurally with CSD: B_1 is the dollars that would discharge the realized-harm half of total severance.

Mechanisms in cluster. Restitution payments; reparations programs; welfare-state instruments (universal healthcare, social-safety-net programs, public pensions); environmental-justice damages compensation; toxic-tort litigation; class-action damages; truth-and-reconciliation processes.

Why "Restitution" over "Reparations." Legal-register-clear; cross-political-tradition adoption potential. "Reparations" carries political-loading that narrows framework adoption to politically-progressive audiences; "Restitution" preserves the same restoration-of-what-was-taken meaning while reaching center-right, libertarian, and politically-cautious institutional audiences who would disengage at "Reparations." Restitution-law scope is broader than damages-law (encompasses monetary + legal + structural + rights-based restoration), fitting CSD's broader scope.

Lineage. Restitution-law tradition (general legal vocabulary); Darity & Mullen 2020 *From Here to Equality* (UNC Press) primary reparations-economics methodology source; Bullard 1990 *Dumping in Dixie* (Westview) environmental-justice damages cross-reference; toxic-tort + class-action damages tradition; truth-and-reconciliation processes (South Africa post-apartheid + analogous infrastructures); Esping-Andersen 1990 *The Three Worlds of Welfare Capitalism* (Princeton) welfare-state design lineage; Hicks 1932 compensation principle in cross-context-translation form.

Status of B_1 globally. Underdeveloped. Restitution-and-reparations is the active scholarly territory; less politically settled than B_2; varies widely across welfare-state architectures; environmental-justice damages compensation is litigation-driven and incomplete.

Positioning relative to the Darity-Mullen reparations typology. Darity & Mullen 2020 *From Here to Equality* structures reparations as a three-component program: (1) acknowledgment-and-atonement; (2) *redress*, which itself decomposes into two sub-types — symbolic redress and material restitution — of which Darity and Mullen explicitly declare in favor of restitution; and (3) closure (the recipient community's standing to make further claims if subsequent harms occur). In the Darity-Mullen typology, "restitution" is therefore not a concept alongside-and-distinct-from reparations — it is the material sub-type of the redress component within their reparations program. The framework's Restitution Bond (B_1) is positioned as the instrument that operationalizes the restitution component of redress in Darity-Mullen's typology, extended to the broader case-class the framework engages (commons-extraction cases beyond the Black-American-descendants-of-US-slavery context Darity and Mullen specifically scope). Darity (2026) did not view the framework's calculations across its broader case-class — cost-severance and extraction-community contexts beyond the reparations apparatus Darity and Mullen specifically scope — as underestimating what is due. The framework's Method 1 + Method 2 + Method 3 triangulation discipline (§3) is the framework's structural pledge against underestimation. Where the framework writes "Restitution Bond" without qualifier, the reader should read it as *the instrument operationalizing restitution-as-redress within the Darity-Mullen reparations framework*, not as a separate accounting object.

§5.1.2 — B_2 Foreclosure Bond

Definition. B_2 — Foreclosure Bond is the forward-looking sub-instrument in the framework's two-instrument decomposition of Accountability Bond (B = B_1 + B_2). It denotes the cluster of accountability mechanisms that close the Residual Commons Value (RCV) gap by forcing extractors to internalize the dollar quantification of permanent foreclosure imposed on future generations. In the decomposed equation total CS = (CSD − B_1) + (RCV − B_2), B_2 pairs structurally with RCV: B_2 is the dollars that would discharge the prospective-foreclosure half of total severance.

Mechanisms in cluster. Hartwick savings (sovereign wealth funds — Norway's GPFG = canonical existing exemplar); reclamation bonds (SMCRA 1977 surface mining); Environmental Impact Bonds (Balboa 2016); Pigouvian carbon taxes; cap-and-trade schemes; decommissioning bonds for oil and gas (BLM 43 CFR Part 3162; BSEE 30 CFR Part 250 + 254; ~36 state plugging/abandonment bond regimes); CERCLA / RCRA financial-assurance requirements.

Why "Foreclosure Bond" over "Resource-Foreclosure Bond." Term-pair coherence with Foreclosure Cost (C_1 in RCV integrand) — *Foreclosure Cost is what's lost; Foreclosure Bond is what would internalize it*. Shorter; preserves descriptive content via framework context; dinner-table-strong via mortgage-foreclosure analogy (in mortgage-foreclosure, ownership permanently severed from borrower; in resource-foreclosure, future-use permanently severed from future generations).

Lineage. Hotelling 1931 *Journal of Political Economy* — foundational resource-economics; Hotelling rent under honest accounting flows to B_2 instrument. Hartwick rule 1977 — invest resource rents in reproducible capital. Solow 1974 — intergenerational equity and exhaustible resources. OSMRE Reclamation Bonds + GAO-17-207R + Yang & Davis 2021. Balboa 2016 *Accountability of Environmental Impact Bonds*. Pigou 1920 *Economics of Welfare*.

§5.2 Naming outcome

Earlier working labels were "Reparations Bond" for B_1 and "Resource-Foreclosure Bond" for B_2. For B_1, "Reparations Bond" was reviewed against "Restitution Bond." For B_2, "Resource-Foreclosure Bond" was reviewed against "Foreclosure Bond," "Future-Options Bond," "Long-Horizon Bond," "Intergenerational Resource Bond," "Hartwick Bond," "Sovereign Reserve Bond," and "Forward Accountability Bond." The current names were selected on three criteria.

Cross-political-tradition adoption breadth. "Reparations" carries US-political-loading specific to the Black-American descendants-of-US-slavery context that Darity and Mullen specifically scope. "Restitution" carries broader cross-tradition adoption — legal-scholarship, civil-law, international-law, and classical-philosophy traditions all use the term — which suits the framework's commons-extraction case-class extending beyond that population.

Term-pair coherence. "Restitution Bond" + "Foreclosure Bond" reads as a natural pair: both are short two-word labels denoting the backward-and-forward sub-instruments of the same Accountability Bond aggregate.

Descriptive content preserved via framework context. The shorter "Foreclosure Bond" over "Resource-Foreclosure Bond" recovers its domain meaning from the surrounding framework apparatus rather than from internal label-length; §5.1.2's *Why "Foreclosure Bond" over "Resource-Foreclosure Bond"* note supplies the B_2-specific rationale (Foreclosure Cost / Foreclosure Bond term-pair coherence; dinner-table-strong via mortgage-foreclosure analogy).

§5.3 Independence note

B_1 and B_2 are **structurally independent** — a context with high CSD but low forward-looking RCV (e.g., post-mining Appalachian sites where commons largely already lost) carries high B_1 obligation but low B_2 obligation. A context with low CSD but high RCV (e.g., asteroid mining proposed but not yet executed) carries low B_1 obligation but high B_2 obligation. The aggregate B captures the total bond posture; the decomposition captures the temporal structure of the obligation.

Norway as canonical-existing-B_2-exemplar: Norway's GPFG operationalizes Hartwick's rule (1977) by capturing extraction rent for future-generation accounts. This is canonical existing B_2, not aggregate B — Norwegian welfare-state is approximately B_1-for-Norwegian-citizens but does not extend to non-Norwegian populations affected by Norwegian oil's climate externalities. B_1 globally is underdeveloped.

§5.4 Intergenerational moral grounding (Parfit 1984 standing + Sen 1985/1999 valuation; B_2 Foreclosure Bond direction)

> The framework's intergenerational moral grounding follows Parfit (1984) Part IV. Specifically: the framework adopts the impersonal-outcomes evaluation framework (the goodness-of-future-states-for-whoever-arises-in-them) rather than person-affecting evaluation (the wronging-of-specific-future-individuals). The two-instrument architecture maps onto this distinction: B_1 Restitution Bond addresses already-realized person-affecting harm (CSD; the historical extraction-portfolio); B_2 Foreclosure Bond addresses prospective impersonal-outcomes loss (forward-looking RCV; the commons not available to whatever future generations arise). Parfit's non-identity problem is not solved by the framework but is operated-within: the framework's claim is that *the goodness of future-generation outcomes* is the morally-relevant variable, not *which specific future persons are wronged* . This positioning legitimates IPG as a moral accounting object rather than as a free-standing posit.
>
> The intergenerational moral grounding pairs two traditions, each answering a distinct question. Parfit (1984) Part IV grounds the *standing* question (developed above) — whether future generations have moral standing whose extraction-foreclosed welfare counts in present accounting. Amartya Sen's capability-and-social-welfare-function framework (Sen 1985; Sen 1999) grounds the *valuation* question — how that standing should be operationalized in measurement. The framework's Residual Commons Value (§3) is, in Sen's vocabulary, a measurement of the capability set extraction forecloses on whoever-the-future-people-turn-out-to-be: the substitutability function $S(\tau \mid t_0)$ captures the probability that a given capability remains accessible at future time τ given foreclosure decisions at t_0, and the externality-tail and foreclosure-cost integrands measure the welfare reduction across capability-foreclosed states. The Parfit / Sen pairing — Parfit grounding the standing, Sen grounding the valuation — is what makes the framework's intergenerational-pricing claim defensible across the moral-philosophy and welfare-economics traditions simultaneously rather than within only one (Darity 2026).

Bibliography: Parfit, Derek. *Reasons and Persons* . Oxford: Oxford University Press, 1984. Part IV (intergenerational moral status; non-identity problem). Sen, Amartya. *Commodities and Capabilities* . Amsterdam: North-Holland, 1985 (capability-set valuation foundations); *Development as Freedom* . New York: Knopf, 1999 (capability-and-social-welfare-function framework).

§5.5 — Bidirectional application of the framework apparatus

The framework's apparatus — Commons Inversion Test (§6), Four Gates admission (§7), and Three Ways of Counting triangulation (§3) — is direction-agnostic in its formal statement. Theorem 10.3 (Abundance Masking) operates whether the abundance being stripped is forward-projected or backward-realized: as abundance approaches its scarcity threshold, costs that were masked by that abundance become visible. The same methodology applies in both temporal directions; what differs is the cost-stream being priced.

Forward application (this volume's empirical scope). The framework's documented case studies — McDowell coal, Norway petroleum, Deepwater Horizon, Libby vermiculite, Baotou rare earths — apply the Four Gates and Three Ways of Counting to price what extraction events forward of analysis time would foreclose. The published RCV figures and the implied Cost Severance figures (§11.1–§11.6) operate in this register. The canonical equation `CS = RCV − B` (§1.7) is the forward-pricing form used throughout this volume's empirical computations.

Backward application (structurally identical; outside this volume's empirical scope). The same apparatus applies in reverse to price backward damages — the legacy effects of extraction that has already occurred. Just as the abundance of clean air masks atmospheric externality costs until air becomes scarce (Mars colony habitability per Scenario 1 of §12; climate-crisis legibility on Earth), the abundance of freedom masks coercion costs until freedom becomes scarce. The Commons Inversion Test's structural move — invert abundance to reveal masked costs — runs both directions. Stripping atmospheric abundance reveals atmospheric externality costs (forward, documented in §11); stripping the abundance of Autonomy commons (Definition 1.10) reveals coercion costs (backward, outside this volume's empirical scope). The Three Ways of Counting estimate value from three independent epistemological angles in both directions: *Method 1* prices substitution forward (cost of engineering a substitute commons function) and remediation/compensation backward (cost of repairing or compensating realized damage). *Method 2* prices revealed restraint forward (Norway's GPFG as a polity's revealed valuation of forward foreclosure-prevention) and revealed restitution backward (Darity & Mullen 2020 reparations-economics methodology + the case lineage of Holocaust reparations, the 1988 Civil Liberties Act for Japanese-American internment redress, Black Lung Trust Fund payouts, the South African Truth and Reconciliation Commission). *Method 3* prices scarcity-adjusted option value forward (the option value of preserving forward optionality) and the same option value extinguished at the time of past extraction backward (evaluable retrospectively from parameters known at analysis time, applied to the realized-extraction case).

Four Gates direction-agnosticism (gate-by-gate). The Four Gates Admission Apparatus (§7) admits a candidate C_i on identical structural requirements in both directions; only Gate 3 carries an operational asymmetry favoring the backward direction. Gate 1 (CIT) tests scarcity-grounding under the same counterfactual structure — a backward-direction C_i (longevity gap; wealth gap; cultural-transmission disruption) reduces to zero under counterfactual abundance of the foreclosed commons just as a forward-direction C_i does. Gate 2 (Units Consistency) admits a backward C_i if and only if its units satisfy [$\$ \cdot \text{resource-unit}^{-1} \cdot \text{time-unit}^{-1}$], identical to the forward requirement (or, for measured legacy-effect quantities, a dollar-converted snapshot form at analysis time). Gate 3 (Boundedness) is the operational asymmetry: forward-direction integration runs over $t \to \infty$ and requires convergence analysis under D's asymptotic decay (§16.1); backward-direction pricing operates on legacy-effect quantities bounded by construction (the longevity gap is bounded in years; the wealth gap is bounded in dollars; the historical extraction-portfolio is bounded in documented production records), so Gate 3 is typically satisfied trivially for backward C_i. Gate 4

(Independence) tests non-overlap with existing terms in the integrand identically across directions; the directional flip does not alter the disjointness requirement.

Epistemic asymmetry across directions. The structural symmetry of the apparatus does not entail epistemic symmetry of the cost-estimation work. Backward-direction C_i is often empirically anchored: the longevity gap is measured in years (vital-statistics data); the wealth gap is measured in dollars (Federal Reserve Survey of Consumer Finances + adjacent series); the historical extraction-portfolio is documented in production records. Forward-direction C_i requires modeling under counterfactual scenarios — utility a future generation derives from in-situ stock; externality flow across the externality tail; substitutability function $S(t \mid t_0)$ trajectory. The structural symmetry of the gates and the counting methods is preserved; applied work in the two directions draws on different evidence bases and is subject to different sensitivity-analysis disciplines — backward work to data-quality and methodological-choice sensitivity (cf. Darity & Mullen 2020 reparations-debate methodology); forward work to discount-rate, S(t) trajectory, and horizon-truncation sensitivity (§16.1 + §11.8).

Total CS, decomposed. Within the full bidirectional framing, total cost severance for any extraction context is:

$$\text{total CS} = (CSD - B_1) + (RCV - B_2)$$

where the first term prices backward (already-realized damages minus realized restitution) and the second term prices forward (forward-projected foreclosure minus forward bonds posted). The high-level form `CS = RCV − B` used throughout this volume's chapter prose (Ch 6 line 304 et seq.) is exact in cases where the backward severance has been substantially restituted ($CSD - B_1 \approx 0$) or in pure forward-looking analysis bracketing backward damages. This volume operates in the latter mode by design; the equation it carries is the forward-pricing form.

Example backward variables (cited; not computed in this volume). The framework's methodology accommodates backward-direction variables this volume does not price. The principal examples in the contemporary literature:

- *The racial wealth gap as legacy-effect pricing.* Darity & Mullen 2020 *From Here to Equality* applies Method 1 + Method 2 + (implicit) Method 3 reasoning against the racial wealth gap accumulated through American policy choices since 1619. Their resulting price-of-repair estimate is the most developed backward-direction CSD computation in the contemporary literature. The framework's apparatus accommodates this work directly; the methodology is the framework's, applied backward, with Darity and Mullen as the primary developers.
- *The longevity gap as legacy-effect pricing.* Darity and colleagues' more recent work prices the 6–7 year average life-expectancy differential between Black-American and white-American populations as a measurable legacy effect of cumulative policy. McDowell County's 13-year life-expectancy gap (cited Ch 2 + Ch 6) is the same legacy-effect pricing variant applied to coal-extraction-affected populations. The framework's apparatus accommodates this work directly.
- *The coercion vector itself.* Slavery and forced-labor cases are structurally distinct from the framework's information-asymmetry analytical case-class (per §1.10 Autonomy commons "Scope-of-applicability boundary"). The reparations-economics field flags the direct pricing of coercion-as-cost-dimension as methodologically unresolved (Darity 2026). The framework names the variable and extends to coerced-

extraction cases through the legacy-effects pathway above; it does not claim to have priced the coercion vector directly.

- *Indigenous land dispossession; cultural-knowledge severance; intergenerational trauma transmission.* Cross-domain backward variables operating through the Kindred, Cohesion, and Epistemic commons (Definition 1.10). Each is structurally accessible to the framework's methodology applied backward; computed prices for these variables sit outside this volume's empirical scope.

Sibling-section structure (§5.3 / §5.5). §5.3 (Independence note) and this section (§5.5) are sibling structural consequences of the decomposition $B = B_1 + B_2$. §5.3 covers *structural independence* — B_1 and B_2 can vary independently context-by-context (high-CSD-low-RCV in post-mining Appalachia; low-CSD-high-RCV in pre-extraction asteroid mining; Norway's paid-for-citizens B_1 alongside globally-underdeveloped B_1 and B_2). This section covers *structural symmetry* — the apparatus admitting C_i into either direction runs identically. Together they specify the decomposition's two consequences for applied work: the two sub-instruments vary independently across contexts (§5.3), and the framework's methodology applies symmetrically across the two directions (§5.5).

Empirical scope discipline. This volume's empirical computations price forward foreclosure (RCV-side) for the case studies in §11. The backward application is structurally identical methodology — the example variables above demonstrate the framework's reach — with field-extending papers and third-party applications applying the methodology backward to the cases this volume cites but does not compute. The framework's reach is bidirectional by construction; this volume's empirical scope is forward by design.

Coordination note (B_1-direction moral grounding). §5.4 (Intergenerational moral grounding) is currently scoped to the B_2 direction with Parfit 1984 grounding standing and Sen 1985/1999 grounding valuation, both mapped to forward-direction RCV. The Parfit framing also carries a person-affecting reading (the wronging of specific persons), which maps to backward-direction realized harm and is developed operationally by the reparations-economics tradition (Darity, Mullen, Hamilton, Coates, Anderson; Method 2 backward-direction exemplars cited in the Backward application paragraph above). Whether a B_1-direction moral-grounding companion should be expanded as an explicit subsection (an extension of §5.4; a new §5.6; or left implicit in this section's Method 2 backward-direction lineage) is a subsequent-revision question. The operational answer is currently delivered through this section's reparations-economics lineage; the structural question of where the B_1-direction moral-grounding should live is logged here without prescribing the resolution.

§6. Commons Inversion Test (CIT)

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§6.1 — Methodological Principle

The Commons Inversion Test is a discovery methodology, not a measurement methodology. Where the RCV formula measures cost severance, the Abundance Test reveals which cost tiers have been systematically hidden by abundance. The two methodologies are complementary: the Abundance Test identifies which terms belong in $E(R, t)$; the RCV formula then prices them.

§6.2 — The Seven Abundance Dimensions

Note on the enumeration below. The seven dimensions enumerated below are the original cohort surfaced through case-study work; the framework's full cohort across the cases this volume examines is ten dimensions (Habitability · Spatial · Temporal · Institutional · Kindred · Ecosystem · Political · Cohesion · Epistemic · Autonomy), with the three additional dimensions (Cohesion, Epistemic, Autonomy) defined per §1.10 above. The dimensions are organizational scaffolding over variables CIT has surfaced across the cases this volume examines — illustrative examples drawn from the empirical record, not a closed ontological taxonomy. Different traditions may surface different categories or refine these; the methodology operates on whatever candidate commons a tradition treats as load-bearing.

Dimension 1: Habitability Abundance → Oxygen/Climate/Radiation Costs

Strip Earth's abundant atmosphere. Oxygen processing, atmospheric regulation, and radiation shielding costs become visible. On Earth, these are absorbed as commons goods at no market cost. Their value is revealed only when absent (Mars, spacecraft), at which point they are priced at life-support system cost — approximately $1M–$10M per person-year in engineered equivalents.

Dimension 2: Spatial Abundance → Spatial Displacement Costs

Strip abundant land/space. Costs of distance from home, transport, inability to return, and community immobility become visible. For McDowell County coal: the coal's value dispersed nationally while health costs and community collapse concentrated locally. Spatial abundance masks the spatial dimension of cost severance. See Cost Severance formalization in §16.3.

Dimension 3: Temporal Abundance → Acceleration Costs

Strip abundant time. Costs of compressing multi-century resource availability into decades become visible — rate-of-extraction premiums, the loss of optionality, the foreclosure of applications not yet invented. For Appalachian coal: at 1960 extraction rates the deposits would have lasted 300+ years. Actual extraction compressed this to decades, foreclosing any 22nd- or 23rd-century use.

Dimension 4: Institutional Abundance → Governance Costs

Strip abundant legal/political infrastructure. Costs of establishing enforcement, maintaining accountability, and ensuring honest accounting become visible. When institutional capacity is absent (failed states, bankruptcy-protected corporations), cost severance operates at maximum scale because $B \approx 0$.

Dimension 5: Kindred Abundance → Knowledge/Cultural Costs

Strip abundant population. Costs of knowledge transmission, cultural continuity, and maintaining community expertise below critical thresholds become visible. McDowell County's population collapse destroyed three generations of technical mining knowledge, traditional ecological knowledge, and social capital accumulated over a century. These costs do not appear in any ledger.

Dimension 6: Ecosystem Abundance → Ecosystem Service Costs

Strip abundant ecosystem health. Costs of pollination, hydrological cycles, soil formation, and waste processing become visible. Acid mine drainage eliminates aquatic food chains. Mountaintop removal disrupts hydrological networks serving downstream communities for generations after mining ends.

Dimension 7: Political Abundance → Governance Capture Costs

Strip abundant democratic and regulatory infrastructure. Costs of maintaining honest accounting systems, regulatory enforcement, and institutional integrity become visible. When political abundance is present, extraction industries can invest a fraction of captured value in suppressing honest pricing of their own costs — lobbying, regulatory capture, suppression of scientific findings. When political abundance is absent (failed states, captured regulators), accountability bonds collapse toward zero and cost severance operates at maximum scale. Political capture is both a consequence of cost severance and a cause of its persistence.

§6.3 — Relationship to the RCV Formula

Each commons category maps to one or more terms in the RCV formula:

Abundance Dimension Stripped	RCV Term Illuminated
Habitability	E(R, t) — atmospheric externality component
Spatial	Cost Severance integral (§16.3)
Temporal	[1 − S(t)] · U(R, t, Q(t)) — foreclosure component
Institutional	B term in CS = RCV − B
Kindred	U(R, t, Q(t)) — option value and knowledge component
Ecosystem	U(R, t, Q(t)) — ecological function component; E(R,t) — remediation tail
Political	B term in CS = RCV − B — political capture suppresses B toward zero; also E(R,t) — costs of institutional corrosion are externality tail items

§6.4 — Two sub-forms

Commons-Absence Inversion (CAI). Given a cost claim that names a commons, ask: *would the extraction price differently if the commons claim did not exist at all?* If yes → cost claim is commons-grounded. If no → cost claim is private-property externality (different framework applies).

Commons-Consumption Inversion (CCI). Given a commons-grounded cost claim, ask: *would the extraction price differently if the commons were inexhaustible / self-restoring at zero marginal cost?* If yes → cost claim is consumption-of-finite-commons (Cost Severance proper). If no → cost claim is coordination-of-mutually-acceptable-use (Ostrom-tradition commons-management; framework's CS does not apply).

The two sub-forms run in sequence: CAI first (is this commons-grounded at all?), then CCI (is this consumption-grounded?). Only claims passing both qualify as Cost Severance.

§6.4.1 — Level-of-claim specification

Discipline: CIT's outcome depends on the level at which the cost claim is specified. The same extraction relationship can produce a CIT-pass at one level of aggregation and a CIT-fail at another, and this is not a defect of the test — it is a methodological feature requiring the analyst to be explicit about the level the cost claim is named at.

Formal rule: Before running CAI + CCI on a candidate cost claim, specify the level at which the commons is asserted to be finite. The specification must name: (1) **aggregation scope** (individual-user / community / population / cohort / generational / civilizational / planetary); (2) **temporal scope** (instantaneous / cumulative-over-period / lifetime / intergenerational / open-ended); (3) **substitutability scope** (single-substitute / class-of-substitutes / no-substitute-available).

Worked example — platform-economy attention extraction: consider the candidate cost claim that platform attention-extraction (e.g., recommendation algorithms optimizing for engagement) constitutes Cost Severance.

- **Level A — individual-user instantaneous.** "TikTok extracts attention from individual user X during session Y." CAI passes. CCI **fails** : individual attention regenerates between sessions. **CIT-fail at Level A.**
- **Level B — cumulative-population intergenerational.** "Platform attention-extraction across the entire user population, cumulative across the digital-attention era, depletes a finite collective-cognitive-bandwidth commons whose regeneration capacity is bounded by neurobiological and developmental constraints." CAI passes. CCI **may pass** . **CIT may pass at Level B** , depending on empirical bounds.

Reporting discipline: every CIT application must specify the level-of-claim before reporting the verdict. A bare "CIT passes" without level specification is methodologically incomplete.

§6.5 — Four Gates admission apparatus

A commons-extraction-cost claim is admitted to the framework's CS accounting if and only if it passes all four gates:

1. **CIT (Commons Inversion Test)** — passes both CAI and CCI sub-forms (with level-of-claim specification per §6.4.1).
2. **Units Consistency** — claim's units match the underlying commons's units.
3. **Boundedness** — claim has finite bounds (rules out unbounded "infinity" claims).
4. **Independence** — claim is not double-counted across overlapping commons categories.

Gate-test asymmetry: gates 1 + 2 + 3 + 4 are tests *of* a candidate cost claim. They are NOT *conditions* for what counts as a commons (Option C' political-philosophical-accommodation: the candidate commons comes from the political-philosophical tradition, not from the framework's gates). The gates filter cost *claims* against established commons; they do not filter what counts as a *commons* .

§6.6 — Domain distinction: extraction commons vs. coordination commons

Discipline: CIT's domain is *extraction commons* — commons whose finite stock is consumed by non-shared-stakeholder extractors, with cost borne by stakeholders who do not capture extraction value. CIT is NOT designed for *coordination commons* — commons whose finite stock is shared among coordinating-stakeholder users, with cost borne under coordination breakdown. The two commons-classes are structurally distinct.

Operational filter (CCI sub-form): a coordination commons fails CCI — if the resource were inexhaustible, the coordination structure would be unchanged (the cost is access-management, not consumption-of-finite-commons). An extraction commons passes CCI — if the resource were inexhaustible, extraction would price differently (the cost IS consumption-of-finite-commons by non-shared-stakeholders).

Methodology cross-reference: Ostrom-tradition coordination-commons analysis (Ostrom 1990 *Governing the Commons* ; Hess & Ostrom 2007 *Understanding Knowledge as a Commons* ; the eight design principles literature) is the framework-adjacent apparatus for coordination commons. CIT + Three Ways + B = $B_1 + B_2$ are the framework apparatus for extraction commons. The two methodologies are non-competing; they address structurally distinct commons-classes.

Hybrid cases: real-world cases that carry both elements (e.g., a pastoral commons whose forest is also externally extracted; an aquifer that is both irrigation-coordination-commons and oil-extraction-target) admit parallel analyses — each method addresses the component of the case its domain covers. The framework does NOT subsume Ostrom's apparatus; Ostrom does NOT subsume the framework. Both apply to their respective domains within a hybrid case.

Reporting discipline: every CIT application should specify (per §6.4.1 level-of-claim discipline) whether the cost claim is named at extraction-commons level or coordination-commons level. A CIT-fail at the coordination-commons level is correct routing to Ostrom-tradition apparatus, not a framework limitation.

§6.7 — Worked Examples: Seven Walkthroughs (Four Gates Applied)

The seven cases: McDowell County coal · the commute trade · Norwegian oil · 2008 financial crisis · the asteroid miner (Mars/Europa) · healthcare end-of-life · opioid Appalachia. Each is a different combination of CIT sub-form (Commons-Absence vs Commons-Consumption) and gate-exercise pattern.

Overview. The Four Gates as a system

CIT (Commons Inversion Test) — content gate. Does the cost depend on a commons? Two sub-forms: - **Commons-Absence Inversion:** imagine the commons not existing. Cost = replacement cost. - **Commons-Consumption Inversion:** imagine the commons not being consumed. Cost = opportunity cost.

Units Consistency — mathematical-form gate. Is cost expressed in commensurable units (dollars × time-domain)?

Boundedness — mathematical-form gate. Does the cost integral converge?

Independence — mathematical-form gate. Is the cost not redundantly correlated with already-admitted C_i (no double-counting)?

Structural distinction: CIT is a TEST (procedural methodology); the other three are CONDITIONS (static mathematical properties). Naming asymmetry honors the layer distinction.

Walkthrough 1. McDowell County coal: Black Lung healthcare cost

Candidate cost: Black Lung healthcare expenditures + lost productivity + premature death — borne by miners, families, Medicare/Medicaid, McDowell County's social-service infrastructure.

Gate 1 — CIT: Imagine the habitability commons of McDowell County absent. Picture extraction stripping the place of clean air, healthy bodies, intact community life-support. Does the cost become visible? Yes — Black Lung pathology is exactly what becomes visible when the habitability commons is eroded by the dust the regulators didn't enforce against, the medical supervision the company resisted, the community's transition reserve no one funded. **Commons-Absence Inversion.** PASSES.

Gate 2 — Units Consistency: Healthcare costs in dollars; integrate over years of patient life + cohort effects + intergenerational life-expectancy gap. Dollars × time-domain. PASSES.

Gate 3 — Boundedness: Each patient's healthcare cost over their lifetime is bounded. The cohort effect (13-year life-expectancy gap × population × earnings × healthcare costs) sums to a bounded total. Intergenerational health-effects extension (developmental + grandchildren mortality patterns) extends but doesn't blow up. PASSES.

Gate 4 — Independence: Black Lung healthcare cost is mostly independent of the *ecosystem-remediation cost* (orange creeks, hillside scarring) and the *community-collapse cost* (population loss, school closures). Some correlation exists — community collapse worsens health outcomes — but at the level of C_i admission, healthcare-cost-on-bodies and community-cost-on-population structure are distinct cost streams. PASSES with note: double-count audits at integration level needed.

Verdict: All four gates PASS. Black Lung health cost admits to RCV.

Walkthrough 2. The commute trade (Personal Stories Candidate #1): foregone-life-time cost

Candidate cost: 1,800 hours/year of life-time consumed by 5-hour daily commute — costs incurred by you (foregone language-learning, exercise, family time, professional development, leisure, sleep). Information-asymmetric self-severance: past-you signed the lease without knowing what future-you would pay.

Gate 1 — CIT: Imagine the time-commons UNCONSUMED — picture the 5 hours/day returned to you. Does the cost become visible? Yes — the language you didn't learn, the relationships not built, the rest not taken, the skills not acquired all become visible against the un-consumed-commons counterfactual. **Commons-Consumption Inversion.** PASSES.

Gate 2 — Units Consistency: Time can be valued in dollars (hourly opportunity cost; revealed-preference for course tuition). 1,800 hours × revealed-preference rate = dollar value. Dollars × time-domain. PASSES.

Gate 3 — Boundedness: 1,800 hours × duration-of-lease (1 year). Bounded. PASSES.

Gate 4 — Independence: Cost is independent of other extraction costs in the case (apartment manager's costs; landlord profits — separate C_i). The case is special because value-capturer = cost-bearer (past-you = future-you), but the gates still operate cleanly. Information-asymmetric self-severance is admissible — the framework handles it. PASSES.

Verdict: All four gates PASS. Foregone-life-time cost admits to RCV.

Methodology note: This case demonstrates how CIT's Commons-Consumption sub-form admits opportunity-cost-on-time as a legitimate C_i. Cases where extraction draws down a commons (rather than destroying it) require the Consumption sub-form; AIT's pre-CIT framing obscured this.

Walkthrough 3. Norwegian oil extraction: future-generation climate damage

Candidate cost: Cumulative atmospheric carbon damage from a barrel of Norwegian oil, borne by future generations — sea-level rise; ecosystem disruption; agricultural loss; climate-disaster damages; biosphere-stability costs.

Gate 1 — CIT: Imagine the atmospheric-stability commons absent for future generations. Does the cost become visible? Yes — climate damage is the canonical case of cost severed across generations. **Commons-Absence Inversion across the temporal gap.** PASSES.

Gate 2 — Units Consistency: Climate damage in dollars; integrate forward across centuries with discount-rate convention. Dollars × time. PASSES.

Gate 3 — Boundedness — this is where the gate actively works: A naive integration of climate damage forward across centuries with a low discount rate **doesn't converge** — Stern Review's argument turns on declining-or-zero discount rate which makes the integral effectively infinite over very long horizons. The gate would FAIL without an explicit convention.

Framework's treatment: apply Nordhaus DICE / Stern Review discount-rate convention. With a sensible discount rate (Stern's ~1.4%, or hyperbolic-declining per Weitzman), the integral converges to a bounded value. The discount-rate choice is politically contested and dominates the SCC magnitude — the QUALITATIVE result (CS > 0) is robust across discount rates above zero, but the MAGNITUDE varies by an order of magnitude. **PASSES with explicit convention applied; magnitude depends on discount-rate choice; framework is honest about this.**

Gate 4 — Independence: Climate cost is independent of, e.g., oil-well-decommissioning cost (different physical process), local-pollution cost from refineries (different time scale), labor-market disruption cost from energy transition. PASSES.

Verdict: All four gates PASS (with §3 boundedness condition explicitly stating discount-rate convention). Future-generation climate damage admits to RCV.

Methodology note: This case demonstrates Boundedness as a non-trivial gate that actively works on intergenerational/climate cases. The Norway sovereign wealth fund is the canonical real-world example of B

(Accountability Bond) approaching RCV via commons-management instrument, even though the climate-cost integral has Boundedness-gate complexity.

Walkthrough 4. 2008 financial crisis: institutional-commons damage with multi-stream Independence-gate exercise

Candidate cost streams: - 10M home foreclosures (foreclosure cost) - Retirement savings destroyed for cohort approaching retirement (retirement cost) - Small-business failures from credit contraction (business cost) - State-budget shortfalls + austerity-driven public-service cuts (institutional cost) - Mortgage-system trust damage (institutional commons) - Wealth-destruction concentrated on Black + Hispanic households (kindred + cohesion + institutional commons) - Long-term wage stagnation for the labor cohort that lost employment in 2008-2010 (labor-time cost)

Gate 1 — CIT: Imagine the institutional commons (financial-regulatory architecture; trust in market integrity; mortgage-system reliability; cohort-protective fiscal architecture) absent. Does the cost become visible? Yes. Each cost stream above maps to a different commons-grounding (institutional / kindred / cohesion / autonomy / temporal). **Commons-Absence across multiple commons categories simultaneously.** PASSES per cost stream.

Gate 2 — Units Consistency: Each cost stream in dollars × time-domain. PASSES.

Gate 3 — Boundedness: Each cost stream is bounded — crisis period (2007-2010) + recovery period (2010-2020) + lingering effects (2020-present). Finite range. PASSES.

Gate 4 — Independence — THIS is where the 2008 case earns its complexity:

Cost streams are *correlated* through the crisis itself. The same household may have: - Lost their home (foreclosure cost) - Lost their retirement savings (retirement cost) - Lost their job (labor-time cost) - Watched institutional trust collapse (institutional commons)

Admitting all four as separate C_i would *double-count* — the household experiences a single integrated extraction event; counting each stream as fully independent overstates total severance.

The Independence gate forces careful decomposition. Tech Appendix §L formal specification requires: when admitting multiple C_i from the same case, demonstrate linear independence. For 2008, this means: admit the cost streams at the level where they don't overlap (e.g., foreclosure cost = housing wealth lost, NOT counting retirement-savings-loss separately if those losses are correlated through the same household-balance-sheet collapse).

Operationally: either admit the costs at higher-level categories (household-wealth-destruction as one C_i) and forgo the granularity, OR admit them granularly with explicit correlation-correction (subtract the overlap mathematically).

PASSES with non-trivial Independence-gate work required.

Methodology note: This case demonstrates the Independence gate as the framework's double-counting prevention apparatus. Without it, RCV computations on multi-stream cases would systematically inflate. With it, the framework can be defended at peer-review depth on cost decomposition.

Walkthrough 5. The asteroid miner (Ch 7 Mars habitat / Europa thought-experiment): multi-commons absent simultaneously

Setting: Hypothetical extraction operation in space — asteroid mining, Mars habitat resource extraction, or hypothetical Europa-ice mining. Used in Ch 7 as the framework's universality test ("if it works on Mars, it works on Earth").

Candidate cost streams: - Life-support infrastructure cost (oxygen, food, water production replacement) - Communication-delay cost (relativistic + physical-distance limits on Earth-contact) - Return-trip cost (spatial commons consumed; one-way journey foreclosed) - Psychological-isolation cost (autonomy + kindred commons consumed) - Medical / specialist-care cost (epistemic commons absent at distance) - Habitat-ecosystem cost (no Earth biosphere; closed-loop replacement)

Gate 1 — CIT: Imagine Earth's habitability commons absent — the atmosphere, ecosystem, social network, medical infrastructure, mobility, cohesion-of-Earth-community. Does the cost become visible? Yes — at extreme. Multiple commons absent simultaneously: habitability + ecosystem + spatial + kindred + epistemic + autonomy + cohesion. **Commons-Absence Inversion at maximum.** PASSES — and powerfully, because the case foregrounds what we don't notice on Earth (we don't price oxygen because the atmospheric commons is so abundant).

Gate 2 — Units Consistency: Hypothetical pricing applied (cost-per-pound-to-orbit; replacement-cost of Earth ecosystems; etc.). Dollars × time-domain. PASSES with hypothetical inputs.

Gate 3 — Boundedness: Bounded over the colony lifetime + return-mission window + post-colony recovery period (if any). Finite. PASSES.

Gate 4 — Independence: Multiple commons producing multiple C_i — life-support (replaces atmospheric + ecosystem commons); communication-delay (epistemic + kindred commons); return-trip (spatial); psychological (autonomy + kindred). Some correlation (psychological cost worsens with longer communication delay), but at first order distinct cost streams. PASSES with care.

Verdict: All four gates PASS. The asteroid-miner case admits multiple C_i across multiple commons categories.

Methodology note (universality test function): The asteroid case's pedagogical work is to show CIT operating where Earth readers most viscerally feel the commons-absence (oxygen in space; water on Mars; community in isolation). It doesn't introduce new mechanism — it foregrounds what Earth's commons-abundance hides. **The universality claim becomes legible:** if the framework works in space (where every commons is priced because it must be replaced), it works on Earth (where commons-abundance lets extractors externalize what they would have paid for if the commons weren't free). Strong M5 dinner-table case.

Walkthrough 6. Healthcare end-of-life (Butler pacemaker case): Commons-Consumption applied to medical-commons

Setting: Katy Butler's reporting on her father's pacemaker — a procedure that extended his cognitive decline 5+ years, while Medicare reimbursement architecture systematically routed physicians toward procedures over conversations ($461 procedural payment vs. $54 end-of-life conversation payment historically). Butler's mother caregiver-burnout (~80 hr/week). 25× growth in medical GoFundMe campaigns 2011-2020. 88% medical-GoFundMe failure rate. Racial disparity in crowdfunding success (Ly & Soman 2020).

Candidate cost streams: - Patient's prolonged-cognitive-decline cost (autonomy + kindred + dignity loss) - Family caregiver labor-time cost (Butler's mother's 80 hr/week × years) - Family-financial-resource depletion cost - Lost end-of-life-quality cost (kindred + cohesion + autonomy commons) - Medicare/Medicaid system-distortion cost (institutional commons) - Foregone-conversation cost (epistemic commons — informed consent about end-of-life options)

Gate 1 — CIT: Imagine the medical-commons of dying-with-dignity-with-family-present UNCONSUMED — picture the medical architecture not drawing down patient autonomy, family time, dignified end-of-life process. Does the cost become visible? Yes — the procedural-medical-financial architecture extracts these by incentivizing prolongation over comfort, billable procedure over conversation. **Commons-Consumption Inversion** (the medical commons exists in society but is being drawn down by the extraction architecture). PASSES.

This is a distinctive case — it demonstrates Commons-Consumption Inversion in a non-time-commons context. The medical commons (informed end-of-life choices + family presence + dignified process) was *available in principle* but actively *consumed by extraction* . Different from McDowell (Commons-Absence — habitability destroyed); different from Norway oil (Commons-Absence — atmospheric stability damaged). Here, the commons existed and was drawn down, not destroyed.

Gate 2 — Units Consistency: Healthcare costs in dollars × time. Caregiver-time can be valued (replacement-cost of professional care). Family-financial-depletion already in dollars. PASSES.

Gate 3 — Boundedness: Bounded over patient lifetime (with prolongation) + family caregiving period + financial-resource depletion period. PASSES.

Gate 4 — Independence: Multi-stream — patient's suffering, family caregiver's labor-time, financial resources, system distortion. Some correlation (financial depletion correlates with caregiving-time), but distinct cost streams at first order. PASSES with care.

Verdict: All four gates PASS. End-of-life cost admits to RCV.

Methodology note: This case demonstrates Commons-Consumption Inversion outside the time/attention-economy domain. The medical-commons of dying-with-dignity is consumed by the procedural-reimbursement architecture, not destroyed. The framework's CIT operates on this mode cleanly. Also a key adoption case for medical professionals + healthcare policy reformers (the framework's vocabulary travels into medical-ethics discourse).

Walkthrough 7. Opioid Appalachia (Sackler / Purdue Pharma): documented extraction with multi-commons damage + both CIT sub-forms simultaneously

Setting: Per Patrick Radden Keefe's *Empire of Pain* (2021) + public litigation record. Purdue Pharma deliberately targeted vulnerable communities (including McDowell County) with OxyContin, suppressing internal evidence of addiction risk. Sackler family extracted ~$11B before bankruptcy filing. Big Three distributors (McKesson, Cardinal Health, AmerisourceBergen) settled for $21B in 2022. 141 overdose deaths per 100K in McDowell County. Communities still in crisis.

Candidate cost streams: - Patient deaths + addictive-harm cost (autonomy + epistemic + kindred commons) - Family destruction cost (kindred + cohesion commons) - Community-collapse cost (cohesion + institutional commons) - Healthcare-system overload cost (institutional commons) - Multi-generational trauma cost (intergenerational + temporal commons) - Foster-care-system overload cost (institutional + kindred commons) - Lost-life-trajectory cost (autonomy + temporal commons consumed by addiction)

Gate 1 — CIT — both sub-forms simultaneously:

Commons-Absence sub-form: Imagine the epistemic commons (truthful information about OxyContin's addictive risk) absent — that is what was suppressed. Imagine community-cohesion commons absent — that is what 141/100K overdose deaths produces. Imagine kindred commons of intact families absent — that is what addiction-driven family destruction produces. **Multiple commons in Absence Inversion.**

Commons-Consumption sub-form: Imagine individual life-time UNCONSUMED by addiction — picture the 1990s-2010s cohort of working-age adults in extraction-targeted Appalachia counties not having their lifetime drawn down by pharmaceutical-engineered addiction. Lost-life-trajectory cost becomes visible. **Commons-Consumption Inversion at individual scale operating simultaneously with Commons-Absence at community scale.**

This case is the framework's clearest example of both sub-forms operating simultaneously — the pharmaceutical extraction destroyed community-level commons (Absence) while consuming individual-level life-time commons (Consumption). PASSES at full strength.

Gate 2 — Units Consistency: Statistical-life-value × excess deaths; addiction-treatment costs; foster-care system costs; healthcare-system overload costs. All in dollars × time-domain. PASSES.

Gate 3 — Boundedness: Bounded over the crisis period (1996-present) + community-recovery period (extending; potentially decades). Finite. PASSES.

Gate 4 — Independence: Multi-stream — patient deaths, family destruction, community collapse, healthcare overload, foster-care overload, multi-generational trauma. Significant correlation (community collapse causes family destruction; addiction-deaths cause foster-care overload). The Independence gate forces careful decomposition. The framework can use higher-level admission (community-extraction-cost as one C_i) OR granular admission with explicit correlation-correction. Both operationally valid. PASSES with non-trivial Independence-gate work required.

Verdict: All four gates PASS. Opioid-extraction-Appalachia cost streams admit to RCV.

Methodology note (empirical validation of framework): This case is the framework's strongest empirical validation. The diagnostic the framework makes (extraction operating across multiple commons, severing cost from value extraction) was **already empirically validated by the public litigation record** before the framework was built. Sackler $11B extraction = framework's predicted value extraction; $21B settlement = framework's predicted partial Accountability Bond (B); the $11B + $21B is still vastly less than RCV (deaths × statistical-life-value + community-recovery costs + multi-generational trauma). The framework's prediction matches what the courts found.

Cross-reference: This case is the framework's most-direct refutation of Vance's *Hillbilly Elegy* cultural-pathology framing. The framework demonstrates, via public litigation record, that the "cultural pathology" Vance describes was in significant part a deliberately manufactured pharmaceutical market. Per bibliography §12.

Cross-cutting findings across all 7 walkthroughs

What each gate does in operational practice

CIT does the heaviest lifting consistently. Every case requires identifying the commons + applying the right sub-form. The framework's substantive content gate. Universal application.

Units Consistency is mostly boilerplate in well-formed cases. Catches edge cases like trying to admit a probability or dimensionless ratio as if it were a cost — would FAIL there. In practice, well-formed candidate costs already pass.

Boundedness sleeps on most cases but **wakes up dramatically on climate / intergenerational cases** where the integral can diverge without explicit discount-rate convention. The Norway oil case (§3) is paradigmatic. This is where the gate earns its keep.

Independence sleeps on simple single-stream cases but **wakes up on multi-stream correlated cases** — 2008 financial crisis (§4); McDowell County (§1) where health, community, ecosystem, and institutional costs flow from the same extraction; opioid Appalachia (§7) with multiple cost streams correlating through community-collapse dynamics. Prevents double-counting that would inflate RCV illegitimately.

CIT sub-form distribution across cases

Case	CIT sub-form
McDowell coal	Absence (habitability destroyed)
Commute trade	Consumption (time drawn down)
Norwegian oil / climate	Absence (atmospheric stability damaged for future)
2008 crisis	Absence (institutional commons damaged)
Asteroid miner	Absence (multiple Earth commons absent)
Healthcare end-of-life	Consumption (medical commons drawn down)
Opioid Appalachia	**Both** (community Absence + individual Consumption)

Pattern: physical-resource-extraction cases tend toward Absence; personal-scale + medical / time / attention cases tend toward Consumption; multi-commons-multi-stream cases (opioid; large-scale crisis) can exercise both sub-forms simultaneously.

Cost-stream complexity scales with case scope

- Single-cost-stream cases (commute trade): trivial Independence; gates flow easily.
- Multi-cost-stream cases (McDowell; healthcare): non-trivial Independence; some correlation; manageable.
- Highly-multi-stream cases (2008 crisis; opioid Appalachia; asteroid miner): Independence is decisive; framework's careful decomposition discipline is what makes RCV defensible.

Empirical validation cases

The opioid Appalachia case (§7) is the framework's strongest empirical-validation case — public litigation record predates framework construction; framework's predictions match court findings. The 2008 crisis case (§4) has comparable public-record validation through Financial Crisis Inquiry Commission.

These cases serve as case-study, academic, and critic-engagement evidence together — the framework didn't predict from theory then check; the predictions and the public record converge on the same diagnosis from independent starting points.

Gates as a complete admission apparatus

- CIT determines what's IN.
- Units Consistency keeps the math well-formed.
- Boundedness keeps the integral finite.
- Independence keeps the integral honest.

Together they're a complete system. CIT is the framework's intellectual contribution; the other three are the mathematical hygiene any well-formed cost-integrand needs. Naming asymmetry (test vs. condition-properties) honors the division of intellectual labor.

§7. Four Gates Admission Apparatus

The four gates governing admission of cost terms to the framework's RCV computation: (1) **CIT** — Commons Inversion Test (CAI + CCI sub-forms); (2) **Units Consistency** ; (3) **Boundedness** ; (4) **Independence** .

A framework that prices cost severance by integrating discovered cost terms against a declining discount factor requires a discipline for what counts as an admissible term. Without that discipline, the framework is unbounded — any quantity whose existence can be narrated becomes an RCV contributor, and the computation becomes a vehicle for the analyst's priors. With the discipline, the framework is generative: new variables are admitted on specific grounds, older analyses are revisable as new grounds emerge, and the computation remains a structured argument rather than a collection of preferences.

This section specifies the four-gate discipline formally. A candidate variable C_i enters the RCV integrand if and only if all four gates pass under the stated Context. Gate 1 (§7.1) tests scarcity-grounding (CIT). Gate 2 (§7.2) tests dimensional consistency. Gate 3 (§7.3) tests integral convergence. Gate 4 (§7.4) tests non-overlap with existing terms. The four gates are jointly sufficient: a candidate that passes all four is admissible; a candidate that fails any one is rejected under those specifications and may be re-proposed under different specifications (redefinition, disjoint re-scoping, unit conversion, horizon truncation) that address the failed gate.

The gates are not hierarchical. The sequence below is operational (rejecting on CIT first is the cheapest test; overlap analysis is most expensive because it requires the candidate to be fully specified). A candidate must pass all four in any order. The current canonical two-term RCV integrand in §3.1 is the special case of the general sum-of-costs form under the four gates; both C_1 = foreclosure and C_2 = externality-tail pass all four gates under the framework's standard specifications, as worked below.

§7.1 — Gate 1: Commons Inversion Test (CIT)

Formal statement. A candidate cost term C_i passes the CIT gate if and only if: under a counterfactual in which the scarcity-condition underlying C_i is removed (the relevant resource, substitutability-constrained good, or absorption-capacity is available in non-scarce quantities at the affected times), the cost reduces to zero (or to a cost intrinsic to the activity rather than to the extraction of this resource at this time).

The gate's operational form requires the analyst to specify the abundance counterfactual explicitly before testing. A candidate that persists under abundance is not a cost of extraction in this framework; it may be a cost of the activity in general, a cost of something else, or not an economic cost at all. A candidate that vanishes under abundance is scarcity-grounded and eligible to continue to subsequent gates.

Worked example — foreclosure cost (C_1). Under abundance of substitutes at all future times ($S(t \mid t_0) = 1 \ \forall t \geq t_0$), $[1 - S(t \mid t_0)] = 0$, and the foreclosure contribution vanishes. Foreclosure is scarcity-grounded in the substitutability condition. Pass.

Worked example — externality tail (C_2). Under abundance of the extracted resource itself, $E(R, t)$ is not eliminated — combustion damage, remediation obligation, and community-health cost persist regardless of stock size. The resolution is the gate's precise statement: CIT inverts the scarcity condition underlying the cost, not the physical damage itself. For E, the relevant scarcity is absorption-capacity (atmospheric, hydrological, community). Under infinite absorption capacity, the externality flow does no marginal damage; the cost vanishes. Scarcity-grounded in absorption-capacity. Pass under the absorption-capacity counterfactual.

Rejected candidate — existential-meaning cost. Under abundance of meaning/purpose, the cost does not reduce to zero; meaning is not a resource for which abundance-vs-scarcity is economically well-defined. The counterfactual is incoherent. Fail.

Admissible candidates exist outside the canonical two-term specification. Aesthetic cost (hedonic-priced, under landscape-beauty-scarcity) passes CIT because under abundance of alternative undisturbed landscapes the marginal cost of a specific landscape loss approaches zero. Cultural-pride cost passes CIT under abundance of alternative cultural-pride-producing activities. Both candidates then face the remaining three gates, which determine whether they enter a particular Context's RCV integrand.

§7.2 — Gate 2: Units Consistency

Formal statement. A candidate cost term Ci passes the dimensional gate if and only if Ci has units [dollars · (resource-unit)−1 · (time-unit)−1] under the RCV integrand, such that integration against the dimensionless discount factor D(t, t0) produces a quantity with units [dollars · (resource-unit)−1].

Dimensional requirement for each term in the RCV integrand:

Term	Units	Notes
[1 – S(t | t0)]	dimensionless	probability that no adequate substitute exists
U(R, t, Q(t))	$ · res-unit^{-1} · time^{-1}	stock-dependent utility flow of retained resource
E(R, t)	$ · res-unit^{-1} · time^{-1}	externality flow attributable to the extraction
C_i (candidate)	$ · res-unit^{-1} · time^{-1}	required for admission
D(t, t0)	dimensionless	discount factor (Weitzman-type under declining rate)
∫ · dt result	$ · res-unit^{-1}	RCV per resource unit

Candidates specified in non-dollar units (utility indices, ordinal rankings, hedonic indices without dollar-conversion, quality-of-life scores without monetary equivalents) fail this gate until they are converted to dollar terms with transparent methodology. The gate does not forbid conversion — hedonic pricing, contingent valuation, and revealed-preference methods are all acceptable — but the conversion must be explicit, defensible, and documented in the RCV write-up.

Worked example — community-transition-reserve cost (C3). Empirical allocation from McDowell County and comparable extraction communities places the community-transition cost at roughly $5–$15 per ton of coal extracted, distributed across the transition period following mine closure. Units: [$ · ton−1 · year−1] during the transition window, zero elsewhere. Pass.

Rejected candidate — dignity cost (as proposed, unconverted). Proposed as an ordinal ranking of community dignity loss, with no dollar conversion and no time allocation. Fail as proposed; re-proposable as a contingent-valuation-priced cost with transparent conversion.

§7.3 — Gate 3: Boundedness (Integral Convergence)

Formal statement. A candidate cost term C_i passes the boundedness gate if and only if the integral

$$\int_{t_0}^{\infty} C_i(R, t, \text{Context}) \cdot D(t, t_0)\, dt$$

converges to a finite value under the stated parameter assumptions for D and C_i.

Sufficient condition. If $C_i(R, t, \text{Context})$ is bounded above for all $t \geq t_0$ and decays to zero (or to a constant absorbed by the discount factor) as $t \to \infty$ at a rate at least as fast as $D(t, t_0)$ decays toward its asymptotic value, the integral converges. Weitzman-declining-rate discount factors (§16.1) satisfy this condition for any C_i that is asymptotically bounded and non-growing.

Convergence-condition table for common C_i asymptotic behaviors:

C_i asymptotic behavior	Convergence under Weitzman D	Admissibility
decays to 0 (exponential or polynomial)	yes	admit
bounded, approaches non-zero constant	conditional (constant × asymptotic D integrated)	admit under finite-horizon truncation + sensitivity
grows polynomially	depends on growth rate vs. D decay	case-by-case; typically requires truncation
grows exponentially	typically diverges	reject as proposed
oscillates unboundedly	diverges	reject

The analyst specifies (a) the asymptotic behavior of C_i, (b) the applicable discount-factor specification, and (c) a convergence proof or a numerical verification over a sufficiently long horizon (typically 200–500 years for terrestrial extraction contexts; longer for contexts with century-scale persistence such as atmospheric carbon or deep-aquifer depletion). Conditional convergence cases are handled by finite-horizon truncation with an accompanying sensitivity bound.

Worked example — carbon externality tail. $E_{carbon}(t)$ decays as atmospheric CO_2 is removed by natural sinks, with an effective half-life on the order of 100 years for a substantial fraction and roughly 1000+ years for a residual fraction. Under Weitzman-declining-rate discounting, the integral converges; 500-year horizon captures approximately 95% of the integral value, with extension to 2000 years adding under 5%. Pass with numerical verification; finite-horizon truncation at 2000 years is standard practice.

Rejected candidate — perpetual-dignity-loss (proposed as $C_i = \text{constant}\ \forall\, t \geq t_0$). Under any Weitzman-type discount factor that decays only to a small nonzero asymptotic constant, the integral grows without bound. Fail as proposed; re-proposable with finite-horizon specification or explicit decay.

§7.4 — Gate 4: Independence (Non-Overlap)

Formal statement. A candidate cost term C_i passes the independence gate if and only if C_i does not double-count any component already captured by existing terms $\{C_1, C_2, \ldots, C_{i-1}\}$ in the integrand under the current Context.

For each existing term C_j, the analyst specifies the cost category C_j captures. For the candidate C_i, the analyst specifies its cost category. If the categories are disjoint, independence holds. If they overlap partially, the analyst must resolve the overlap by one of three moves: (a) redefine one of the terms to restrict it to its disjoint portion; (b) merge the two terms into a single broader term; (c) reject the candidate as redundant under the existing specification.

Worked example — community-transition-reserve cost (C_3) versus the standard E term (defined in §1). Canonical E in §3.1 captures atmospheric, hydrological, and environmental-remediation externalities. C_3 captures community-infrastructural-and-demographic transition costs. Categories are disjoint. Pass under the standard E term (defined in §1) specification.

Worked example — requires-merger: carbon-specific externality (C_{carbon}) versus aggregated E. If a candidate C_{carbon} is proposed when E already aggregates atmospheric externalities including carbon (as in the Ch 6 headline computation where E includes \$190/ton CO_2 × 2.32 tons CO_2 per ton coal — EPA emission factor for bituminous coal at short-ton accounting, per EPA AP-42 §1.1), C_{carbon} double-counts. Resolution options: (a) redefine E to exclude carbon (narrowing its scope to non-carbon environmental externalities) and admit C_{carbon} separately; (b) retain aggregated E and reject C_{carbon} as redundant. Either resolution preserves framework consistency. The choice is a reporting-granularity preference, not a structural requirement.

§7.5 — Gate-Application Protocol

The four gates are applied in the following operational sequence for any candidate C_i. The sequence is chosen for efficiency of rejection, not definitional priority; a candidate must pass all four gates in any order to be admitted.

Step 1 — CIT. Specify the abundance counterfactual. Evaluate whether C_i reduces to zero (or to an intrinsic-activity cost) under abundance. If yes, proceed to Step 2. If no, reject; optionally re-propose under an alternative counterfactual specification.

Step 2 — Dimensional. Verify the candidate's units match $[\$ \cdot \text{res-unit}^{-1} \cdot \text{time}^{-1}]$. If yes, proceed to Step 3. If no, reject; optionally re-propose with explicit dollar-conversion methodology.

Step 3 — Boundedness. Verify the candidate's integral $\int C_i \cdot D \, dt$ converges under stated parameters (analytic proof or numerical verification over a 200–500 year horizon, 2000 year for persistence cases). If yes, proceed to Step 4. If no, reject; optionally re-propose with finite-horizon truncation or a decay specification.

Step 4 — Independence. Specify cost categories for all existing $\{C_j\}$ and for the candidate. Verify non-overlap. If disjoint, admit. If overlap, resolve by redefinition, merger, or rejection; re-verify Steps 1–3 for any redefined term.

A candidate that passes all four gates enters the RCV integrand. Admission is provisional in the sense that subsequent candidates may surface overlap that requires re-checking independence; previously-admitted terms are not removed by subsequent additions, but their definitions may be narrowed. The framework is monotonic in admission but not in specification.

§7.6 — Worked Example: Adding Community-Transition-Reserve Cost to McDowell County Coal

Starting RCV (§3.1 canonical):

$$RCV_{McDowell} = \int_{1960}^{\infty} \{ [1 - S(t \mid 1960)] \cdot U(R, t, Q(t)) + E(R, t) \} \cdot D(t, 1960)\, dt$$

Candidate: C_3 = community-transition-reserve cost.

Context: McDowell County, West Virginia, 1960 forward. Population peaked near 100,000 in 1950 and declined to approximately 19,000 by 2020. Transition costs include retraining displacement, infrastructure maintenance under declining tax base, public-service degradation, and intergenerational-mobility foreclosure in the transition window following mine closure.

Gate 1 (§7.1) (CIT). Counterfactual: under a national community-transition-reserve fund fully bonded to extraction at the time of permitting, community-transition costs are absorbed by the fund rather than severed onto residents; the cost to the commons reduces to zero. Scarcity-grounded in transition-capital availability. Pass.

Gate 2 (§7.2) (dimensional). Empirical allocation: \$5–\$15 per ton of coal per year during the transition window, approximately 1990–2050. Units: $[\$ \cdot ton^{-1} \cdot year^{-1}]$. Pass.

Gate 3 (§7.3) (boundedness). C_3 is non-zero on the finite interval [1990, 2050] and zero elsewhere. The integral of a bounded function over a finite interval, discounted by a bounded discount factor, is finite. Present-value contribution (under canonical Weitzman parameters): approximately \$5–\$10 per ton. Pass.

Gate 4 (§7.4) (independence). Foreclosure $[1 - S] \cdot U$ captures future-user value of the retained resource stock; E captures atmospheric and environmental externalities; C_3 captures community-infrastructural-and-demographic transition costs. Categories are disjoint under canonical specifications. Pass.

Updated formula:

$$RCV_{McDowell} = \int_{1960}^{\infty} \{ [1 - S(t \mid 1960)] \cdot U(R, t, Q(t)) + E(R, t) + C_3(R, t, \text{Context}) \} \cdot D(t, 1960)\, dt$$

Numerical update. Present-value contribution of C_3 approximately \$5–\$10/ton. Against the canonical Ch 6 figure of \$550–\$570/ton (dominated by the carbon externality), the updated RCV estimate is \$555–\$580/ton. Market price remains \$4.50/ton (1960 historical) or \$40–\$140/ton (contemporary range by grade). The qualitative result $CS > 0$ is preserved; the pricing gap widens slightly.

Adding C_3 does not destabilize the computation. Convergence is preserved. Dimensional consistency is maintained. No existing term is invalidated. The framework has admitted a new variable via the four gates; subsequent analysts with improved community-transition-cost data can refine the C_3 estimate without disturbing the rest of the integrand. This is the generative property the framework is designed to support.

§7.7 — Relationship to Prior Methodology

The four gates extend, rather than replace, the framework's prior methodological discipline.

CIT (§6) is now explicitly positioned as Gate 1 (§7.1), the first of four gates. The other three gates supply the formal-mathematical constraints that CIT alone did not enforce. CIT's role as the framework's epistemic core is preserved; what changes is that CIT passage alone is no longer sufficient for admission.

Merger gate and primitiveness gate (applied to commons categories in the framework's earlier development) are recognized as the independence gate's dimensional-level application. The merger gate tests whether a proposed dimension is subsumed by an existing one; the primitiveness gate tests whether a proposed dimension is structurally irreducible. Both are cases of the general non-overlap requirement operating at a coarser granularity than variable-level independence.

Convergence analysis (Theorem 10.4 on RCV integral convergence under Weitzman discounting) is the boundedness gate's canonical instance. The gate extends the theorem's coverage from the two-term canonical formula in §3.1 to any finite-sum integrand that satisfies §7.3's sufficient condition.

The canonical two-term RCV formula (§3.1) remains the baseline specification for the published worked cases. The general sum-of-costs form is an extension under the four gates, formalized in §17 (Formula Generalization) below.

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§8. Asymmetric Regret Rule (ARR)

Prior-art citations: Savage 1951 (*Journal of the American Statistical Association* 46(253): 55–67) for minimax-regret foundations; **Loomes & Sugden 1982** (*Economic Journal*) for asymmetric-regret theory; **Lempert, Popper & Bankes 2003** (*Shaping the Next One Hundred Years*) for robust decision making under deep uncertainty; **Rio Declaration 1992 Principle 15** for precautionary-principle tradition. Framework's contribution: specialization with bidirectional flip-by-context + irreversibility-weighted regret-tail framing for commons-extraction decisions. Aphorism: *"We can always extract later. We can never un-extract."*

§8.1 — The Asymmetric Regret Decision Rule (default under uncertainty)

If uncertainty about existential substitutability gap classification: default to the higher classification. Preservation cost ≤ Replacement cost when no substitute exists. "We can always extract later. We can never un-extract."

§8.2 — The rule

When facing a decision whose error has asymmetric regret structure — where one direction of error produces small recoverable cost and the other direction produces large irreversible cost — the framework's decision rule is to choose the direction with smaller maximum regret.

§8.3 — Lineage acknowledgment

ARR draws on: **Savage 1951** ("The Theory of Statistical Decision," *JASA*) — minimax-regret decision theory foundations. **Loomes & Sugden 1982** ("Regret Theory," *Economic Journal*) — regret-based individual decision theory. **Lempert, Popper & Bankes 2003** (*Shaping the Next One Hundred Years*) — robust-decision-making under deep uncertainty.

The framework's contribution: applying minimax-regret discipline specifically to commons-extraction decisions where one direction of regret is *commons-extinction* (irreversible by definition under the CCI sub-form). The asymmetry — commons-loss is structurally irreversible; commons-preservation can be revisited — converts a generic minimax-regret rule into a commons-specific maximum-protective rule.

Cross-reference to Method 3 α-dominance: the empirical companion to ARR. Method 3's irreversibility_premium = $1/(1-\alpha)^{\beta}$ diverges as $\alpha \to 1$, exactly when ARR's asymmetric-regret structure is triggered. ARR is the decision rule; Method 3 α-dominance is the estimation companion. Same structural insight at two levels.

§9. Three-Model Convergence Framework

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§9.1 — Overview and Model Independence

The RCV framework is validated through convergence with two additional independent modeling approaches. The three models are:

1. Damage Function (bottom-up): Audit all identifiable costs tier by tier.
2. Real Options (market-referenced): Price the social option value of keeping the resource in situ.
3. RCV Model (integral formulation): Substitutability-weighted foreclosure plus externality tail.

Model Independence Defense. The sharpest attack on this three-model framework is that Real Options and RCV may share deep mathematical structure — both price irreversibility plus uncertainty. The defense: they package uncertainty differently. Real Options calibrates volatility from market data; RCV calibrates substitutability from engineering and materials science. These calibrations come from genuinely different data sources with uncorrelated error modes. A model that overestimates S(t) (substitutability) does not necessarily overestimate market volatility, and vice versa. The convergence therefore cannot be explained by shared error structure. The Damage Function uses a third, entirely separate data source: documented litigation outcomes, epidemiological studies, and remediation cost estimates. All three sources are uncorrelated in the sense that matters for convergence: correlated measurement error would not consistently produce IPG estimates within one order of magnitude across six tested cases.

§9.2 — Model 1: Damage Function Approach

Bottom-up audit of all identifiable costs in documented categories:

Total Cost = Health + Environmental + Community + Carbon + Infrastructure + Political Capture

Strengths: Auditable by non-economists; each line item can be independently verified; resistant to manipulation because costs are documented in litigation, epidemiological studies, and regulatory filings.

Acknowledged weakness (frame as floor): Systematically understates true cost because it can only capture identified costs. Unknown unknowns — future applications of the resource not yet invented, long-latency health effects not yet diagnosed, ecological cascades not yet documented — are excluded by construction. Label this the floor, not the central estimate. The Abundance Test is the natural extension: the Damage Function asks "what are the costs?"; the Abundance Test asks "what costs have we systematically failed to see?" They are complementary, not redundant.

§9.3 — Model 2: Real Options Approach

Social option value of keeping the resource in situ:

Option Value = Private Option Value + Avoided Externalities + Innovation Time Value

Intuition: A barrel of oil in the ground is a financial option — the right to extract at any future time, but not the obligation. That option has value beyond the present extraction price. Social option value extends this: not extracting today preserves the option to extract when substitutes have developed (increasing S(t)) and when alternative technologies make in-situ preservation more valuable.

Worked example (McDowell County coal, 1960): Coal at $4.50/ton in 1960. Social option value calculated using: (1) volatility of coal prices over 60 years; (2) avoided externality costs using EPA social cost of carbon; (3) innovation time value (the 60 years from 1960–2020 produced solar panels at $0.03/kWh). The social option value in 1960 substantially exceeded the extraction price — meaning socially, extraction was premature. The option would have been worth more kept alive.

The structural-overlap attack — that Real Options and RCV share deep mathematical structure (both price irreversibility + uncertainty) — is acknowledged explicitly rather than buried. The defense rests on different calibration sources with uncorrelated error modes: Real Options calibrates volatility from market data; RCV calibrates substitutability from engineering and materials science. A reader who notices the structural overlap and finds it addressed openly trusts the convergence more than one who notices the overlap was not addressed.

§9.4 — Model 3: RCV Model

See §3 for full formulation. The RCV model's distinctive contribution is pricing the foreclosure component — the cost borne by generations that never exist in market time and cannot bid for the resource. The Damage Function and Real Options can capture this cost only approximately; the RCV integral captures it directly through the substitutability-weighted term.

§9.5 — Three-Model Convergence Table

Case Study	Damage Function (IPG)	Real Options (IPG)	RCV Model (IPG)	Convergence

Case Study	Damage Function (IPG)	Real Options (IPG)	RCV Model (IPG)	Convergence
McDowell County Coal	33–122×	45–89×	67–134×	✓
Deepwater Horizon	15–17×	12–21×	16–19×	✓
Libby, Montana	55–82×	48–76×	61–91×	✓
Baotou Rare Earths	12–35×	18–48×	22–41×	✓

All four cases produce IPG $\gg$ 1 across all three models. The ranges overlap within one order of magnitude in every case, consistent with 10.2 (Cross-Model Convergence). The carbon term dominates the Damage Function IPG for fossil fuels; the Real Options model produces systematically lower estimates because it uses market-derived volatility, which discounts long-horizon uncertainty; the RCV model produces the highest estimates because it uses the Weitzman declining discount rate, which weights long-horizon costs more heavily.

§9.6 — Backtesting Against Implicit Social Pricing

As a defensive stress test for academic readers: the three models can be back-validated against a fourth implicit method — revealed social willingness to pay for remediation after the fact. If post-hoc cleanup, legal settlement, and public health spending at each case site are taken as society's revealed valuation of the severed costs, these figures are consistent with the lower bounds of the three-model estimates. This "implicit pricing" approach is not presented as a fourth model (it has significant selection bias problems) but as a sanity check showing the models are not overestimating by orders of magnitude. The implicit prices suggest the models may, if anything, underestimate — which is consistent with the Damage Function floor labeling.

§10. Theorems and Proofs

Theorem 10.1a (Cost Severance Identity)

Assumption A1 (Cost Severance Definition): $CS(R, t_0) = RCV(R, t_0) - B(R, t_0)$ where R is a non-renewable resource extracted at time t_0; RCV is per-unit Residual Commons Value; B is per-unit aggregate Accountability Bond. (Per Definition 1.7 / §1.7.)

Statement. Under Assumption A1, the sign of CS is determined by the comparison of RCV and B:

- **CS > 0** iff B < RCV;
- **CS = 0** iff B = RCV;
- **CS < 0** iff B > RCV.

Proof. By Assumption A1, CS = RCV − B. Therefore CS > 0 ⇔ RCV − B > 0 ⇔ B < RCV. Similarly for equality and inequality. ∎

Theorem 10.1b (Structural Cost Severance under Current Institutions)

Assumptions A2–A4:

1. **(A2) RCV is non-negative.** RCV(R, t_0) ≥ 0 for all R, t_0 where R is non-renewable. (Empirical claim supported by §3.1 + §11.)
2. **(A3) B is non-negative.** B(R, t_0) ≥ 0 for all R, t_0. (Definitionally; bonds cannot be negative.)
3. **(A4) Aggregate scope.** RCV represents aggregate Residual Commons Value across all severed-cost components (foreclosure cost + externality tail; per §3.1). B represents aggregate Accountability Bond across all instruments operating against R (sum of severance taxes + reclamation bonds + decommissioning bonds + carbon taxes + sovereign-wealth-fund Hartwick coverage + any other accountability instruments).

Premises P1–P3:

1. **(P1) Intergenerational market incompleteness — extension of Coase 1960.** Markets cannot complete bargaining for intergenerational claims because future generations cannot bid in present markets. Per Coase 1960 framework: complete-bargaining requires all affected parties to participate; intergenerational extraction affects parties (future generations) who cannot participate. Therefore market pricing systematically under-internalizes intergenerational components of total cost.
2. **(P2) Partial externality pricing — extension of Pigou 1920.** Existing externality-pricing instruments (Pigouvian taxes; cap-and-trade; reclamation bonds) cover specific externalities partially; comprehensive coverage of all externalities (including intergenerational) is not currently implemented in any jurisdiction. Per Pigou 1920 framework + Stern Review 2007 (climate externality incompleteness exemplar).
3. **(P3) Current accountability bond inadequacy — empirical.** Across major non-renewable extraction jurisdictions in the contemporary decade, the aggregate Accountability Bond B is empirically less than the aggregate Residual Commons Value RCV. Supporting evidence: §11 validation cases (McDowell County coal; Black Lung Trust Fund deficit; EU ETS gaps; Norwegian sovereign-wealth-fund partial-coverage analysis).

Statement. Under Premises P1, P2, P3, for non-renewable extraction R at time t_0 in any jurisdiction observed under current institutions, aggregate B(R, t_0) < aggregate RCV(R, t_0). Therefore by Theorem 10.1a, CS(R, t_0) > 0.

Proof. By Premise P1, the intergenerational components of RCV cannot be priced through market mechanisms; therefore market-based accountability instruments do not internalize these components. By Premise P2, the externality components of RCV are partially priced through existing Pigouvian instruments; the partial coverage means that the externality components of B are strictly less than the corresponding components of RCV in aggregate. By Premise P3 (empirical), across current-institution jurisdictions, neither comprehensive intergenerational coverage nor comprehensive externality coverage is implemented; therefore aggregate B < aggregate RCV.

By Theorem 10.1a, B < RCV implies CS > 0. The conclusion follows: CS > 0 holds for non-renewable extraction under current institutions across the jurisdictions empirically validated in §11. ∎

Symmetric application — backward CSD-side under canonical CS. The theorem extends symmetrically to the backward Cost Severance Damages side under the canonical bidirectional form `total CS = (CSD − B₁) + (RCV − B₂)` per §5.5. The same premise structure operates on the backward instrument: P1 (intergenerational market incompleteness — Coase 1960) establishes $RCV - B_2 > 0$ on the forward instrument as derived above; P2 (partial externality pricing — Pigou 1920) applies symmetrically to backward damages, with past extraction externalities partially priced through restitution mechanisms (B_1) and the partial coverage establishing $CSD - B_1 > 0$ in jurisdictions with incomplete restitution. Sum is strictly > 0, so canonical CS > 0. Under canonical CS, the theorem strengthens: only one of ($CSD - B_1$) or ($RCV - B_2$) needs to be > 0 for canonical CS > 0. Across Book 1's empirical scope this is over-determined — both terms are positive across the §11 case studies.

Falsifiability. Theorem 10.1b is falsifiable by either (a) discovery of a non-renewable extraction case under current institutions where aggregate B ≥ aggregate RCV (single counterexample); or (b) empirical demonstration that across most current institutions, accountability instruments are sufficient (B ≈ RCV) such that Premise P3 fails. Empirical evidence (§11) addresses (b); ongoing monitoring of jurisdictional accountability instruments addresses (a).

Domain of applicability. Theorem 10.1b applies to non-renewable extraction (resources whose substitutability function S(t) does not approach 1 within the relevant time horizon — see §13 existential substitutability gap analysis) under contemporary institutions (defined as the set of accountability instruments in operation in the current decade across jurisdictions where the extraction occurs). Renewable extraction under sustainable management, and non-renewable extraction in hypothetical jurisdictions with comprehensive accountability instruments, are outside the theorem's domain (and would constitute falsifying counterexamples if observed empirically).

Empirical Observation 10.2 (Cross-Model Convergence)

Across the framework's tested calibration case landscape — four empirical cases (McDowell coal; Norway oil; Deepwater Horizon; Libby vermiculite) plus two thought-experiment scenarios (asteroid mining; lunar regolith — see §13): six cases total — the Damage Function approach, Real Options approach, and RCV model produce IPG estimates within one order of magnitude. This empirical regularity is consistent with the underlying cost severance structure being approximately characterized by all three approaches.

Supporting argument (per §9 Model Independence Defense). The three approaches use uncorrelated data sources — Damage Function from documented litigation outcomes + epidemiological studies + remediation cost estimates; Real Options from market volatility data; RCV from materials-science substitutability estimates. Under these data-source-independence conditions and a common target (true social cost), correlated measurement error would not consistently produce IPG estimates within one order of magnitude across the tested case landscape (cf. Hong & Page 2004 *PNAS* "Groups of diverse problem solvers can outperform groups of high-ability problem solvers"; Mosteller & Tukey 1977 *Data Analysis and Regression* on multiple-estimator convergence).

Failure modes (when the empirical regularity may break): (a) the three approaches share systematic bias from common framework choices (e.g., similar discount-rate framework, similar substitutability assumptions); (b) the tested case landscape is too small to surface latent divergence patterns; (c) case selection biases toward cases where convergence is expected. Future empirical work (numerical Monte Carlo + additional case calibrations) may surface failure modes.

Domain of applicability. Tested calibration case landscape per §11. Generalization to additional resource classes, geographies, or extraction regimes requires further empirical work.

Categorization note. Empirical Observation 10.2 is presented as an empirical observation rather than a theorem because its claim form is empirical regularity supported by case-evidence and data-source-independence reasoning, not derivation from primitive premises. The framework's Theorems 10.1, 10.3, 10.4, 10.5 are derivable theorems with mathematical-derivation proofs; 10.2 is a categorically different claim form. The categorization difference is honored in the labeling.

Theorem 10.3 (Abundance Masking under Stock-Flow Framework)

Assumptions A1–A4:

1. **(A1) Stock-flow framework.** The commons under analysis is a non-renewable depletable resource with abundance $A(t) \in (0, \infty)$ where A is monotonically non-increasing under continuous extraction.
2. **(A2) Cost-function structure.** The cost function $c: (\tau, \infty) \to \mathbb{R}_{\geq 0}$ is continuous, non-negative, monotonically decreasing in A, with the limit behavior:
   - $\lim_{A \to \infty} c(A) = 0$ (or bounded above by a small constant).
   - $\lim_{A \to \tau^+} c(A) = \infty$ (or grows faster than any polynomial in $1/(A - \tau)$).

   The framework adopts (without loss of generality) the explicit functional form $c(A) = c_0 \cdot (\tau/(A - \tau))^\xi$ for some $\xi \geq 1$; alternative forms with the same limit behavior are admissible.
3. **(A3) Standard supply-elasticity behavior.** Supply elasticity $\eta_S(A)$ approaches zero as $A \to \tau^+$, consistent with standard microeconomic supply theory. Equivalently, marginal extraction cost approaches the resource's reservation price near scarcity threshold.
4. **(A4) Domain restrictions.** Theorem 10.3 applies to non-renewable depletable commons under stock-flow framework with standard supply-elasticity. The theorem does NOT apply to:
   - Renewable commons where regeneration ≥ extraction rate (cost-function does not exhibit the limit behavior).
   - Information / non-rivalrous commons where use does not deplete the stock.
   - Network-effect commons where cost dynamics depend on coordination effects.
   - Multi-threshold commons (separate analysis required for each transition; Theorem 10.3 applies locally to each transition).

Abstract premise framing (complementary to A1–A4): Assumption A2's explicit functional form realizes four abstract cost-function properties: (P1) monotonicity in abundance; (P2) abundance limit $\lim_{A \to \infty} c(A) = 0$; (P3) convexity in scarcity ratio; (P4) nonlinear growth near scarcity threshold. The two framings are complementary: A1–A4 are concrete premises with explicit functional form realization; P1–P4 are abstract cost-function properties A2's functional form realizes.

Notation note. τ (tau) denotes scarcity threshold throughout this theorem; ξ (xi) denotes the cost-function curvature parameter. The substitutability function $S(t)$ appearing in the framework's RCV integrand (§3.1; Theorem 10.4) is distinct and disjoint in usage; S_{max} denotes the long-run substitutability limit (existential substitutability gap; §13).

Statement. Under Assumptions A1–A4, for any non-renewable depletable commons with cost function $c(A)$ per A2:

- **(i) As abundance $A \to \infty$, cost $c(A) \to 0$; equivalently, for any $\varepsilon > 0$ there exists A_ε such that $c(A) < \varepsilon$ for $A > A_\varepsilon$** — cost is masked by abundance to within any specified tolerance once abundance is sufficiently large relative to scarcity threshold.
- **(ii) As abundance $A \to \tau^+$ (approaches scarcity threshold from above), cost $c(A) \to \infty$** — cost becomes visible and grows nonlinearly.

Proof of (i). By Assumption A2, $\lim_{A \to \infty} c(A) = 0$. Therefore for any $\varepsilon > 0$, $\exists A_\varepsilon$ such that $c(A) < \varepsilon$ for $A > A_\varepsilon$. In particular, when A is sufficiently large compared to τ, cost is masked by abundance to within any specified tolerance. ∎

Proof of (ii). By Assumption A2, $\lim_{A \to \tau^+} c(A) = \infty$. Therefore for any $M > 0$, $\exists \delta > 0$ such that $c(A) > M$ for $\tau < A < \tau + \delta$. Equivalently, as abundance approaches scarcity threshold from above, cost grows without bound.

The functional form $c(A) = c_0 \cdot (\tau/(A - \tau))^\xi$ with $\xi \geq 1$ satisfies these limit conditions explicitly. Two complementary academic lineages support the functional form:

- **Commodity-economics + supply-elasticity lineage (price-discovery framing):** By Assumption A3 (standard supply-elasticity behavior — Marshall 1890 §V; modern microeconomics), the hyperbolic-type functional form is consistent with empirical commodity supply behavior near binding constraints. Hotelling 1931 rent dynamics + commodity-shock empirical patterns (Hamilton 2009 *Brookings Papers on Economic Activity*; Kilian 2009 *American Economic Review*; Pindyck 2008 *Energy Journal*) display the abundance-masking pattern in priced markets: prices rise nonlinearly as binding-constraint regimes are approached.
- **Resource-economics + tipping-point lineage (biophysical/structural framing):** Stock-dependent cost-function behavior (Pindyck 1978 *Journal of Political Economy*; Dasgupta & Heal 1979 *Economic Theory and Exhaustible Resources*) provides the resource-economics foundation. Ecological tipping-point dynamics (Barnosky et al. 2012 *Nature* "Approaching a state shift in Earth's biosphere"; Lenton et al. 2008 *PNAS* "Tipping elements in the Earth's climate system") provide the same nonlinear-growth-near-threshold behavior in ecological systems.

The two lineages are complementary: commodity-economics analyzes abundance-masking via price-discovery mechanisms; resource-economics + tipping-point analyzes it via biophysical state-shift mechanisms. Together they establish the phenomenon as both market-economics-recognizable and ecological-economics-recognizable. ∎

Empirical illustration. §13 demonstrates the abundance-masking pattern across illustrative cases: asteroid iron ($A \gg \tau$) has RCV ≈ 0; lunar helium-3 near the fusion-deployment threshold (A approaching τ) has high existential substitutability gap. §11 validation cases provide additional empirical support: McDowell

County coal (post-peak; $A < \tau$ regime); rare-earth minerals (China export-restriction price spikes 2010–2012); oil-shock dynamics (Hamilton 2009; Kilian 2009).

Operational corollary. The Commons Inversion Test (CIT; Definition 1.8) is the methodological protocol that detects the abundance-masking phenomenon established analytically here. Stripping away dimensions of abundance and observing which cost components transition from $C \approx 0$ to $C \to$ large is the empirical signature of Theorem 10.3's analytic prediction. CIT's validity as a discovery methodology follows from Theorem 10.3.

Theorem 10.4 (RCV Integral Convergence under Weitzman Declining Discount)

Assumptions A1–A4:

1. **(A1) Utility growth.** $U: \mathbb{R} \times [t_0, \infty) \times \mathbb{R} \to \mathbb{R}_{\geq 0}$ satisfies polynomial growth: $\exists\, U_0 \geq 0, k \geq 0$ such that $U(R, t, Q(t)) \leq U_0 \cdot (1 + t)^k$ for all $t \geq t_0$.
2. **(A2) Externality growth.** $E: \mathbb{R} \times [t_0, \infty) \to \mathbb{R}_{\geq 0}$ satisfies polynomial growth: $\exists\, E_0 \geq 0, m \geq 0$ such that $E(R, t) \leq E_0 \cdot (1 + t)^m$ for all $t \geq t_0$.
3. **(A3) Substitutability function.** $S: [t_0, \infty) \to [0, 1]$ is non-decreasing (substitutability is monotonically improving over time as substitute technologies mature). $S(t) \to S_{max} \in [0, 1]$ as $t \to \infty$.
4. **(A4) Discount factor — Weitzman 2001 declining-rate framework.** $D(t, t_0) = \exp(-\int_{t_0}^t r(s)\, ds)$ where $r: [t_0, \infty) \to \mathbb{R}_{\geq 0}$ is bounded. Per Weitzman 2001, $r(s) \to r_\infty \geq 0$ as $s \to \infty$ (long-run certainty-equivalent rate).

Statement. Under Assumptions A1–A4, the RCV integral $RCV(R, t_0) = \int_{t_0}^\infty \{[1 - S(t)] \cdot U(R, t, Q(t)) + E(R, t)\} \cdot D(t, t_0)\, dt$ converges (is finite) when at least one of the following sufficient conditions holds:

- **(SC1) Positive long-run discount:** $r_\infty > 0$.
- **(SC2) Sufficient substitutability convergence:** $\int_{t_0}^\infty [1 - S(t)] \cdot U(R, t, Q(t))\, dt < \infty$ AND $\int_{t_0}^\infty E(R, t) \cdot D(t, t_0)\, dt < \infty$.

Knife-edge divergence corollary. $RCV(R, t_0) = \infty$ when $S_{max} < 1$ AND $\int_{t_0}^\infty r(s)\, ds < \infty$ AND $\int_{t_0}^\infty [1 - S_{max}] \cdot U(R, t, Q(t))\, dt = \infty$. The summable-decay hypothesis $\int r(s)\, ds < \infty$ is strictly stronger than $r_\infty = 0$; it guarantees $D_\infty = \exp(-\int_{t_0}^\infty r(s)\, ds) > 0$ (required for the lower bound used in the proof). Explicit cases satisfying the strengthened hypothesis include $r(s) = 0$ for s sufficiently large and $r(s)$ integrable on $[t_0, \infty)$; the hypothesis excludes slow-decaying r such as $r(s) = 1/s$ where $r_\infty = 0$ but $\int r(s)\, ds = \infty$ and $D_\infty = 0$. Under the strengthened hypotheses, permanent non-substitutable foreclosure under summable discount-rate decay accumulates unbounded utility-weighted cost.

Proof. $RCV(R, t_0) = \int_{t_0}^\infty f(t)\, dt$ where $f(t) = \{[1 - S(t)] \cdot U(R, t, Q(t)) + E(R, t)\} \cdot D(t, t_0)$ is non-negative under A1, A2, A3 ($S(t) \in [0, 1] \Rightarrow 1 - S(t) \geq 0$; $U \geq 0$ by A1; $E \geq 0$ by A2; $D > 0$).

Case (SC1): $r_\infty > 0$. Then $\exists\, T \geq t_0$ such that for $s \geq T$, $r(s) \geq r_\infty/2 > 0$. For $t \geq T$:

$$D(t, t_0) = \exp(-\int_{t_0}^t r(s)\, ds) \leq \exp(-\int_T^t (r_\infty/2)\, ds) \cdot \exp(-\int_{t_0}^T r(s)\, ds) = K \cdot \exp(-(r_\infty/2) \cdot (t - T))$$

for constant $K = \exp(-\int_{t_0}^T r(s)\, ds)$. The integrand is bounded by:

$$f(t) \leq [U_0 \cdot (1 + t)^k + E_0 \cdot (1 + t)^m] \cdot K \cdot \exp(-(r_\infty/2) \cdot (t - T))$$

Polynomial growth dominated by exponential decay ⇒ integrand integrable on $[T, \infty)$. Continuous on $[t_0, T]$ ⇒ integrable on $[t_0, T]$. By Lebesgue dominated convergence theorem (Royden 1988 §4.4; Folland 1999 §2.4), $\int_{t_0}^\infty f(t)\,dt < \infty$. ∎ Case SC1.

Case (SC2): $\int_{t_0}^\infty [1 - S(t)] \cdot U(R, t, Q(t))\,dt < \infty$ AND $\int_{t_0}^\infty E(R, t) \cdot D(t, t_0)\,dt < \infty$ (the latter independent assumption since E's convergence does not depend on substitutability).

$\int_{t_0}^\infty f(t)\,dt = \int_{t_0}^\infty [1 - S(t)] \cdot U \cdot D\,dt + \int_{t_0}^\infty E \cdot D\,dt$

Since $D(t, t_0) \leq 1$ (D is a decay factor), $[1 - S(t)] \cdot U \cdot D \leq [1 - S(t)] \cdot U$ pointwise. By monotonicity of the Lebesgue integral for non-negative integrands and the pointwise bound, $\int_{t_0}^\infty [1 - S(t)] \cdot U \cdot D\,dt \leq \int_{t_0}^\infty [1 - S(t)] \cdot U\,dt < \infty$ by the first SC2 hypothesis. The second SC2 hypothesis directly gives the second integral finite. Sum is finite. ∎ Case SC2.

Knife-edge corollary proof. Suppose $S_{max} < 1$ and $\int_{t_0}^\infty r(s)\,ds < \infty$. Then:

- $1 - S(t) \geq 1 - S_{max} > 0$ for all $t \geq t_0$ (since $S(t) \leq S_{max}$).
- $D(t, t_0) \to D_\infty = \exp(-\int_{t_0}^\infty r(s)\,ds) > 0$ (from the summable-decay hypothesis $\int r(s)\,ds < \infty$; this is strictly stronger than $r_\infty = 0$ alone, which permits slow-decaying r such as $r(s) = 1/s$ where $D_\infty = 0$).
- Hence $f(t) \geq [1 - S_{max}] \cdot U(R, t, Q(t)) \cdot D_\infty$.
- If $\int_{t_0}^\infty [1 - S_{max}] \cdot U\,dt = \infty$, then $\int_{t_0}^\infty f(t)\,dt \geq D_\infty \cdot [1 - S_{max}] \cdot \int_{t_0}^\infty U\,dt = \infty$.

Hence $RCV(R, t_0) = \infty$ under the knife-edge conditions. ∎ Corollary.

Interpretation note. The knife-edge divergence is not a pathological failure of the model — it correctly reflects that permanent non-substitutable foreclosure under zero long-run discount with cumulative utility is genuinely infinite cost. A civilization that values future generations equally to the present (zero pure-time-preference) and faces permanent loss of something irreplaceable ($S_{max} < 1$) with cumulative welfare stakes ($\int U = \infty$) confronts infinite expected cost. The model's divergence is the correct answer; the policy implication is "do not extract under these conditions."

Theorem 10.5 (Substitution Dominance)

Premises P1–P4:

1. **(P1) Social opportunity cost exceeds market price.** $RCV(R, t_0) > P$ (equivalently, $IPG(R, t_0) > 1$ per §3.2). Under Theorem 10.1b's empirical premise ($B < RCV$ across current institutions), this condition is empirically supported across §11 case studies (the $IPG \gg 1$ finding). Note: $CS > 0$ (i.e., $RCV > B$ per Theorem 10.1a) is a related but distinct condition; the substitution-dominance result here requires $RCV > P$ (social opportunity cost greater than extractor revenue), not $CS > 0$ (social cost greater than posted bond). The two conditions coincide only when $B = P$, which generally does not hold under current institutions per Theorem 10.1b.
2. **(P2) Substitutability-improvement structural premise.** There exists a substitute technology S^* such that investment in S^* shifts $S(t)$ upward: $S^*(t) > S(t)$ for all $t \geq t_0$.
3. **(P3) Future-generation positive weight.** The social welfare aggregation assigns positive weight to future generations: $\partial SW/\partial U(t) > 0$ for some $t > t_0$.
4. **(P4) Investment-cost opportunity-cost framing.** $Investment\_cost \leq P$ (the resources funding substitute investment are the opportunity-cost-equivalent of extraction-revenue resources).

Statement. Under Premises P1–P4, for any non-renewable resource R, investment in a substitute technology S* (that increases S(t)) is weakly dominant over extraction of R from a social welfare perspective.

Proof. Consider the social welfare comparison between two paths:

- **Path A:** Extract R at t_0, receiving market price P.
- **Path B:** Invest equivalent resources (per P4) in substitute technology S*, delaying or foregoing extraction of R.

Under Path B (per P2), $S^*(t) > S(t)$ for all $t \geq t_0$, which:

- *(i)* Reduces the foreclosure component: $[1 - S^*(t)] \cdot U(R, t, Q(t)) < [1 - S(t)] \cdot U(R, t, Q(t))$ for all t. ΔForeclosure_cost > 0 (welfare-positive change).
- *(ii)* Preserves the strategic option to extract later if needed, maintaining full option value (cf. Dixit & Pindyck 1994 *Investment under Uncertainty*). ΔOption_value > 0.
- *(iii)* Reduces the externality tail E(R, t) by the amount of extraction deferred or avoided. ΔExternality_avoided > 0.

Welfare-comparison algebra, tracking three distinct quantities — RCV (social opportunity cost), B (existing accountability bond), P (extractor revenue):

SW(Path B) − SW(Path A) = ΔForeclosure_avoided + ΔOption_value + ΔExternality_avoided − (Investment_cost − P).

Each Δ is non-negative under P2 (substitutability improvement; reduces foreclosure), Dixit & Pindyck 1994 option value (preserved by deferral), and externality avoidance respectively. Under P4 (Investment_cost ≤ P), the parenthesized term (Investment_cost − P) is ≤ 0. Therefore SW(Path B) ≥ SW(Path A) whenever P2–P4 hold (weak dominance).

The specific magnitude argument requires RCV > P (per the revised P1): the foregone extraction revenue P is less than the social opportunity cost RCV of letting the resource be extracted. This is a distinct claim from CS > 0: CS = RCV − B measures the gap between social cost and existing bonds (Theorem 10.1a); RCV > P measures the gap between social cost and extractor revenue (the IPG > 1 condition per §3.2). CS > 0 holds whenever B < RCV; RCV > P holds when social opportunity cost exceeds market value, which is empirically supported across §11 case studies. Adding positive option value and positive externality avoidance can only strengthen the SW(Path B) ≥ SW(Path A) inequality; strict inequality follows when RCV > P strictly (per P1).

This holds:

- **Under any discount rate consistent with P1 (CS > 0):** Weitzman 2001 declining-rate framework establishes CS > 0 conservatively; high constant discount rates that drive CS toward zero violate P1 and are out of scope.
- **Under any ethical framework satisfying P3 (positive weight to future generations):** pure-presentism (zero weight to future) violates P3 and is out of scope.
- **Under any S_max where P2 holds (investment can improve S(t)):** technological floors that prevent S(t)-improvement violate P2 and are out of scope.

Lineage. Theorem extends Hartwick 1977 *American Economic Review* sustainable-investment rule (substitute-investment-from-rents) under Weitzman 2001 *American Economic Review* declining-discount + Dixit & Pindyck 1994 option-value + Solow 1974 *Review of Economic Studies* intergenerational-equity baseline. Welfare-comparison formalism follows Stiglitz 2000 *Economics of the Public Sector*. Climate-economics application precedented in Stern 2007 *The Economics of Climate Change: The Stern Review*. ∎

Failure modes (when the dominance result breaks): (a) **Discount-rate failure:** under sufficiently high discount rate that drives CS to zero or below, P1 fails. (b) **Substitutability-investment failure:** if no substitute technology can meaningfully improve S(t), P2 fails. (c) **Ethical-framework failure:** under pure-presentism (zero weight to future generations), P3 fails. (d) **Investment-cost failure:** if substitute-investment is genuinely more expensive than extraction-revenue (Investment_cost > P), P4 fails; weak dominance becomes a strict comparison rather than guaranteed dominance.

Corollary 10.5.1 (Optimal Extraction Sequencing). Under Theorem 10.5's premises, when extraction is undertaken, social-welfare-optimal sequencing extracts from sources where RCV is lowest first.

Proof of Corollary 10.5.1. Direct application of Theorem 10.5 across resource-class comparisons. Among extraction-eligible resources, those with lower RCV violate the dominance result less severely; among substitution-eligible resources, those with higher CS strengthen the dominance result.

The framework therefore prioritizes:

- *(i)* Asteroid mining (RCV ≈ 0; substitutability ≈ 1; minimal externality tail) over Earth's critical mineral deposits (high existential substitutability gap, high RCV).
- *(ii)* Renewable energy investment over fossil extraction (CS → ∞ as climate externalities are fully priced under policy-completion regimes). ∎

Interpretation note. The framework is a navigation system, not a prohibition. Extraction is permitted; the framework directs extraction toward low-RCV sources first.

§11. Empirical Validation Cases

The framework's backtesting across historical cases demonstrates consistent cost severance patterns and validates the three-model convergence. All four cases show IPG ≫ 1; all four show the same structural pattern (value dispersed, costs concentrated); all four involve B ≈ 0 relative to RCV.

§11.1 — McDowell County Appalachian Coal (1940-2020)

Period: 1940–2020 (primary extraction 1940–1980; legacy costs ongoing)

Market price: $4.50–$18/ton (mine mouth, nominal, period average)

Damage Function estimate (floor):

- Health costs nationally: $2–4/ton (respiratory, occupational disease)
- Environmental remediation: $1–3/ton (acid mine drainage, surface restoration)
- Community costs: $5–15/ton (economic multiplier collapse, outmigration, social services)
- Carbon at EPA Social Cost of Carbon ($190/ton CO_2): ~$544/ton (dominant term for fossil fuels)
- Total without carbon: $8–22/ton
- Total with carbon: $552–566/ton

RCV model estimate: $580–620/ton (using Weitzman declining rates, $S_{max} \approx 0.85$ for electricity generation)

IPG: 33–122× (without carbon) / substantially higher with full carbon pricing

Observed cost severance: $3.7–$6B reclamation shortfall (publicly documented); population decline from 100,000 to 18,000; 13-year life expectancy gap vs. national average; highest per-capita opioid death rate in the United States.

Spatial cost severance. The Damage Function IPG reflects national per-ton underpricing. The community-level IPG is dramatically higher because costs are geographically concentrated in McDowell County while value is dispersed nationally. The $544/ton carbon term is a global atmospheric externality; the community costs are localized. Both belong in E(R,t); they operate at different spatial scales.

§11.2 — Deepwater Horizon: Macondo Well (2010)

Well lifetime revenue (projected): ~$1.1B (40–60M barrels at 2010 prices)

Total documented costs:

- BP legal settlements: $18.7B (Clean Water Act, economic damages, NRD)
- Federal criminal penalty: $4B
- Ongoing ecosystem and fisheries losses (through 2025): estimated $8–12B additional
- Total documented: $20–30B

IPG: 15–17× (documented costs only, lower bound)

RCV model estimate: 16–19× (including ongoing Gulf ecosystem externalities)

Convergence: ✓ All three models agree within a factor of 1.5.

Observed CS: Gulf Coast fishing communities bore costs; BP shareholders received pre-spill profits. Post-spill BP stock partially recovered; Gulf fisheries remain impaired 15 years later.

§11.3 — Libby, Montana: W.R. Grace Vermiculite Mine (1963-1990)

Lifetime extraction revenue: ~$100M (80% of world vermiculite supply)

Total documented costs:

- Federal Superfund cleanup (ongoing): $600M+ committed

- Legal settlements (asbestos wrongful death/injury): $250M+
- Ongoing medical monitoring and treatment: $50M+/year ongoing
- Lost property values, economic disruption: estimated $500M+
- Total documented: $4B+ (40:1 cost-to-revenue ratio, minimum)

IPG: 55–82× (Damage Function) / 48–76× (Real Options) / 61–91× (RCV)

Key CS feature: W.R. Grace knew about asbestos contamination from 1963 and concealed it for 27 years — a textbook case of Political Capture Cost. The concealment extended CS by three decades. The accountability bond B ≈ 0 throughout the operating period because the externality was hidden.

§11.4 — Baotou Inner Mongolia: Rare Earth Processing (1980-present)

Context: Baotou is the dominant rare-earth mining and processing complex within China; China accounts for approximately 69% of global rare-earth mine production (2024 per USGS *Mineral Commodity Summaries 2025*), and the Bayan Obo deposit within Baotou's prefecture holds approximately 37–40% of global rare-earth reserves (within the published-estimate range; USGS reports China holds ~49% of global reserves, of which Bayan Obo holds ~80%) and is the single largest rare-earth deposit in the world. The toxic tailings lake (Weikuang Dam) covers 11 km² and receives ~10M tons of radioactive tailings annually.

Value captured: Chinese rare earth export revenue ~$2–4B/year; downstream electronics manufacturing value estimated $1T+ in global consumer goods enabled by Baotou-sourced rare earths.

Costs concentrated locally:

- Elevated cancer rates in surrounding villages (lung, stomach, skin)
- Agricultural land rendered unusable within 10km radius
- Groundwater contamination of the Yellow River basin (serves 100M people)
- Forced relocation of entire villages
- Remediation cost estimated at $5–15B for tailings lake alone

IPG: 12–35× (Damage Function) / 18–48× (Real Options) / 22–41× (RCV)

CS structure: Global electronics consumers capture value; Baotou residents bear costs. This is the most explicit case of spatial cost severance operating at international scale — the geographic distance between beneficiary (Silicon Valley consumer) and cost-bearer (Inner Mongolian villager) is ~10,000 km. Cost severance scales with distance.

§11.5 — Norway petroleum calibration (Three Ways of Counting executed)

Worked execution of Methods 1+2+3 against Norway 1971–present petroleum extraction data. Numerical anchors below are the framework's empirical operationalization.

Norway calibration — Three Ways of Counting executed

Empirical anchors (Norway petroleum extraction, 1971–present)

Anchor	Value	Source
Cumulative oil + gas production through 2024	~55 billion BOE through 2024 (~8.9 billion Sm^3 o.e. cumulative per Norwegian Offshore Directorate latest resource accounting; ~57% of estimated total recoverable resources of ~15.6 billion Sm^3 o.e.)	Norwegian Offshore Directorate cumulative-production data; Norwegianpetroleum.no historical production page
GPFG (Government Pension Fund Global) value, end-2025	~$2.0 trillion (21,268 billion NOK per NBIM end-2025 figures; ~$2.0T at year-end FX); reached $2.2 trillion peak during 2025	Norges Bank Investment Management (NBIM) Annual Report 2025; Ministry of Finance
Per-citizen GPFG share	~$356,000–$391,000 / Norwegian citizen (range from $2.0T end-2025 value to $2.2T 2025-peak value, divided by population ~5.62M per Statistics Norway 2025)	NBIM Annual Report 2025; Statistics Norway population estimates
State capture rate (B / petroleum wealth)	~70-80% (direct taxation + production licenses + Equinor state ownership + fund contributions)	Norwegian Petroleum Directorate; Norway Ministry of Finance
Annual fiscal-rule withdrawal	~3% expected real return	Norway fiscal rule (2001)
Carbon intensity (combusted)	~0.43 tons CO_2 per barrel of oil; ~0.31 tons CO_2 per BOE-equivalent natural gas (per EPA / EIA / IPCC AR6 standard combustion factors; equivalently 2.75 t CO_2 per ton CH_4 via combustion stoichiometry, applied to natural-gas energy content of ~52 mmBtu/ton and BOE energy content of 5.8 mmBtu)	EPA emissions factors; EIA CO_2 Emissions Coefficients by Fuel; IPCC AR6 standard combustion factors
Cumulative emissions from Norwegian production (combusted)	~20.35 billion tons CO_2 (mixed oil-and-gas, weighted)	Calculated under cumulative-historical-average production split (approximately 50% oil / 50% gas per Norwegian Offshore Directorate): 27.5B BOE oil × 0.43 t/BOE + 27.5B BOE gas × 0.31 t/BOE = 11.825 + 8.525 ≈ 20.35B t CO_2
Norwegian production-side emissions	~3-5% of combustion-side emissions	Norwegian Environment Agency reports

Anchor	Value	Source
(extraction + processing)		

Method 1 — Replacement Cost (lower-bound floor for Habitability commons)

Logic: Method 1 prices the engineering cost to replace the substitutable component of the commons. For Norwegian oil, the substitutable component on the Habitability axis is the atmospheric CO_2-removal function the natural sinks provide for free; the engineering substitute is direct-air-capture (DAC). Three-horizon DAC range:

- **Conservative (current operational, first-of-a-kind DAC):** \$600–\$1,000/ton CO_2 × 20.35B tons = **\$12.2–\$20.35T total atmospheric replacement cost** (weighted oil + gas).
- **Mid-range (at-scale commercial within next decade):** \$300–\$600/ton × 20.35B tons = **\$6.1–\$12.2T** .
- **Optimistic (2050 nth-of-a-kind):** \$100–\$300/ton × 20.35B tons = **\$2.04–\$6.1T** .

Per-BOE conversion:

- Conservative DAC: \$12.2–\$20.35T / 55B BOE = **\$222–\$370/BOE atmospheric-only Method 1 floor** .
- Mid-range DAC: \$111–\$222/BOE.
- Optimistic DAC: \$37–\$111/BOE.

Scope of this floor. Method 1 floor as constructed prices ONLY the Habitability commons (atmospheric CO_2-removal substitute). It does NOT price the foreclosed-non-energy-use of the petroleum hydrocarbons (which Norway acknowledges is a remaining commons-loss per case file: "the barrel is still gone"). Adding Foreclosure-of-non-energy-use raises the Method 1 floor by an additional \$50-200/BOE depending on substitute-availability assumptions for petrochemical feedstock + specific-hydrocarbon-application uses.

Method 1 anchor for Norway (Habitability + Foreclosure-of-non-energy-use): \$161–\$422/BOE at mid-range DAC (\$111–\$222/BOE atmospheric Habitability + \$50–\$200/BOE moderate Foreclosure-of-non-energy-use estimate).

Method 2 — Norway Revealed Preference (anchor; what Norway actually internalized)

Logic: Method 2 reads the per-BOE value Norway's polity revealed it considered worth internalizing through the GPFG accumulation + tax architecture. Under the framework's epistemic-humility discipline, this is a **lower-bound on true RCV** (Norway internalized real value but did NOT internalize all of it — atmospheric externality + non-Norwegian-population climate costs + non-energy-use foreclosure are not in the GPFG).

Calculation:

- GPFG value \$2.0T (end-2025) at ~75% capture rate of net petroleum wealth (mid-point of 70-80% range).

- Implied total petroleum wealth: \$2.0T / 0.75 ≈ **\$2.67T** .
- Per-BOE value: \$2.67T / 55B BOE ≈ ~**\$48/BOE Method 2 anchor** (down marginally from prior \$50/BOE anchor at \$1.9T fund value / 50B BOE; updated production denominator drives a small directional shift).

Caveats (epistemic-humility):

- Norway's capture rate INCLUDES the foreign-economy reinvestment compounding effect — the GPFG's growth is partly the global-economy productive output reinvested. Stripping the foreign-compound-interest component down to extraction-time rent capture, the per-BOE rent-capture is smaller (~\$30-40/BOE).
- \$50/BOE is in 2025 dollars; older extraction-era rent was lower in nominal terms. Inflation-adjusted, the inflation-corrected per-BOE accumulation is roughly \$50/BOE.
- Norway's capture is per-Norwegian-citizen + per-Norwegian-future-generation accounting. It does NOT account for non-Norwegian populations affected by Norwegian oil's climate externalities. Per the case file: "Norwegian-welfare-state is approximately B1-for-Norwegian-citizens but doesn't extend to non-Norwegian populations." So Method 2 anchor is **B_1-for-Norwegians-only** , not aggregate B for the global commons.

Method 2 anchor for Norway: ~\$48/BOE (B_1-for-Norwegians-only; B_2 globally underdeveloped).

Method 3 — Scarcity-Adjusted Option Value (upper bound, Norway)

Logic: Method 3 prices the scarcity-adjusted option value per the cross-case dominance pattern and worked-example application in §11.8 (calibrated parameters for Norway).

Norwegian parameter calibration:

- $\sigma \approx 50$–200 (oil reserves recoverable via secondary extraction; functionally non-renewable on civilizational scale but partially substitutable)
- $\alpha \approx 0.50$–0.75 (institutional architecture moves Norway out of α-dominance regime — restoration optionality preserved through GPFG)
- V_option ≈ \$30–200/BOE (climate-policy + future-substitutability uncertainty)
- $\beta = 1$ (default conservative-irreversibility regime)

Calculation (mid-range parameter choice — σ=100, α=0.65, V_option=\$80, β=1):

- scarcity_multiplier(σ=100) = 1 + log(101) × Hotelling_anchor (using 5%/yr Hotelling-rate proxy) ≈ 1 + 4.6 × 0.05 ≈ 1.23
- irreversibility_premium(α=0.65, β=1) = 1/(0.35) ≈ 2.86
- **Method 3 RCV ≈ \$80 × 1.23 × 2.86 ≈ \$281/BOE**

Sensitivity range across parameter ranges:

- Low parameters (σ=50, α=0.50, V_option=\$30): \$30 × 1.20 × 2.0 ≈ **\$72/BOE**

- High parameters (σ=200, α=0.75, V_option=$200): $200 × 1.27 × 4.0 ≈ **$1,016/BOE**

Method 3 range for Norway: ~$70–1,000/BOE; mid-range ~$280/BOE.

Triangulation findings for Norway

Method	Estimate (per BOE)	What it prices
M1 — Replacement Cost (Habitability + non-energy-use Foreclosure)	~$300–$650	Atmospheric DAC + petrochemical-feedstock substitution
M2 — Norway Revealed Preference	~$48	B1-for-Norwegians-only (extraction rent captured)
M3 — Scarcity-Adjusted Option Value	~$70–$1,000 (mid: $280)	Combined scarcity + irreversibility + option-value

Convergence assessment:

- **M1 (full scope) and M3 mid-range estimates converge at ~$250–$300/BOE.** M1 mid-range full-scope spans ~$161–$422/BOE (atmospheric Habitability $111–$222 + Foreclosure-of-non-energy-use $50–$200); M3 mid-range ~$281/BOE. Both methods cluster in similar order-of-magnitude RCV territory when calibrated to Norway's parameters.
- **M2 is the OUTLIER LOWER-BOUND at ~$48/BOE** — exactly as the epistemic-humility constraint predicted. Norway's revealed preference is a LOWER bound on true RCV, not a middle estimate. The institutional architecture moved Norway out of the α-dominance regime BUT did not capture the full atmospheric externality OR the non-Norwegian-population climate cost.
- **Implied Cost Severance (CS = RCV − B) for Norway:** if RCV ~$300/BOE and B (per-Norwegian) ~$48/BOE, then CS per BOE ~$250/BOE. **Cumulative Norwegian CS ≈ 55B BOE × $250 ≈ $13.75T.**

Structural finding: Even Norway, the framework's canonical-existing-B2 exemplar, carries a residual CS of approximately $13–$15 trillion when triangulated across Three Ways. The framework's claim — that Norway has dramatically reduced CS but not eliminated it — is empirically supported. Norway's CS-reduction is ~84% (1 − $48/$300) relative to a hypothetical no-architecture baseline; American extraction regimes' CS-reduction is approximately 0%. The 84-percentage-point gap between Norway's CS and American CS is the framework's load-bearing empirical claim.

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§11.6 — Appalachian coal (McDowell County) calibration

Worked execution of Methods 1+2+3 against McDowell County coal-extraction data 1900–2020. Companion to §11.5 Norway.

Appalachian coal calibration — Three Ways of Counting executed

Empirical anchors (McDowell County, West Virginia, 1900-2020)

Anchor	Value	Source
Cumulative coal production (McDowell + adjacent counties, peak production era)	~600 million tons	West Virginia Geological Survey + USGS coal-production statistics
1960 average coal price (Ch 6 convergence-table anchor)	$4.50/ton	Ch 6 convergence table
Documented IPG — canonical range	~33–122×	Case-study audit §2.1 + terms_index Severed Cost record
Carbon intensity (combusted)	~2.32 tons CO_2 per ton coal (EPA emission factor for bituminous coal at short-ton accounting)	EPA AP-42 §1.1 Bituminous and Subbituminous Coal Combustion (93.28 kg CO_2/mmBtu) × EIA-published heat content for bituminous coal (~24.93 mmBtu/short ton) ≈ 2,324.56 kg/short ton ≈ 2.32 mt CO_2/short ton
Cumulative emissions from McDowell coal (combusted)	~1.4 billion tons CO_2	Calculated: 600M short tons × 2.32 mt CO_2/short ton ≈ 1.39B mt CO_2
Black Lung Trust Fund cumulative payouts (1969-present); outstanding debt	~$44 billion cumulative payouts; $5.1B outstanding debt as of September 2024	DOL Office of Workers' Compensation Programs; Black Lung Disability Trust Fund annual reports
Reclamation Bond shortfalls	$4–6 billion gap (industry-vs-actual)	OSMRE; GAO-17-207R
Bankruptcy-transferred liabilities	~$865M to taxpayers (2014-2016 alone)	OSMRE bankruptcy filings
McDowell County life-expectancy gap	13 years vs. national-average county	CDC + Robert Wood Johnson Foundation county health rankings

Method 1 — Replacement Cost (lower-bound floor)

Habitability commons (atmospheric component) — per §3.3 canonical DAC bands:

- **Conservative DAC ($600–$1,000/ton CO_2; first-of-a-kind, Climeworks Orca/Mammoth):** 1.4B tons CO_2 × $600–$1,000/ton = **$836B–$1.39T atmospheric replacement cost** (cumulative); per-ton coal at 2.32 mt CO_2/short ton × DAC range = **$1,392–$2,320/ton coal** .
- **At-scale DAC ($300–$600/ton CO_2; commercial within next decade):** 1.4B tons CO_2 × $300–$600/ton = **$418B–$836B atmospheric replacement cost** ; per-ton coal **$696–$1,392/ton coal**

.

- **Optimistic 2050 DAC (\$100–\$300/ton CO_2; nth-of-a-kind, IEA / IPCC AR6 / US National Academies):** 1.4B tons CO_2 × \$100–\$300/ton = **\$139B–\$418B atmospheric replacement cost** ; per-ton coal **\$232–\$696/ton coal** .

Habitability + Ecosystem (mountaintop-removal habitat + watershed):

- Mountaintop-removal restoration estimates (where engineering substitution is partial): ~\$50,000–\$200,000/acre disturbed; cumulative McDowell disturbed area ~150,000 acres → **~\$7.5B–30B Ecosystem Method 1 floor** (substitutable component only; non-substitutable component priced higher in Method 3).
- Per-ton coal: ~\$12–50/ton.

Cohesion + Kindred + Spatial (community-replacement):

- Cohesive-community-replacement is the least Method-1-tractable. Conservative engineering-substitute estimate: ~\$200,000-500,000/displaced-resident × 50,000 net out-migration ≈ \$10–25B. Per-ton coal ~\$17–42/ton.

Method 1 anchor for McDowell coal (combined Habitability + Ecosystem + Cohesion floors), per §3.3 canonical DAC bands: ~\$1,421–\$2,412/ton coal at conservative DAC (\$600–\$1,000/ton CO_2); ~\$725–\$1,484/ton at at-scale DAC (\$300–\$600/ton CO_2); ~\$261–\$788/ton at optimistic 2050 DAC (\$100–\$300/ton CO_2).

Method 2 — Revealed Preference (anchor)

Logic: Method 2 for McDowell reveals what existing accountability regimes captured, NOT what was wrongly extracted. The expected result is that B << RCV by orders of magnitude — the "accountability gap" the framework names.

Calculation:

- Black Lung Trust Fund cumulative payouts: \$44B over 56 years
- Reclamation bonds posted (net of shortfalls): ~\$3-5B
- Bankruptcy-transferred-to-taxpayer liabilities: ~\$0.9B (recent era only)
- Other miner-compensation + medical-cost-shifting: estimated ~\$5-10B (uncovered medical care; disability programs)
- **Total realized B for McDowell-coal context: ~\$53-60B (50+ years aggregate)**

Per-ton coal: \$53B / 600M tons ≈ \$88/ton (2025\$ adjusted; nominal-era figures are lower).

Caveat. This is an upper-bound on realized B because some of these dollars (Black Lung Trust Fund) are paid by the federal general fund, not by the extracting industry — meaning B as captured-by-the-extraction-bond-architecture is even lower. Industry-paid component is roughly \$5-8B (reclamation bonds + private medical settlements). **Industry-paid B per ton ~\$8-15/ton.**

Method 2 anchor for McDowell coal: ~\$8–88/ton (low end: industry-paid only; high end: societally-paid including Black Lung).

Method 3 — Scarcity-Adjusted Option Value (upper bound, McDowell coal)

McDowell parameter calibration:

- $\sigma \approx 200$–500 (coal regenerates over geological time; functionally non-renewable; high effective scarcity)
- $\alpha \approx 0.85$–0.95 (mountaintop-removal habitat irreversible at finite engineering cost; cultural-and-community commons irreversible without targeted intervention; atmospheric carbon-cycle disruption irreversible at climate-tipping-point boundary)
- V_option ≈ \$50–500/ton (per-ton coal option value of preserving the resource for future-non-energy-uses + climate-stability)
- $\beta = 1$ (conservative; $\beta = 2$ produces order-of-magnitude amplification per §3.4 sensitivity)

Calculation (mid-range parameter choice — σ=300, α=0.9, V_option=\$200, β=1):

- scarcity_multiplier(σ=300) = 1 + log(301) × 0.05 ≈ 1.29
- irreversibility_premium(α=0.9, β=1) = 1/(0.10) = 10
- **Method 3 RCV ≈ \$200 × 1.29 × 10 ≈ \$2,580/ton**

Sensitivity range:

- Low parameters (σ=200, α=0.85, V_option=\$50): \$50 × 1.27 × 6.67 ≈ **\$420/ton**
- High parameters (σ=500, α=0.95, V_option=\$500): \$500 × 1.31 × 20 ≈ **\$13,100/ton**
- Conservative β=2 at α=0.9: irreversibility_premium = 100; Method 3 ≈ \$25,800/ton (precaution-regime)

Method 3 range for McDowell coal: ~\$420–\$13,100/ton at β=1; ~\$5,000–\$130,000/ton at β=2 precaution-regime; mid-range β=1 ~\$2,500/ton.

Triangulation findings for McDowell coal

Method	Estimate (per ton coal)	What it prices
M1 — Replacement Cost (Habitability + Ecosystem + Cohesion floors)	~\$261–\$2,412 (DAC-horizon-dependent, per §3.3 canonical bands: \$261–\$788 optimistic 2050; \$725–\$1,484 at-scale; \$1,421–\$2,412 conservative)	Engineering-substitute floors for substitutable components
M2 — Revealed Preference (existing accountability regime)	~\$8–\$88 (industry-paid: \$8-15; societally-paid including Black Lung: \$50-88)	What existing B has actually captured
M3 — Scarcity-Adjusted Option Value (β=1 default)	~\$420–\$13,100 (mid: \$2,500)	Combined scarcity + irreversibility + option-value (α-dominated)

Convergence assessment:

- **M1 and M3 mid-range converge at ~$725–$2,500/ton coal.** Triangulation finding: when α is near 1 (the McDowell case), M3 dominates and exceeds M1 substantially. M1 is the floor (at-scale-DAC mid-range $725–$1,484/ton); M3 sets the upper bound ($2,500/ton mid-range, up to $13,100/ton at high parameters); the gap between them is informative — it's the irreversibility premium.
- **M2 is dramatically below M1 and M3 — by 1-3 orders of magnitude.** This is the framework's load-bearing empirical claim: existing American accountability regimes capture ~1-3% of true RCV for McDowell coal. The "accountability gap" is real, large, and triangulation-confirmed.
- **Implied Cost Severance:** at RCV ~$2,500/ton (M3 mid-range) and B ~$50-88/ton (M2 societally-paid), CS ≈ $2,400-2,450/ton. **Per-ton CS ~50× the 1960-era market price; per-ton CS dwarfs the realized B by 25-50×.**
- **IPG calculation:** if IPG = (RCV / market price), then McDowell IPG ≈ $2,500 / $4.50 (1960 nominal) ≈ **555×** — but this is mixing 1960 nominal with 2025 RCV, which inflates the ratio. Inflation-corrected (1960 $4.50 ≈ $50 in 2025), McDowell IPG ≈ $2,500 / $50 ≈ **50×** . This is roughly consistent with the canonical 33-122× range and confirms the compression at the lower end.

Structural finding: McDowell coal triangulates to an RCV in the $1,000-3,000/ton range (2025$), against realized B of $50-88/ton (societally-paid; $8-15/ton industry-paid). The 50× IPG ratio is empirically grounded by triangulation across three independent methods. The framework's "accountability gap" claim is structurally supported.

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§11.7 — CSD-RCV correlation hypothesis test

The two-case Norway + McDowell test reframes the framework's initial CSD-RCV correlation hypothesis: CSD and RCV do not invert as the original hypothesis assumed. They positively correlate (extraction-severity signature) while remaining structurally orthogonal (per the two-instrument architecture, §2). Both findings are framework-coherent.

CSD-RCV correlation hypothesis test

Hypothesis under test: *contexts with high CSD (large past extraction) tend to have low RCV (commons largely consumed); contexts with low CSD (early-stage extraction) tend to have high RCV (commons largely intact).*

The two-case test

Norway:

- CSD: relatively low (Norwegian welfare-state outcomes good; Norwegian population per-capita well-served by GPFG; Norwegian land minimally damaged by extraction; CSD is concentrated in non-Norwegian populations affected by climate externalities, which is large in absolute terms but distributed globally rather than localized).

- RCV: moderate-to-high. Norway captured ~17% of true RCV (M2 / M3 mid-range; $50/BOE / $300/BOE); the remaining 83% is uncaptured.
- **Norway = moderate CSD + high uncaptured RCV.**

McDowell coal (Appalachian):

- CSD: VERY high. 13-year life-expectancy gap; population collapse; cultural commons hollowed out; Black Lung mortality cumulative across generations; community-transition costs uncompensated.
- RCV: very high (the commons-loss continues even after extraction ends; mountaintop-removal habitat irreversible; carbon externality persists for centuries).
- **McDowell = very high CSD + very high uncaptured RCV.**

What the two-case test shows

The hypothesis predicted **inverse correlation** (high CSD → low RCV; low CSD → high RCV). The two cases show:

- Norway: moderate CSD + moderate-high uncaptured RCV.
- McDowell: very high CSD + very high uncaptured RCV.

This is POSITIVE correlation, not inverse. Both Norway and McDowell carry uncaptured RCV; McDowell carries more on both axes simultaneously.

Reframed hypothesis: the original hypothesis ("inverse correlation") may be wrong. The empirical pattern in the two-case test is **positive correlation: contexts with more extraction-history have BOTH more CSD (more harm done to commons-bearers) AND more uncaptured RCV (more commons-loss not yet priced).**

This is actually structurally consistent with the framework: extraction does not subtract CSD from RCV; it adds to BOTH. Past extraction harms the commons-bearers (CSD ↑) AND consumes the commons (RCV ↑ where uncaptured). The two move TOGETHER, not in opposition.

Implications (CSD-RCV correlation test)

For the framework:

- The **two-instrument architecture remains structurally sound** but the correlation hypothesis as initially specified should be reframed.
- The reframed hypothesis: *contexts with more extraction history carry both more CSD and more uncaptured RCV; the framework's two-instrument architecture is the apparatus that makes both visible simultaneously.*
- This is a stronger structural claim than the inverse-correlation hypothesis, because it means CSD-and-RCV-co-elevation is a SIGNATURE of severity-of-extraction rather than an independence finding.

Hypothesis status:

- **Initial hypothesis: NOT SUPPORTED by two-case test.**

- **Reframed hypothesis (positive correlation as severity-signature):** preliminarily supported; needs broader case-set verification.
- **Future case-set extension:** test on additional cases — opioid Appalachia (very high CSD + moderate RCV); Chesapeake fisheries (moderate CSD + boundary-case RCV); indigenous land dispossession (very high CSD + cross-domain RCV that's about enclosure not depletion).

Methodological importance: CSD and RCV are not redundant measurements (independence preserved); but they are co-driven by extraction-history (positive correlation expected for the framework's primary case-class).

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§11.8 — Method 3 sensitivity analysis: α-dominance finding + cross-case dominance map

Sensitivity analysis of Method 3 $RCV_{M3} = V_{option}$ × scarcity_multiplier(σ) × irreversibility_premium(α) across realistic σ + α + V_{option} ranges + functional-form specifications. Key finding: **α (irreversibility) dominates the cross-case output range** . The framework debate about RCV is therefore a debate about irreversibility — the empirical companion to the Asymmetric Regret Rule's irreversibility-weighting (§8 decision-rule level; §11.8 estimation-method level). Same structural insight expressed at decision-rule level (ARR) and estimation-method level (Method 3).

Sensitivity findings

Dominance map

Regime	σ	α	V_option	Dominant parameter	Why
Commons-extinction-class	any	0.9–1.0	any	**α**	irreversibility_premium → ∞ as α → 1; nothing else matters at order-of-magnitude scale
Scarce-but-restorable	100+	0.1–0.5	any	**σ**	scarcity_multiplier carries the work; irreversibility_premium ≈ 1
Abundant-restorable	0.1–10	0–0.1	varies	**V_option**	both multipliers ≈ 1; V_option drives the result
Hybrid (most real cases)	10–100	0.5–0.9	\$100–\$10,000	**mixed; α → σ → V_option order**	α matters most because it's near-1; σ second because it's the visible scarcity; V_option third because it's the dimensional anchor

Worked-example application (using framework's three calibration cases)

Method 3 estimates for the framework's three named calibration anchors:

McDowell coal (Appalachian extraction context):

- $\sigma \approx 200–500$ (coal regenerates over geological time; functionally non-renewable on civilizational scale)
- $\alpha \approx 0.85–0.95$ (mountaintop-removal habitat is mostly irreversible; community-and-cultural commons partially recoverable through targeted intervention but largely lost in current US policy regime)
- V_option ≈ $500–$2,000/ton (energy-system substitution + climate-stability optionality)
- **Dominance: α dominates** (irreversibility_premium ≈ 7–20 at $\alpha = 0.85–0.95$; scarcity_multiplier ≈ 6–7 via log(1+200..500); V_option ≈ $500–$2,000). The framework's RCV estimate for McDowell is driven primarily by irreversibility — the framework's "the coal is gone" / "the mountains are cut" claim is the load-bearing structural finding.

Norwegian oil (Method 2 calibration anchor):

- $\sigma \approx 50–200$ (oil reserves; recoverable secondary extraction reduces effective σ)
- $\alpha \approx 0.50–0.75$ (Norwegian extraction is partially recoverable via sovereign-wealth-fund accumulation that could fund DAC-class restoration; the resource itself is lost but the value is preserved through the institutional architecture)
- V_option ≈ $30–$200/barrel (climate-policy + future-substitutability uncertainty)
- **Dominance: σ + α near-equal; V_option third.** Norway's institutional architecture has reduced effective irreversibility (by funding restoration optionality through GPFG accumulation) — so α is moderate rather than near-1, and σ + α both matter. This is the structural reason Norway is the framework's canonical-existing-B2 exemplar: institutional architecture moves the case out of the α-dominance regime.

Asteroid mining (Ch 7 thought-experiment):

- $\sigma \approx 1–10$ (cosmic-scale stocks; substitutability across asteroids ≈ 1; effective scarcity is low)
- $\alpha \approx 0.05–0.2$ (asteroid extraction is recoverable through extraction of substitutes; minimal habitat / community / atmospheric loss)
- V_option ≈ $0.01–$10/ton (low option value because substitution is plentiful)
- **Dominance: V_option dominates by default.** Both multipliers ≈ 1; V_option × small-multiplier produces small RCV; framework's "RCV approaches market price for asteroid iron" claim follows. This is the framework correctly producing market-price-equivalent answers where market pricing is approximately right — exactly the calibration the asteroid case is designed to demonstrate.

Cross-case dominance pattern

The dominance map across the three calibration cases reveals a structural finding:

No single parameter universally dominates Method 3 across the framework's case landscape. Different cases activate different parameters as the dominant driver:

- **Earth-bound civilization-scale extractions** (McDowell coal; Appalachian habitat; atmospheric stability): α dominates because irreversibility is what makes commons-extinction the load-bearing concern.

- **Institutional-architecture-mediated extractions** (Norway oil; well-managed renewables): $\sigma + \alpha$ moderate; V_option third. Institutional architecture is the lever that shifts cases out of the α-dominance regime.
- **Substitutable / cosmic-scale extractions** (asteroid; cosmic resources): V_option dominates because both multipliers approach 1.

This is itself the framework's load-bearing structural claim about Method 3: RCV is driven by *whatever is closest to its range-extreme* , and the framework's primary cases (extraction of resources with high existential substitutability gaps) cluster in the α-dominance regime. The framework's "irreversibility is the heart of the matter" prose claim is empirically supported by this sensitivity pattern.

Sensitivity to β (irreversibility-aversion calibration)

Switching from $\beta = 1$ to $\beta = 2$ in irreversibility_premium amplifies the α-dominance regime by additional orders of magnitude near $\alpha = 1$. For the McDowell case at $\alpha = 0.9$, irreversibility_premium grows from 10 ($\beta = 1$) to 100 ($\beta = 2$), shifting the framework's RCV by an order of magnitude.

Methodological recommendation: $\beta = 1$ is the conservative default. Sensitivity to $\beta = 2$ is reported as an alternative precaution-regime per the Method 3 specification above. Analysts applying the framework can calibrate β to the case's risk-posture (climate-tipping-point cases may warrant $\beta > 1$; well-restorable cases warrant β closer to 1).

Implications for framework debate-narrowing

The sensitivity analysis was motivated by the hypothesis: *if one parameter dominates, the real debate about RCV becomes a debate about that parameter, narrowing the framework's contested surface.*

Sensitivity finding for that motivation:

- The framework's **Earth-bound primary cases** (McDowell, Deepwater, Libby, atmospheric-carbon, biodiversity, indigenous-dispossession) all cluster in the **α-dominance regime** .
- For those cases, the real debate about RCV is **a debate about irreversibility** — not a debate about discount rates, not a debate about scarcity, not a debate about option-value. **Whether commons-loss is irreversible is the load-bearing empirical question.**
- This is **the framework's optimal contested-surface positioning** : irreversibility is empirically tractable (extraction either is or isn't reversible at finite engineering cost), philosophically aligned with the framework's Asymmetric Regret Rule (irreversibility weights the rule), and academically aligned with the IPCC AR6 / climate-tipping-point literature.

Framework-positioning implication: The Method 3 specification foregrounds the α-dominance finding and frames the framework's contested surface as "is this commons-loss irreversible?" rather than the broader "is the RCV estimate correct?" The narrower question is more empirically tractable and more aligned with established climate-economics + biodiversity-economics + cultural-preservation literature.

This narrowing is itself a methodological contribution: by surfacing α-dominance, the framework gives reviewers + opponents + adopters a focused empirical question rather than an uncontainable methodological debate.

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§11.9 — DAC three-horizon range: Method 1 Habitability commons anchor specification

Operationalizes Method 1 Replacement Cost for the Habitability commons (atmospheric CO_2) via a three-horizon DAC cost range — present operational facilities (\$600–1,200/tCO_2); 2030–2040 projections (\$150–500/tCO_2); long-run scaled-engineering targets (\$100–200/tCO_2). The three-horizon framing replaces single-point anchors with time-bounded ranges that honor scale-up uncertainty. Distinct from Social Cost of Carbon (SCC): SCC measures damages-avoided whereas DAC measures direct-replacement cost — both inputs to RCV but at different methodological layers.

Direct-air-capture (DAC) cost per ton CO_2 — Method 1 Replacement Cost anchor for the Habitability commons

Method 1 (Replacement Cost; canonically defined in §3.3) uses DAC cost-per-ton as the worked-example anchor for the Habitability commons (atmospheric ecosystem). The Method 1 logic: the floor on Habitability-commons Residual Commons Value is the engineering cost to replace the substitutable component of the commons (the CO_2-removal function the atmosphere provides at zero marginal cost via natural sinks).

Operational DAC facilities (2024-2025 cost figures)

Facility	Operator	Operational since	Capture capacity	Cost per ton CO_2 (current)	Cost per ton CO_2 (target at scale)	Source
Orca (Iceland)	Climeworks	2021	~4,000 tons/year	~\$600–\$1,000/ton	n/a (pilot)	Climeworks public reporting; IEA *Direct Air Capture 2022*
Mammoth (Iceland)	Climeworks	2024	~36,000 tons/year (target full scale)	~\$600–\$1,000/ton initial; targeting ~\$300–\$400/ton at full operation	\$300–\$400/ton	Climeworks public reporting (2024 commissioning); MIT Technology Review coverage

Facility	Operator	Operational since	Capture capacity	Cost per ton CO_2 (current)	Cost per ton CO_2 (target at scale)	Source
Stratos (Texas, US)	Carbon Engineering / 1PointFive (Occidental)	construction; targeting 2025 commissioning	~500,000 tons/year (target)	n/a (pre-operational)	$94–$232/ton (Carbon Engineering 2018 *Joule* paper basis); ~$300–$600/ton more conservative third-party estimates	Keith et al., "A Process for Capturing CO2 from the Atmosphere," *Joule* (2018); 1PointFive announcements

Provisional consolidated range for Habitability-commons Method 1 anchor:

- **Conservative (current operational, first-of-a-kind):** $600–$1,000/ton CO_2.
- **Mid-range (declared targets at-scale, next 5–10 years):** $300–$600/ton CO_2.
- **Optimistic (industry-target nth-of-a-kind by 2050):** $100–$300/ton CO_2.

The framework's Method 1 anchor should report all three ranges with explicit time-horizon attribution, not a single point estimate.

Authoritative review-literature ranges

Source	DAC cost range	Time horizon	Notes
IEA, *Direct Air Capture: A Key Technology for Net Zero* (2022)	$125–$335/ton CO_2 for early commercial-scale; $100/ton achievable by 2030 in optimistic scenarios	2025–2030 commercial-scale; 2030+ optimistic	IEA's reference framing. https://www.iea.org/reports/direct-air-capture-2022
IEA, *Net Zero by 2050* (2021)	$130–$330/ton CO_2	2030 commercial-scale	Net-Zero scenario assumption.

Source	DAC cost range	Time horizon	Notes
IPCC AR6 Working Group III (2022), Chapter 12	\$100–\$300/ton CO_2 in optimistic deployment scenarios; >\$1,000/ton in pessimistic	2030–2050	https://www.ipcc.ch/report/ar6/wg3/
National Academies (US), *Negative Emissions Technologies and Reliable Sequestration* (2019)	\$100–\$600/ton CO_2 over a wide deployment-scenario sensitivity	2030–2050	Pre-Stratos; figures may underrepresent recent learning curves.
Realmonte et al., *Nature Communications* (2019)	\$100–\$600/ton CO_2 at multi-gigaton scale	2050	Modeled at-scale figure.
House et al., *PNAS* (2011)	\$1,000+/ton CO_2 early estimate	2010s pessimistic	Early-literature pessimistic floor; useful for steelman engagement on academic-credibility positioning.

Method 1 anchor specification

Anchor language:

> The Habitability-commons Method 1 anchor is the engineering cost to remove and sequester one ton of CO_2 via direct-air-capture (DAC). Operational DAC facilities (Climeworks Orca and Mammoth in Iceland, 2021 and 2024) currently report capture costs in the range of \$600–\$1,000 per ton CO_2. Targets for at-scale commercial deployment over the next decade are in the \$300–\$600/ton range (Climeworks Mammoth full-operation target; Carbon Engineering / 1PointFive Stratos pre-operational projection). Authoritative review literature (IEA *Direct Air Capture 2022* ; IPCC AR6 Working Group III, 2022; US National Academies 2019) places the achievable cost by 2050 in the \$100–\$300/ton range under optimistic deployment scenarios. The Method 1 anchor accordingly reports a three-horizon range: \$600–\$1,000/ton (current operational), \$300–\$600/ton (at-scale commercial within the next decade), \$100–\$300/ton (optimistic 2050 nth-of-a-kind). The atmospheric-commons CS arithmetic uses the appropriate-horizon figure for the comparison being made; the McDowell-coal carbon term (Ch 6 convergence table) uses the IPCC-aligned \$190/ton SCC for a reason distinct from DAC replacement-cost: SCC prices the damage caused, DAC prices the engineering reversal of that damage, and the two should not be conflated.

Distinction discipline:

- **Social Cost of Carbon (SCC):** prices the damage that one ton of CO_2 emission causes (current US EPA central estimate: $190/ton CO_2 at 2% discount rate; Rennert et al. 2022 *Nature*).
- **DAC cost-per-ton:** prices the engineering cost to remove one ton already emitted.
- **Method 1 (Replacement Cost) Habitability anchor:** uses DAC, NOT SCC. The Method 1 logic is "what would it cost to engineer a substitute for the commons function" — the substitute function for the atmospheric CO_2-removal capacity is DAC. SCC is the damage-side number; Method 1 is the substitution-side number. They appear together in the convergence-table presentation but answer different questions.

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§11.10 — Space-economics anchors: Ch 7 (asteroid miner) calibration

Numerical anchors for the Ch 7 asteroid-miner thought experiment — launch costs (Earth-to-LEO, Earth-to-GTO), asteroid sample-return per gram, commercial-mining proposal estimates (cited as actor-stated projections, not validated), Mars surface-operations cost reference. Grounds the Mars/Europa thought experiments in plausible-economics terms while preserving the boundary-awareness register (per §12).

Space-economics numerical anchors — Ch 7 (asteroid miner) calibration

Ch 7 uses thought-experiment-style scenarios (Mars colony water; asteroid iron; lunar helium; comet volatiles; Europa ice; exoplanet hypotheticals) to demonstrate that CIT operates across parameter ranges. The chapter is intentionally light on numerical anchoring — the demonstration is that CIT produces different verdicts at different parameter values, not that any specific numerical estimate holds. Selected numerical anchors nonetheless strengthen academic-credibility positioning, particularly for the asteroid-iron extrapolation where the "RCV approaches market price" claim depends on substitutability and externality being approximately zero.

Launch-cost anchors (Earth-to-LEO and Earth-to-GTO)

Vehicle	Operator	Cost per kg to LEO (current, 2024–2025 estimates)	Notes
Falcon 9	SpaceX	~$2,700/kg (LEO; full-stack rate based on $67M/22.8t)	Operational since 2010; reusable booster.
Falcon Heavy	SpaceX	~$1,400/kg (LEO; $90M/63.8t LEO capacity)	Operational since 2018.
Starship (early-operational target)	SpaceX	~$200–$1,000/kg (LEO; targeted; not yet validated)	Pre-operational as of 2024–2025; figures are SpaceX-stated targets, not third-party-validated.

Vehicle	Operator	Cost per kg to LEO (current, 2024–2025 estimates)	Notes
Vulcan Centaur	ULA	~$5,000–$8,000/kg (LEO)	Operational 2024.
New Glenn	Blue Origin	~$2,000–$6,000/kg (LEO; targeted)	First flight 2025; cost figures are pre-operational.

Source: FAA/AST commercial-space launch reports; Aerospace Corporation Center for Space Policy and Strategy briefings; SpaceX public statements; *Space News* coverage.

Order-of-magnitude framing: Earth-launch infrastructure has reduced cost-per-kg by approximately one order of magnitude (from ~$30,000/kg Space Shuttle era to ~$2,000–$5,000/kg current commercial-launch baseline) over 2010–2025; a further order-of-magnitude reduction to $200–$500/kg is plausible-but-not-validated for the next decade dependent on Starship operational maturity.

Asteroid sample-return mission costs (per gram returned)

Mission	Operator	Sample mass returned	Mission cost (USD, mission-vintage)	Cost per gram returned	Notes
OSIRIS-REx → Bennu	NASA	121.6 g (final accounting; 2023 capsule return)	~$1.16B (lifecycle; 2016 launch through 2023 return)	~$9.5M/g	Single-mission, single-asteroid; primarily science-mission, not optimized for cost-per-gram.
Hayabusa2 → Ryugu	JAXA	5.4 g (2020 capsule return)	~$300M (lifecycle, 2014 launch through 2020 return)	~$56M/g	Smaller mission profile; technology-demonstration-and-science.
Hayabusa → Itokawa	JAXA	1500 microscopic particles, ~mg total (2010 return)	~$170M (lifecycle)	n/a (sample-mass too small for meaningful $/g framing)	First-generation; mission profile not optimized for sample mass.

Source: NASA OSIRIS-REx mission lifecycle accounting; JAXA Hayabusa/Hayabusa2 mission accounting; *Nature Astronomy* and *Science* coverage of sample-return missions.

Framing for Ch 7 calibration: science-mission sample-return at ~$9.5M/g (Bennu) is approximately **5–7 orders of magnitude** above the per-gram cost framework that asteroid-mining commercial proposals operate in (which assume per-gram costs in the $1–$100 range at scale, contingent on (a) full-mass extraction not single-sample-return; (b) substantial Starship-class or post-Starship launch-cost reduction; (c) in-situ resource utilization eliminating the return-to-Earth requirement for many use-cases). The 5–7 order-of-magnitude gap is the framework's relevant calibration point: Ch 7's "RCV approaches market price for asteroid iron" claim depends on cost-per-kg dropping by ~6 orders of magnitude relative to current science-mission economics, AND on substitutability remaining high (which is the core CIT claim, not a launch-cost-dependent claim).

Asteroid-mining commercial-proposal cost estimates (for context)

Source	Asteroid-mining cost per ton extracted material (target / projection)	Notes
Keck Institute Asteroid Retrieval Mission Study (2012)	~$2.6B for retrieval of a 7-ton asteroid to lunar orbit (~$370,000/kg retrieved)	Pre-Starship; mission profile assumes existing launch infrastructure.
Planetary Resources commercial estimates (2013–2018, pre-acquisition)	$50–$500/kg at scale (water; iron); not validated by operational missions	Company-stated targets; broadly skeptical reception in space-economics literature.
Deep Space Industries commercial estimates (2013–2018, pre-acquisition)	Comparable order of magnitude to Planetary Resources estimates	Same caveat.

Source: Keck Institute for Space Studies, *Asteroid Retrieval Mission Study* (2012); industry-presentation materials from Planetary Resources and Deep Space Industries; *Space Resources Roundtable* proceedings.

Caveat for Ch 7: commercial asteroid-mining cost projections are actor-stated targets, not validated operational figures. The 5–7 order-of-magnitude reduction implicit in commercial projections has not been demonstrated by any operational mission. The framework's Ch 7 thought-experiment use of "asteroid iron is approximately market-priced under any reasonable accountability bond" is robust to wide variance in this projection because the structural argument is about substitutability and externality (both ~0 for asteroid iron), not about specific cost-per-kg numbers.

Mars surface-operations cost reference (for thought-experiment grounding)

The Mars-colony water-extraction thought-experiment in Ch 7 does not depend on specific per-kg cost numbers, but the chapter's plausibility benefits from a one-paragraph reference that current operational Mars surface costs (Curiosity, Perseverance, Ingenuity, Mars Exploration Rovers) run in the order of $1B+ per mission for ~1-ton-class delivered surface mass. This anchors the thought-experiment's "small colony" framing in plausible-not-fantasy economics — the colony exists at the boundary of what operational Mars-surface presence has achieved, not at orders of magnitude beyond it.

Mission	Operator	Delivered Mars surface mass	Mission cost (USD, mission-vintage)	Cost per kg surface-delivered
Curiosity (Mars Science Laboratory)	NASA	899 kg rover (2012 landing)	~$2.5B lifecycle	~$2.8M/kg surface-delivered
Perseverance (Mars 2020)	NASA	1,025 kg rover (2021 landing)	~$2.7B lifecycle	~$2.6M/kg surface-delivered
InSight	NASA	358 kg lander (2018 landing)	~$830M lifecycle	~$2.3M/kg surface-delivered

Source: NASA mission lifecycle accounting; *Planetary Society* mission cost analyses.

Order-of-magnitude framing: current Mars surface-presence economics are ~$2–3M per kg delivered, dominated by interplanetary-cruise + entry-descent-landing infrastructure rather than per-kg launch from Earth. A small colony (~10 person-class, ~10–100 ton delivered surface mass) under current operational economics would carry order-of-magnitude $30B–$300B delivery cost, before any consideration of in-situ resource utilization, Starship-class delivery cost reduction, or the speculative-but-plausible-decade-scale developments Ch 7's thought-experiment assumes.

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§11.11 — IPG-table reconciliation (33-122x canonical → 50-555x triangulated range)

The framework's earlier IPG (Intergenerational Pricing Gap) reports in Ch 6 + Ch 8 cited a 33–122× range for the McDowell coal IPG (the ratio of true RCV to market-priced cost). The Three Ways calibration (§11.6) produces a triangulated range of **50–555×** across Methods 1+2+3 with Method 3 α-dominance applied. Reconciliation finding:

- **The two ranges are not in tension.** The 33–122× canonical reports reflect Method-1-anchored estimates with conservative time-horizon attribution. The 50–555× triangulated range reflects Methods 1+2+3 with the upper bound carrying Method 3's α-irreversibility-premium for the multi-generational time scale.
- **The variance reflects time-horizon attribution, not framework imprecision.** Method 1 (replacement-cost) anchors at engineering-time-scale (decades); Method 3 with α-dominance reflects multi-generational-time-scale irreversibility-premium. Both are correct RCV measurements at their respective time scales. The IPG range widens with time horizon — that's a feature, not a defect.
- **Pedagogical implication for Ch 6:** when Ch 6 walks through the McDowell convergence table, the reconciliation should be made explicit — *"the canonical 33–122× report and the triangulated 50–555× range both correctly describe McDowell's IPG, at different time-horizons. The framework is not committed to a single number; it is committed to a method that makes the time-horizon attribution explicit."*
- **Pedagogical implication for Ch 8:** Ch 8's 22-vs-1× tier-by-tier walkthrough (per the Ch 8 worked-example reframing) sits within the lower-bound (Method 1, near-term) end of the range. Ch 8's claim is

conservative; the wider range is the framework's full accounting honesty.

The reconciliation establishes that the IPG variance across reports is not framework-imprecision but time-horizon-attribution discipline.

§12. Boundary-Awareness Scope Claim (Mars/Europa illustrations)

The Mars/Europa scenarios below stand as boundary-awareness illustrations of where the framework reaches and where its boundary conditions surface.

The framework must produce intuitively correct results across all extraction scenarios — on Earth and beyond. Seven cross-scenario tests validate universality. The framework passes all seven, and correctly identifies its own boundary condition in scenario 6.

Scenario 1: Mars Colony Water (10,000 people, 50-year supply, three deposits)

Parameters: S ≈ 0 (no water substitute at required scale), U enormous (survival-critical), E(t) ≈ 0 (water waste is recyclable), Q is small and fixed.

Framework output: RCV very high. Strategic reserves mandatory. Progressive pricing to manage consumption rate. Aggressive investment in recycling to increase effective Q.

Administrator's obvious decision: Identical — conserve with reserves, price progressively, invest in recycling.

Validation: ✓ Framework produces obviously correct answer. Key insight: cost severance goes to zero because the administrator is the community bearing the cost. No distance between extractor and cost-bearer → no CS. Cost severance requires distance.

Scenario 2: Asteroid Iron Mining (abundant, lifeless, remote)

Parameters: S → 1 (iron has abundant substitutes and alternatives), U low (iron not scarce), E(t) ≈ 0 (no communities, no atmosphere to damage), Q effectively unlimited.

Framework output: RCV ≈ 0. Extract freely.

Validation: ✓ Framework correctly reduces to near-market pricing when externalities are zero and stocks are abundant. This is the model's key falsifiability feature — it does not always disagree with the market. When market pricing is correct, the model confirms it.

Scenario 3: Lunar Helium-3 (potentially critical for fusion energy)

Parameters: S uncertain (depends on fusion development timeline), U potentially enormous (civilizational energy technology), E(t) ≈ 0 (no biosphere), Q limited in accessible deposits.

Framework output: High existential substitutability gap ranking. Preserve as insurance until fusion viability clarified. Insurance cost (foregone revenue from strategic reserve designation) is near zero — lunar helium-3 is not currently commercially extracted.

Validation: ✓ Framework correctly handles uncertain criticality: when the cost of preservation is near zero and the potential cost of premature extraction is enormous, preserve. Asymmetric regret principle applies.

Scenario 4: Comet Flyby Volatiles (access window closes permanently)

Parameters: S depends on resource; Q abundant in the comet; $E(t) \approx 0$; but access is a one-time event — the comet passes once.

Framework output: Extract aggressively during the access window. The asymmetric regret principle flips — you cannot un-miss the comet. The option to extract later does not exist.

Validation: ✓ Framework correctly identifies that the source geometry changes the calculus. Not anti-extraction — extract when RCV is low and preservation is impossible. The model differentiates correctly between "leave it in the ground" (Earth rare earths) and "extract now before the window closes" (comet flyby).

Scenario 5: Mars Colony Rare Earth Mining (same population extracts and bears cost)

Parameters: The colony mines its own subsurface rare earths. The extractor and the cost-bearer are the same people.

Framework output: $CS \to 0$. No cost severance possible.

Deepest structural insight: Cost severance requires distance between value-capturer and cost-bearer — geographic, temporal, or social. When that distance collapses to zero (the Mars colony mines its own resources, the Norwegian government owns its own oil fund), cost severance approaches zero automatically. On Earth, distance — maintained by geography, power, and legal structure — enables the ignorance that enables severance. The model prices what distance hides.

Validation: ✓ Framework reveals the mechanism's structural prerequisite.

Scenario 6: Europa Ice (potential independent biosphere)

Parameters: Europa's subsurface ocean may harbor independent life. $E(R, t)$ potentially includes the destruction of an independent biosphere — a cost that is incommensurable with human utility scales.

Framework output: The model reaches its boundary. The externality of potentially destroying an independent biosphere cannot be priced in commensurable units with $U(R, t, Q(t))$. The framework honestly reports: "I cannot price this. The Asymmetric Regret Rule applies: wait, study, decide later. The insurance cost of non-extraction is near zero. The potential cost of extraction is incommensurable."

Validation: ✓ A framework that correctly identifies its own limits is more credible than one that claims universal applicability. The Europa case establishes where the model stops — which by implication tells you that everywhere inside that boundary, the model works.

Scenario 7: Exoplanet Rare Earths (independent ecosystem, necessary resource)

Parameters: The exoplanet has a biosphere (non-zero E), but civilization genuinely needs a specific resource ($S \approx 0$, U is very high). Genuine tension between preservation and necessity.

Framework output: Extract if necessary, but with extreme accountability, minimum footprint, and maximum substitution investment. The model does not resolve the incommensurability — it cannot — but it does clarify the decision structure: establish what "necessary" means (cannot S(t) be improved?), minimize extraction footprint, price all identifiable externalities, and invest in alternatives aggressively. The model provides a decision framework even at the boundary.

Validation: ✓ Framework handles genuine value conflict by clarifying the decision structure without pretending to resolve the philosophical tension.

§13. Substitutability Function (S) + Industrial-Existential Substitutability Gap

The framework does not name a derived scalar for the difference between industrial-context and existential-context substitutability ceilings (S_max(industrial) – S_max(existential) is a difference between two S-evaluations, on parsimony grounds). The concept is preserved in prose as the *existential substitutability gap* (lowercase descriptive phrase).

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§13.1 — Resource Classification by existential substitutability gap

existential substitutability gap Level	Resource Examples	Policy Response
existential substitutability gap ≈ 0	Coal, sand, iron ore, common stone	Price externalities; no strategic reserve
Low existential substitutability gap	Conventional oil, copper, zinc	Full externality pricing; moderate reserve consideration
Moderate existential substitutability gap	Uranium, lithium, cobalt	Externality pricing + strategic reserve assessment
High existential substitutability gap	Rare earth elements, platinum group metals	Mandatory strategic reserves; export controls; substitution investment
Conditionally High existential substitutability gap	Helium-3, phosphorus, fossil aquifers	Preserve pending technology development; reassess on defined schedule

§13.2 — The Insurance Cost Argument

For high-existential substitutability gap resources: in-situ preservation (leave it in the ground) is nearly free. The ground is free storage. The opportunity cost of designating strategic reserves is foregone extraction revenue — typically single-digit percentages of total extraction value when applied to a fraction of known deposits. The insurance payoff is access to critical materials at a moment when non-reserved stocks have been depleted and substitutes have not yet fully developed — precisely when the need is greatest. In financial terms: a hedge that costs almost nothing to hold and pays off exactly when needed.

§13.3 — The Four-Step Implementation Sequence (Chapter 9 connection)

1. Classify: Assign existential substitutability gap rankings to all major resource classes using materials science, energy modeling, and aerospace engineering inputs.
2. Reserve: Designate in-situ strategic reserves for high-existential substitutability gap resources. Do not extract and warehouse — the ground is free storage.
3. Invest: Direct pricing surplus simultaneously into renewable substitute development (increases $S(t)$) and space access technology (opens alternative supply sources with lower RCV).
4. Reassess: Adjust reserve levels dynamically as $S(t)$ improves and as new resource deposits or substitutes are identified. The framework is not static — it updates as parameters change.

§14. Engagement with Existing Literature

This section, entirely absent from the prior Technical Appendix, is the most important addition for academic credibility. Reviewers and economists will ask how RCV relates to existing frameworks. This section answers that question directly and honestly — acknowledging what each framework gets right and where each one stops short.

§14.1 — Hotelling (1931): Optimal Depletion and Scarcity Rent

What Hotelling gets right: Resource prices should rise at the rate of interest as a non-renewable stock depletes. Owners of finite resources who optimize over time will not sell too fast — they will hold the resource until the price has risen enough to justify extraction. The "scarcity rent" — Hotelling's price premium above extraction cost — is a genuine contribution that RCV builds on.

Where Hotelling stops: Hotelling's model optimizes the extraction path for the resource owner, not for humanity. It asks "when should I sell my oil?" not "what does permanent oil depletion cost everyone else?" The scarcity rent is a private benefit (the owner captures it); the intergenerational foreclosure cost is a public harm (borne by future generations who cannot bid). Hotelling prices the former; RCV prices the latter. They are not alternatives — RCV adds to Hotelling, pricing what Hotelling's model structurally excludes.

Formal relationship: The Hotelling price path is $P(t) = P(0) \cdot e^{r \cdot t}$. The RCV augments this with the foreclosure cost $\int[1-S(t)]U(R,t,Q(t))D(t)dt$, which Hotelling's private-owner optimization cannot capture because future-generation utility is not represented in the owner's objective function.

§14.2 — Hartwick (1977): Invest the Rents

What Hartwick gets right: Hartwick's Rule states that a non-declining consumption path is achievable if a society invests all rents from exhaustible resource extraction in reproducible capital (machines, infrastructure, knowledge). Norway's sovereign wealth fund is the cleanest real-world application: extract the oil, invest the revenue, maintain the capital stock for future generations.

Where Hartwick stops: Hartwick's Rule requires that the resource be priced at its Hotelling rent — that the rent be fully captured, not severed. In practice, under-pricing ($P <$ true rent) means the investment fund receives less than Hartwick requires. More fundamentally, Hartwick assumes that the physical resource can be fully substituted by reproducible capital — that a future economy without oil can use machines instead. This is the strong substitutability assumption. The RCV model relaxes it: for high-existential substitutability gap resources, $S_{max} < 1$, and Hartwick's substitution is incomplete. Even perfect investment of the Hartwick rent cannot replace a resource for which no substitute exists.

Formal relationship: Hartwick's Rule is the investment policy that makes $CS = 0$ under the assumption $S_{max} = 1$. RCV generalizes Hartwick to $S_{max} \leq 1$, showing that when $S_{max} < 1$ (imperfect substitution), no investment policy can fully compensate for foreclosure — strategic preservation is the only complete solution.

§14.3 — Weitzman (2001): Declining Discount Rates

What Weitzman gets right: Under uncertainty about the appropriate long-run discount rate, the certainty-equivalent rate declines toward the lowest plausible rate. This is not merely a policy preference — it is a mathematical result. If we are uncertain whether the appropriate rate is 1%, 2%, or 4%, we should behave as if the rate approaches 1% over long horizons.

RCV adoption: The RCV model adopts Weitzman declining discount rates directly. This is the most significant methodological choice in the model — it biases RCV estimates downward relative to fixed high discount rates, making the results conservative. If anything, the Weitzman rate understates how much we should weight the future, because the case for near-zero long-run discounting grows stronger with each decade of climate damage evidence.

Engagement with Nordhaus (2007): Nordhaus used a 5.5% discount rate to recommend moderate climate action (DICE model). Stern (2006) used 1.4% and recommended aggressive action. The RCV model does not resolve this debate — it acknowledges that the output is sensitive to discount rate assumptions (as the McDowell County IPG range of 33–122× demonstrates). However, the Weitzman result means that rational agents facing uncertainty about the discount rate should act as if the low rate applies. $CS > 0$ holds across the entire plausible discount rate range; only the magnitude varies.

§14.4 — Rawls (1971): Veil of Ignorance and the Just Savings Principle

What Rawls gets right: Principles of justice should be chosen from behind a veil of ignorance — without knowing your position in society. His Just Savings Principle extends this across time: each generation owes something to those that follow, and we should choose only principles we would wish all preceding and succeeding generations to have followed.

RCV operationalization: The RCV model is the Rawlsian veil applied to resource extraction. If you did not know which generation you would be born into — whether 2024 or 2224 or 2524 — would you consent to an extraction regime that treats the planet's non-renewable endowment as the property of whoever gets to it first? Rawls explicitly stated: "there are no grounds for discounting future well-being on the basis of pure time preference." The RCV model implements this as the stock-dependent utility term $U(R, t, Q(t))$, which gives future generations' welfare equal standing in the integral.

Where Rawls is limited: Rawls's framework is primarily synchronic (within a generation) and only partially extends across time. His Just Savings Principle was acknowledged by Rawls himself to be underdeveloped. Brian Barry (later) sharpened it: the overall range of opportunities open to successive generations should not be narrowed. RCV prices the narrowing of opportunities directly — as the foreclosure cost.

§14.5 — Ostrom (1990): Governing the Commons

What Ostrom gets right: Common-pool resources are not inevitably subject to the "tragedy of the commons." Communities can and do develop governance institutions that sustainably manage shared resources. Her eight design principles (clear boundaries, rules fit to local conditions, collective choice arrangements, monitoring, graduated sanctions, conflict resolution, recognition of rights, nested governance) identify the conditions under which commons governance succeeds.

Where Ostrom stops: Ostrom's framework was designed for renewable commons — fisheries, forests, irrigation systems, where the challenge is managing a sustainable harvest rate. Her design principles work beautifully for resources that regenerate. They give us governance for sustainable fishing in the Chesapeake Bay. They give us nothing for the permanent farewell to a barrel of oil. Non-renewable extraction is a one-time event; there is no harvest rate to manage, no regeneration to protect. Ostrom's governance framework is a necessary but insufficient complement to the RCV model — necessary for the renewable commons (Chesapeake Bay fisheries, forests, groundwater above depletion rate), insufficient for the permanent foreclosure case.

RCV as extension: The RCV model can be read as Ostrom's design principles translated into a pricing mechanism for the non-renewable case. Where Ostrom asks "how do we manage this so it lasts?" RCV asks "what does it cost when management fails and it doesn't last?"

§14.6 — Daly (1990s): Ecological Economics and Strong Sustainability

What Daly gets right: "Weak sustainability" (Hartwick's approach: substitute capital for natural capital) is insufficient. "Strong sustainability" requires maintaining the natural capital stock itself — particularly for critical natural capital with no adequate substitute. GDP as a measure of welfare is deeply misleading because it counts environmental destruction as economic activity.

RCV relationship: The RCV model operationalizes Daly's strong sustainability criterion mathematically. Where Daly argues normatively that S_{max} must equal 1 for strong sustainability to be achievable, RCV asks empirically: what is S_{max} for each resource class? For high-existential substitutability gap resources (rare earths, platinum group metals, phosphorus), $S_{max} < 1$ — Daly's strong sustainability criterion cannot be met by substitution alone. The RCV model quantifies the resulting cost, complementing Daly's qualitative argument with a measurable framework.

§14.7 — Harvey (2003): Accumulation by Dispossession

What Harvey gets right: Harvey names the broader mechanism of which cost severance is a specific instance: capital accumulates by dispossessing communities of their assets, commons, and resources through enclosure, privatization, and force. Cost severance is the pricing dimension of this — the financial mechanism by which dispossession is made to appear economically rational.

RCV as measurement tool: Harvey diagnoses the mechanism; RCV measures its magnitude. "Accumulation by dispossession" tells you it happened; the IPG tells you by how much. A McDowell County miner can say "we were exploited" — the RCV model says "you absorbed $552 of cost for every $4.50 you were paid." The measurement does not replace the political analysis; it gives it a number.

§15. Limitations and Boundary Conditions

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1. Substitutability Function Estimation: S(t) requires empirical input from materials science, energy engineering, and industrial ecology. Current estimates carry ±30–50% uncertainty in S_{max} for most resource classes. This uncertainty is reflected in the IPG ranges; the model is presented as a range estimator, not a point estimator.
2. Strategic Behavior: Extractors will attempt to game S(t) estimates (by exaggerating substitutability) and will engage in jurisdiction shopping (extracting where B is lowest). The model predicts this behavior and argues for international coordination as the structural response — not because coordination is easy, but because CS scales with the distance between extraction and governance.
3. Cross-Resource Interactions: Portfolio effects — rare earth extraction enabling solar panel manufacturing enabling coal displacement — require additional modeling beyond per-resource RCV. The model treats resources independently as a first approximation; a full portfolio analysis is a significant further research program.
4. Incommensurable Externalities: The Europa case (Scenario 6) identifies the boundary: the framework cannot price externalities that involve potential destruction of an independent biosphere. Where externalities are incommensurable with human utility scales, the model correctly reports its own limit and defaults to the Asymmetric Regret Rule.
5. Technological Discontinuities: The substitutability function assumes gradual, monotonic improvement in substitutes. A sudden technological breakthrough (e.g., room-temperature superconductivity eliminating rare earth demand) would shift S(t) discontinuously. The model's sensitivity analysis shows that even large positive discontinuities in S(t) do not reverse the CS > 0 finding for resources already extracted — the externality tail continues regardless.
6. Discount Rate Uncertainty: The IPG is sensitive to discount rate assumptions. This is a feature, not a bug — it honestly reflects that how much we weight future generations is a normative choice, not a technical one. The Weitzman result provides the theoretically grounded solution: use declining rates. The sensitivity analysis showing CS > 0 across all plausible rates (from the Nordhaus 5.5% to the Stern 1.4%) demonstrates the finding is robust.

§15.1 Methodology Defense Consolidation — Tech Appendix companion (academic-reviewer depth)

A reader who has followed the framework's apparatus this far is entitled to ask why each piece of it is the piece chosen. The Ch 6 main-text companion carries this defense at reader-facing depth (~2,000 words); the Tech Appendix elaboration below carries it at academic-reviewer depth: lineage citations per defense, technical specification of the rejected alternative. Both registers preserve the same configuration claim (the framework's contribution is the specific configuration of choices, not novelty in any single choice). Each of the eight choices below was alternative-tested; the rejected alternative for each is specified, and the academic lineage the framework's choice extends is cited per intellectual-honesty discipline.

§15.1.1 — CIT versus revealed preference

Rejected alternative — Revealed preference as discrimination gate. The first available alternative for filtering candidate cost claims was revealed preference: read what polities, markets, or established institutions have already paid as a measure of what a commons is worth. Lineage: Samuelson 1948 (revealed-preference axioms in consumer theory); Hicks 1939 *Value and Capital* ; subsequent revealed-preference econometrics (cf. Bockstael & McConnell 2007 *Environmental and Resource Valuation with Revealed Preferences*). The methodology is empirically grounded: data exist, valuation procedures are established, and contingent-valuation literature has refined the techniques over decades.

Why CIT instead. Revealed preference is institutionally captured: it reveals what current institutional architectures have priced, not what the underlying commons is worth absent that architecture. Norway's revealed-preference per-barrel value reveals what Norwegian institutional architecture captured (per §11.5 Norway calibration); it does not reveal what the atmospheric externality from combusting Norwegian oil costs the populations bearing it. CIT, by contrast, runs an institutional-architecture-independent counterfactual (Lewis 1973 *Counterfactuals* + subsequent philosophy-of-action literature). The framework retains revealed preference as one estimation input — Method 2 in the Three Ways triangulation (§3 + §11.5) — but uses it for estimation, not discrimination. Conflating discrimination and estimation produces institutionally-captured results: cost claims that should be admitted on commons-grounding grounds get filtered out because no polity has yet priced them.

§15.1.2 — Two-instrument architecture (CSD + RCV) versus single-instrument

Rejected alternative — Single Cost Severance accounting object. A simpler framework would have specified one CS measurement: backward-looking and forward-looking together, undifferentiated. Standard environmental-economics frameworks (Pigou 1920 *Economics of Welfare* ; Coase 1960 "The Problem of Social Cost"; subsequent damage-function literature including Nordhaus DICE) operate roughly this way — externality is one accounting object, with discounting handling the temporal-structure question implicitly through rate choice.

Why two instruments instead. Past extraction harms specific people whose cost-bearing is empirically documented; the methodology for pricing this is reparations economics (Darity & Mullen 2020 *From Here to Equality* ; Bullard 1990 *Dumping in Dixie* ; Chetty et al. intergenerational-mobility econometrics; toxic-tort + class-action damages tradition; truth-and-reconciliation processes including South Africa post-apartheid). Future extraction forecloses options for people whose moral standing is real but whose specific identities are not knowable in advance; the methodology for pricing this is option-value economics (Dixit & Pindyck 1994 *Investment under Uncertainty* ; Hartwick 1977 sustainable-investment rule; Solow 1974 intergenerational equity; Brennan & Schwartz 1985 commodity-investment real options). The two methodologies do not collapse into a single accounting object without losing analytical leverage. The framework specifies CSD (backward-looking; reparations-economics lineage) and RCV (forward-looking; option-value lineage) as structurally independent instruments because the accounting work each requires is different. The two instruments share the central equation CS = RCV − B; they answer different questions on opposite sides of that equation. Total CS = (CSD − B_1) + (RCV − B_2), connecting the two-instrument architecture (§2) with the B = B_1 + B_2 decomposition (§5).

§15.1.3 — Triangulation (Three Ways) versus single-method

Rejected alternative — Single estimation method for RCV. A single estimation method for Residual Commons Value would have been simpler to teach, simpler to implement, and simpler to defend. Each of the three methods has standing as a primary estimation methodology in its tradition: Replacement Cost (Costanza 1997 *Nature* ecosystem-services valuation; standard regulatory cost-benefit analysis; eminent-domain takings law); Revealed Preference (Hartwick 1977; Solow 1974; subsequent revealed-preference resource economics); Real-Options Scarcity-Adjusted Value (Dixit & Pindyck 1994; Brennan & Schwartz 1985; Black-Scholes 1973 option-pricing foundation; subsequent applications to natural-resource extraction).

Why triangulation instead. Each method has a strength the other two lack and a blind spot the other two illuminate. Replacement Cost prices substitution but cannot price irreplaceability — it anchors only a floor (per §3 Method 1). Revealed Preference prices what one polity actually internalized but inherits that polity's institutional architecture and tradition's commitments — Norway-anchored, the methodology fails for cases without Norway-equivalent revealed-preference data (epistemic-humility constraint). Scarcity-Adjusted Option Value prices irreversibility through the $\sigma + \alpha + V_{option}$ parameter set but requires calibration that ranges across two orders of magnitude in plausible values (per §11.8 Method 3 sensitivity analysis). Triangulation is the standard scientific-method response to estimation under uncertainty: produce three independent estimates from three independent epistemological angles; report convergence range; treat divergence as informative rather than as failure (cf. Campbell & Fiske 1959 multitrait-multimethod matrix in psychometric methodology; subsequent triangulation discipline in qualitative + quantitative research traditions). The framework's reporting discipline — every RCV estimate reports all three methods individually plus the convergence range plus identified divergence sources — is the operationalization of this commitment. No single-method RCV claim has standing in the framework's accounting.

§15.1.4 — Hotelling Identity versus Pigou-only or Hotelling-only

Rejected alternatives — Pigou alone or Hotelling alone. Two adjacent traditions had standing claims to the territory the Hotelling Identity occupies. Pigou (1920, *The Economics of Welfare*) priced costs imposed on third parties through pricing instruments — the externality framework. Hotelling (1931, "The Economics of Exhaustible Resources," *Journal of Political Economy* 39(2): 137–175) priced the rent rule for private extraction of finite resources — Hotelling rent equals the price minus the marginal extraction cost, rising at the rate of interest under competitive extraction equilibrium. Each tradition does what it does well within its scope.

Why the Hotelling Identity instead. Neither tradition alone prices what commons-extraction actually costs when both extraction-rent dynamics and externality-on-cost-bearers operate simultaneously. Pigou-only treats the resource-rent component as private revenue and prices only the externality tail; Hotelling-only prices the rent rule and stops short of the externality. The Hotelling Identity is the framework's bridge: *RCV − Hotelling rent = CS per unit* . The identity does not replace Hotelling 1931 (which prices private exhaustibility correctly within its assumptions: private extractor; finite stock; private extraction-cost only) or Pigou 1920 (which prices externalities correctly within its scope). It establishes the structural relationship between them — surfacing that commons-extraction cost is precisely the residual value beyond standard Hotelling rent, when "value" is defined as RCV (forward-looking commons cost; substitutability + externality + abundance-grounded costs admitted via Four Gates). The framework cites Hotelling 1931 as foundational (Hotelling's part: rent rule for exhaustible resources) and presents the identity extension as the framework's contribution (framework's part: defining RCV as commons' true cost; recognizing Hotelling rent under honest accounting represents commons' scarcity value being appropriated; articulating the identity).

§15.1.5 — Four Gates versus CIT alone

Rejected alternative — CIT as sole admission gate. CIT is necessary for the framework's accounting but not sufficient. A framework that ran CIT alone would admit cost claims that are commons-grounded but unit-incoherent (a claim that mixes monetary, physical, and qualitative units fails Units Consistency); finite-bounded but double-counted (a claim that prices a single harm both as Habitability and as Ecosystem inflates the aggregate via Independence violation); or independent but unit-inconsistent. The aggregate Cost Severance accounting depends on the gates being run together.

Why Four Gates instead. Each gate does work the others can't. **CIT** screens for commons-grounding (§6); the rejected alternative was unfiltered admission, which would treat any cost as commons-grounded and dissolve the framework's discriminating function. **Units Consistency** screens for unit coherence (§7); the rejected alternative was unit-flexibility, which would allow [$/ton] cost claims to aggregate with [QALYs] cost claims without commensuration discipline. The naming asymmetry between CIT (a methodological TEST; action-form name) and the other three gates (mathematical CONDITIONS; property-form names) honors a real structural asymmetry — naming uniformity would falsely conflate kinds. **Boundedness** screens for finite ranges; the rejected alternative was admitting "infinity" claims that dominate any aggregation and produce useless results. **Independence** screens for non-double-counting; the rejected alternative was loose category-membership that allowed a single severance to be counted multiple times via overlapping commons categories. The configuration of all four together filters the candidate-cost-claim space against established commons in a way none of the four alone could.

§15.1.6 — Option C' (commons-as-examples) versus legislative

Rejected alternative — Legislative posture on commons enumeration. The framework names ten commons categories — Habitability, Spatial, Temporal, Institutional, Kindred, Ecosystem, Political, Cohesion, Epistemic, Autonomy — that have surfaced across the eighteen cases of this volume's research. The available alternative was legislative: declare these ten as the canonical complete enumeration of commons, and route any candidate eleventh through a framework-controlled admission process — treating the list as a metaphysical claim about what commons exist rather than as a reproducibility record of what surfaced empirically across the cases examined.

Why Option C' (examples-not-canonical) instead. The framework's choice joins an established tradition of presenting category-lists as illustrative examples drawn from cases rather than as exhaustive metaphysical enumerations. Ostrom's eight design principles (Ostrom 1990 *Governing the Commons*) succeeded precisely by being honest about what they had identified — empirical regularities across hundreds of coordination-commons cases, not a closed taxonomy of what governance principles must exist. Sen and Nussbaum's capabilities approach (Sen 1999 *Development as Freedom* ; Nussbaum 2000 *Women and Human Development*) elaborates ten central capabilities while explicitly acknowledging the list is open to refinement. Rawls's primary goods (Rawls 1971 *A Theory of Justice*) operates similarly. Each of these frameworks succeeded by being honest about what they had identified: examples drawn from cases, not metaphysical claims about what exists. The framework's ten commons categories follow the same discipline.

This matters because what counts as a commons is politically-traditionally contested, and the contestation is not philosophical disagreement that careful argument can resolve — it is structurally divergent commitment-spaces operating from different anthropological + political starting points. **Classical liberalism** treats autonomy as individual natural-right; supporting infrastructure is enabling but not commons. **Civic republicanism** (Pettit 1997 *Republicanism* ; Skinner 1998 *Liberty Before Liberalism* ; Anderson 2017 *Private Government*) treats autonomy as non-domination requiring shared institutions; the institutions function as commons. **Socialist + communitarian traditions** treat autonomy as enabled by shared infrastructure; the enabling conditions ARE commons. **Marxist analysis** treats autonomy as material-conditions dependent. **Lived-oppression perspectives** (slave, colonized, repressed worker) treat autonomy as something extracted, not shared commons. **Indigenous-tradition political philosophies** treat ancestral lands as relationally-personed entities, not Western-property commons.

A framework that legislates the ten as canonical declares one tradition's commitments correct and the others incorrect, and surfaces the framework's own political commitments as a precondition for using its accounting. Option C' political-philosophical-accommodation discipline keeps the framework usable across traditions: the ten are presented as *examples for illustrative purposes* surfaced across the eighteen cases examined — not an exhaustive list. The methodology (CIT + Four Gates) operates on whatever candidate commons a tradition treats as load-bearing; readers applying the framework to their own cases may surface different categories or refine these. The framework's universality is about method, not about enumeration.

§15.1.7 — Asymmetric Regret Rule versus Cost-Benefit Analysis

Rejected alternative — Cost-Benefit Analysis under risk. Standard policy-evaluation methodology under uncertainty is some variant of Cost-Benefit Analysis: estimate expected costs (with probability distributions where uncertainty is parameterizable); estimate expected benefits (same); choose the higher-expected-value action. Lineage: Pareto-Hicks-Kaldor compensation principle (Hicks 1939; Kaldor 1939); Mishan 1971 *Cost-Benefit Analysis* ; subsequent regulatory-impact-analysis literature (OMB Circular A-4); Nordhaus DICE/RICE integrated assessment models. CBA performs well when costs and benefits are symmetrically reversible, when expected values are reliable, and when the time-horizon is short enough that probability distributions are calibratable from comparable historical cases.

Why ARR instead. Commons-extraction decisions violate all three CBA performance conditions. **Reversibility:** the cost of irreversible commons-loss is structurally not reversible (atmospheric carbon-stability cannot be re-engineered at any finite cost above some threshold; species extinction cannot be reversed; cultural-knowledge-loss with last-speaker death cannot be recovered). **Expected-value**

reliability: the relevant probability distributions are subject to deep uncertainty over century-scale time horizons (Knight 1921 risk-vs-uncertainty distinction; Lempert, Popper & Bankes 2003 *Shaping the Next One Hundred Years: New Methods for Quantitative, Long-Term Policy Analysis*). **Symmetry:** the asymmetry between extracting (mostly irreversible) and not-extracting (revisitable) makes the regret structure asymmetric in a way CBA does not capture. ARR specializes Savage 1951 minimax-regret discipline ("The Theory of Statistical Decision," *Journal of the American Statistical Association* 46(253): 55–67) for commons-extraction decisions where one direction of regret is commons-extinction. Loomes & Sugden 1982 ("Regret Theory: An Alternative Theory of Rational Choice under Uncertainty," *Economic Journal*) developed regret-based individual decision theory; ARR specializes this for collective-decision contexts under irreversibility. The Rio Declaration 1992 Principle 15 precautionary-principle tradition is philosophically aligned but unidirectional; ARR is bidirectional flip-by-context. Method 3's α-dominance finding (§11.8) is the framework's empirical evidence that the asymmetric-regret structure is not an edge case for commons-extraction; it is the dominant cross-case driver of RCV variance for Earth-bound civilization-scale extractions.

§15.1.8 — CIT two sub-forms (CAI + CCI) versus single-form

Rejected alternative — Single-form CIT as one inverted question. The framework's earliest working name for the discrimination gate was the Abundance Inversion Test (AIT, since superseded by CIT), treated as a single-form thought-experiment that asked one inverted question: "would the cost vanish under abundance of the relevant condition?" Single-form clarity has methodological appeal: one operation, one filter, one verdict. Standard counterfactual-reasoning methodology (Lewis 1973; subsequent philosophy-of-action literature) typically operates with one counterfactual at a time.

Why two sub-forms instead. The commons-as-structural-identity framing surfaces two distinct operations the single-form had been collapsing. **Commons-Absence Inversion (CAI)** asks: imagine the commons not existing at all. *Would the extraction price differently if the commons claim did not exist?* Yes → cost claim is commons-grounded. No → cost claim is private-property externality (different framework applies — Pigou-tradition). **Commons-Consumption Inversion (CCI)** asks: imagine the commons not being consumed. *Would the extraction price differently if the commons were inexhaustible / self-restoring at zero marginal cost?* Yes → cost claim is consumption-of-finite-commons (Cost Severance proper). No → cost claim is coordination-of-mutually-acceptable-use (Ostrom-tradition commons-management; framework's CS does not apply). The two sub-forms run in sequence: CAI first (is this commons-grounded at all?), then CCI (is this consumption-grounded?). Only claims passing both qualify as Cost Severance.

A single-form CIT would collapse these distinctions and produce ambiguous results when CAI passes but CCI fails — exactly the case for Ostrom-tradition coordination commons (Ostrom 1990 *Governing the Commons* ; Hess & Ostrom 2007 *Understanding Knowledge as a Commons*). The two-sub-form architecture makes the routing explicit, and §6.6 (Domain distinction: extraction commons vs. coordination commons) operationalizes the routing discipline. The §6.4.1 level-of-claim specification operates within both sub-forms.

§15.1.9 — Configuration claim

None of these eight choices is novel as choice-architecture. Counterfactual reasoning (Lewis 1973), externality theory (Pigou 1920), exhaustible-resource economics (Hotelling 1931), reparations economics (Darity & Mullen 2020), option-value economics (Dixit & Pindyck 1994), commons-governance (Ostrom

1990), minimax-regret decision theory (Savage 1951; Loomes & Sugden 1982; Lempert et al. 2003), and political-philosophical accommodation (Anderson 2017; Pettit 1997) are all established traditions with substantial scholarly footprints.

The framework's contribution is the specific configuration in which these choices appear together, applied to commons-extraction contexts: CIT discriminates (counterfactual reasoning specialized to commons-extraction); Four Gates admit (filter-cohort with naming asymmetry honoring methodological-vs-mathematical structure); Three Ways estimate (triangulation under deep uncertainty); Two Instruments account (CSD + RCV partition the temporal structure of obligation; $B = B_1 + B_2$ partitions the closing instrument); Hotelling Identity bridges to standard resource economics with explicit attribution; Option C' accommodates political-philosophical diversity without legislating commons-ontology; ARR specializes asymmetric-regret discipline for commons-extinction decisions. The configuration is what does the work the framework claims to do. Each choice's rejected alternative is named and tested; each choice's lineage extension is cited in honest acknowledgment of prior work; the configuration as a whole is what carries the framework's integrated argument.

§15.2 — Methodological framing footnotes

Three framework terms with high cross-audience reach — *Cost Severance* + *Severed Cost* , *Foreclosure Bond* , and *Intergenerational Option Value* — carry methodological-framing commitments that warrant explicit positioning at academic-reviewer depth. Each was alternative-tested alongside lineage citations and cross-political-tradition handling. The footnotes below preserve that structure: framework choice + rejected alternatives + lineage citations per cluster + cross-political-tradition disambiguation. The choices honor the intellectual-honesty discipline that lineage citations + framework-extension positioning replace any claim of original coinage where established literature exists.

§15.2.1 — Cost Severance + Severed Cost methodological framing across political traditions

Rejected alternative — Rename to avoid HR/accounting "severance costs" lexical collision. Six rename candidates were alternative-tested at multi-audience-register depth (Cost Bifurcation, Cost Diversion, Cost Externalization, Cost Offloading, Cost Displacement, Cost Disposal). All six failed at least one of: (a) dinner-table accessibility — the HR-severance bridge is the term's pedagogical asset, not its liability; (b) cross-political-tradition adoption-travel anchor — US severance-tax statutes (Texas Tax Code Title 2 Subtitle I; Alaska AS 43.55; Wyoming Statutes Title 39 Ch 14; North Dakota Century Code Ch 57-51; Montana Code Annotated Title 15 Ch 36; ~36 states levying severance taxes on natural-resource extraction) is the legal-vocabulary foundation that alternatives could not match; (c) Ch 1 Option C bridge plan converts the HR collision into a teaching on-ramp rather than a renaming move.

Lineage clusters (cross-political-tradition cohort load-bearing for the term's reach across socialist/communitarian + lived-oppression registers):

- **Decolonial / colonial extraction:** Coulthard 2014 *Red Skin, White Masks* ; Tuck & Yang 2012 "Decolonization is not a metaphor"; Wolfe 1999 *Settler Colonialism and the Transformation of Anthropology* ; Whyte (indigenous environmental ethics).

- **Marxist-feminist / reproduction theory:** Federici 2004 *Caliban and the Witch* — primitive accumulation extends through the body and care work, not just land.
- **Race + housing-economics scholarship:** Mian & Sufi 2014 *House of Debt* ; Rothstein 2017 *The Color of Law* ; Taylor 2019 *Race for Profit* ; Coates 2014 "The Case for Reparations"; Hyman 2011 *Debtor Nation* ; Sugrue 1996 *The Origins of the Urban Crisis* ; Desmond 2016 *Evicted* .
- **Anti-political-economy / decolonial theory:** Patel & Moore 2017 *A History of the World in Seven Cheap Things* ; Hickel 2020 *Less Is More* .
- **Reparations economics:** Darity & Mullen 2020 *From Here to Equality* (UNC Press) — reparations economics methodology.
- **Neo-republican / civic ethics:** Anderson 2017 *Private Government* ; Pettit 1997 *Republicanism* .
- **Established US legal vocabulary:** ~36 state severance-tax statutes (cited above) — severance-tax precedent is the existing US legal vocabulary nearest to what *Cost Severance Damages* + *Restitution Bond* + *Foreclosure Bond* architecture extends.

Framework positioning: *Cost Severance* specializes the broader extraction-as-dispossession diagnosis (Mazzucato 2018 + Harvey 2003 lineage) for the specific value-capture-vs-cost-bearing separation in resource-extraction contexts; the HR-severance bridge converts the lexical collision into a teaching on-ramp.

§15.2.2 — Foreclosure Bond methodological framing — housing-crisis affect handling

Rejected alternative — Rename to avoid 2008 housing-crisis affect. Seven rename candidates were alternative-tested (Resource-Foreclosure Bond, Future-Options Bond, Long-Horizon Bond, Intergenerational Resource Bond, Hartwick Bond, Sovereign Reserve Bond, Forward Accountability Bond). All failed at least one of: (a) term-pair coherence with *Foreclosure Cost* (the cost variable that appears in the RCV integrand) — the framework architecture pairs *Foreclosure Cost is what's lost; Foreclosure Bond is what would internalize it* , and that pairing is load-bearing for Ch 9 *Pricing Honestly* 's teaching arc; (b) dinner-table accessibility via mortgage-foreclosure analogy — mortgage-foreclosure forecloses homeowners' equity-of-redemption; resource-foreclosure forecloses future generations' options-substitution; (c) cross-political-tradition adoption-travel anchor in US regulatory precedent (BLM 43 CFR Part 3162 federal oil-and-gas decommissioning bonds; BSEE 30 CFR Parts 250 + 254 offshore drilling decommissioning + oil-spill-response bonds; ~36 state plugging/abandonment bond regimes; CERCLA / RCRA financial-assurance requirements; SMCRA 1977 reclamation bonds; Environmental Impact Bonds per Balboa 2016).

The 2008 housing-crisis affect is the term's structural feature, not its liability: the trauma-affect on jacket-marketing register is honored via Ch 9 bridge prose at first-introduction (~250–350 words at canonical *Foreclosure Bond* first-mention) that surfaces the race-and-housing-economics scholarship, then names the framework's structural commitment: *the bond is forced from extractors to internalize the foreclosure-cost they would otherwise impose on future generations; it is not capital-deployment but capital-constraint* .

Lineage clusters:

- **Resource economics — Hotelling/Hartwick lineage:** Hotelling 1931 *Journal of Political Economy* — Hotelling rent under honest accounting flows to the foreclosure-bond instrument; Hartwick rule 1977 — invest resource rents in reproducible capital (Norway sovereign-wealth-fund operationalizes); Solow 1974 — intergenerational equity and exhaustible resources.

- **Race + housing-economics:** Mian & Sufi 2014 *House of Debt* ; Rothstein 2017 *The Color of Law* ; Taylor 2019 *Race for Profit* ; Coates 2014 "The Case for Reparations"; Hyman 2011 *Debtor Nation* ; Sugrue 1996 *The Origins of the Urban Crisis* ; Desmond 2016 *Evicted* .
- **Decolonial / colonial-foreclosure** (cross-reference §15.2.1): Wolfe 1999; Coulthard 2014; Tuck & Yang 2012; Whyte.
- **Reparations economics — forward-looking analog:** Darity & Mullen 2020 *From Here to Equality* — symmetric application to *Foreclosure Bond* as forward-looking analog of backward-looking restitution.
- **US regulatory precedent:** BLM 43 CFR Part 3162; BSEE 30 CFR Parts 250 + 254; ~36 state plugging/abandonment bond regimes; CERCLA / RCRA financial-assurance; SMCRA 1977 reclamation bonds; Environmental Impact Bonds (Balboa 2016).
- **Pigouvian carbon-taxation:** Pigou 1920 *Economics of Welfare* .

Framework positioning: *Foreclosure Bond* is a cluster-name + principled extension of existing US regulatory practice that 36+ states + federal agencies recognize. Norway's GPFG is the canonical existing exemplar (operationalizes Hartwick rule 1977 via sovereign-wealth-fund); Norway's GPFG is canonical specifically as a forward-looking foreclosure-bond instrument, not as canonical aggregate accountability-bond architecture; Norwegian welfare-state is approximately a backward-looking restitution-bond for Norwegian citizens but does not extend to non-Norwegian populations affected by Norwegian oil's climate externalities.

§15.2.3 — Intergenerational Option Value methodological framing — framework specialization

Rejected alternative — Defer to established option-value literature without dedicated framework term. The academic-borrowed-uncoined alternative was considered: Henry 1974 + Arrow & Fisher 1974 + Dixit & Pindyck 1994 *Investment under Uncertainty* — established option-value-under-irreversibility literature — provide close-fit academic anchors, raising the question whether a dedicated framework term overreaches. The decisive criterion was argument-load-bearing: Ch 4 *The Existence Proof* frames the entire chapter argument via option-value reasoning. The framework specializes generic option-value-under-irreversibility for the intergenerational temporal-scope axis: **the *Intergenerational Option Value* is the value that future generations forgo when present extraction forecloses options-substitution paths they have not yet had the opportunity to develop** . Without naming this specialization, Ch 4's argument relies on reader-side translation from the established option-value literature, which the chapter's audience does not uniformly carry.

Lineage clusters:

- **Foundational option-value-under-irreversibility:** Henry 1974 (quasi-option value); Arrow & Fisher 1974 (uncertainty + irreversibility + welfare).
- **Real-options theory:** Dixit & Pindyck 1994 *Investment under Uncertainty* (Princeton) — irreversibility-as-decision-axis.
- **Intergenerational equity + ecological-economics specialization:** Howarth & Norgaard 1990 *American Economic Review* ; Howarth & Norgaard 1992 *Environmental and Resource Economics* — intergenerational-equity option-value treatment in environmental economics.

- **Cross-tradition reference for socialist/communitarian + lived-oppression registers:** Brown Weiss 1989 *In Fairness to Future Generations* (Transnational Publishers) — international-law tradition of intergenerational equity that frames the same content in justice-vocabulary register; Pettit 1997 *Republicanism* + Anderson 2017 *Private Government* (cross-reference §15.2.1) — civic-republican / neo-republican vocabulary for honoring present-suffering claims alongside future-generation claims.

Framework positioning: *Intergenerational Option Value* specializes the Henry / Arrow-Fisher / Dixit-Pindyck option-value tradition for the intergenerational temporal-scope axis. The specialization names what is otherwise reader-translated: when Ch 4 frames future-generation foreclosure via option-value reasoning, the named specialization preserves the argument without forcing every reader to pattern-match to the underlying option-value literature.

§16. Mathematical Extensions

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§16.1 — RCV with Variable Discount Rates (Weitzman Specification)

$RCV = \int_{t_0}^{\infty} \{ [1 - S(t)] \cdot U(R, t, Q(t)) + E(R, t) \} \cdot \exp(-\int_{t_0}^{t} r(u)du)\, dt$

where $r(t) = r_0 \cdot e^{-\delta t} + r\_min$

The declining discount rate function r(t) starts at r_0 (conventional current rate, ~4–5%) and declines to r_min (~1%) over long horizons, reflecting Weitzman's certainty-equivalent rate under discount rate uncertainty. This specification is conservative relative to a zero or near-zero pure time preference — it biases RCV downward.

§16.2 — Stock-Dependent Substitutability

$S(t \mid t_0, Q(t)) = S\_base(t) \cdot [1 - \alpha \cdot (Q\_critical / Q(t))^{\beta}]$

where α, β > 0 capture scarcity effects on innovation incentives

When stock Q is large, innovators have low incentive to develop substitutes — resources appear abundant, prices are low. As Q approaches Q_critical (the level at which scarcity becomes economically visible), innovation pressure increases and S_base rises faster. This creates a potential feedback loop: very rapid extraction can keep prices low long enough to delay substitute development, resulting in a lower achieved S_max when depletion eventually forces the issue. This is the mechanism behind "stranded asset risk" — industries that extracted too fast find themselves without substitutes when they need them.

§16.3 — Spatial cost severance formalization

$SCS = \int_{E\setminus K} \int_{time} C(x, t)\, dx\, dt$

where x represents geographic location, E is the extraction region, K is the consumption region (lettered K to avoid collision with the cost function C(x,t)), and C(x,t) is the cost burden at location x and time t. Where value-received-at-location appears in expanded forms or extensions, the framework denotes it V(x,t) — a distinct symbol from the framework's Accountability Bond B used elsewhere.

Spatial cost severance measures the cost borne in the extraction region outside the consumption region. When the extraction region is contained within the consumption region ($E \subseteq K$; Mars colony mining its own rare earths; Norway's sovereign wealth fund returning extraction rent to the Norwegian polity that bears extraction's domestic costs), SCS = 0 because no cost falls outside the value-receiving region. When E and K are geographically separated (Appalachian coal, Baotou rare earths, Niger Delta oil), SCS > 0 and scales with the cost density in the non-overlapping region $E \setminus K$.

This formalization explains why cost severance produces systematic geographic inequality: extraction communities absorb environmental, health, and economic disruption costs while consuming communities receive energy and materials at artificially low prices. The 13-year life expectancy gap between McDowell County and the national average is the human face of SCS.

§16.4 — Sensitivity Analysis Summary

The following table shows how the McDowell County IPG (central case) responds to parameter variation:

Parameter	Baseline	Low Case	High Case	IPG Range
Discount rate r_0	4%	1.4% (Stern)	5.5% (Nordhaus)	33× to 122×
S_max (electricity)	0.85	0.95	0.70	±20%
Carbon SCC	\$190/ton CO_2 (EPA)	\$51 (conservative)	\$300+ (damage function)	±60%
Externality tail length	200 years	100 years	Indefinite	±15%

Robustness finding: CS > 0 across all parameter combinations tested. Even under the most conservative assumptions (high discount rate, high S_max, low SCC, short externality tail), the McDowell County IPG ≥ 5× — the market price covers at most 20% of the true cost. The central finding is robust to parameter uncertainty; only the magnitude varies.

§17. Formula Generalization — Sum-of-Costs Form

§17.1 — Overview

§7 specifies the four gates governing admission of cost terms to the RCV integrand. This section presents the corresponding mathematical generalization of the RCV formula — the sum-of-costs form — and derives the properties (convergence, dimensional consistency, CS = RCV − B preservation) that must hold for the generalization to be sound. The generalization is the extension that makes the framework generative: new variables can be admitted as the four gates reveal them, without destabilizing the mathematical architecture. The two-term RCV formula of §3.1 is recovered as the $n = 2$ special case of the general form.

§17.2 — The Generalization: Formal Statement

The two-term RCV formula (§3.1):

$$RCV(R, t_0) = \int_{t_0}^{\infty} \{ [1 - S(t \mid t_0)] \cdot U(R, t, Q(t)) + E(R, t) \} \cdot D(t, t_0)\, dt$$

is replaced by the general sum-of-costs form:

$$RCV(R, t_0, Context) = \int_{t_0}^{\infty} \{ \Sigma_{i=1}^{n} C_i(R, t, Context) \} \cdot D(t, t_0)\, dt$$

where:

- n is the number of cost terms admitted into the integrand under the current Context;
- each C_i has passed the four gates (§7.1-§7.4) under stated specifications;
- Context encodes the specific extraction situation (who captures value, who bears cost, what scarcity operates, what substitutability landscape exists, what externality profile, what community relationships are affected);
- $D(t, t_0)$ is the declining-rate discount factor per §16.1.

The two-term form is recovered as the special case $n = 2$ with $C_1(R, t, Context) = [1 - S(t \mid t_0)] \cdot U(R, t, Q(t))$ (foreclosure) and $C_2(R, t, Context) = E(R, t)$ (externality tail). This equivalence is worked explicitly in §17.6 below.

§17.3 — Convergence Preservation

Claim. If each C_i satisfies the boundedness gate (Gate 3, §7.3) individually under the stated discount factor, and n is finite, then the sum-of-costs integral converges.

Proof. The integrand is a finite sum of functions, each of which yields a convergent integral under D by Gate 3 (§7.3). By linearity of integration:

$$\int_{t_0}^{\infty} \Sigma_{i=1}^{n} C_i(R, t, Context) \cdot D(t, t_0)\, dt = \Sigma_{i=1}^{n} \int_{t_0}^{\infty} C_i(R, t, Context) \cdot D(t, t_0)\, dt$$

Each term on the right is finite (Gate 3 passes individually). A finite sum of finite terms is finite. The integral of the sum converges. Q.E.D.

Remark on countable sums. For countably many C_i (a theoretical extension), uniform convergence of the partial sums plus absolute convergence of the sum of integrals is required. In practice, n is bounded by the empirical scope of CIT-discoverable variables in a given Context (observed bound ≤ ~20 across 18 case studies). Countable-sum extensions are not required for the framework's current applied scope. Future research may address formal boundedness of n as a function of Context complexity; the current framework operates under empirical finiteness.

Interchange of sum and integral. For finite sums, interchange is unconditional. For countable sums, the interchange requires a dominating function by the dominated convergence theorem or a monotone structure by the monotone convergence theorem. In the framework's applied scope (finite n), no dominating function is needed; the interchange is a standard result.

§17.4 — Units Consistency

Claim. If each C_i satisfies the dimensional gate (Gate 2, §7.2) (units [$\$ \cdot \text{res-unit}^{-1} \cdot \text{time-unit}^{-1}$]), then the sum-of-costs integral produces a quantity with units [$\$ \cdot \text{res-unit}^{-1}$] — the same units as the two-term RCV.

Proof. Each $C_i \cdot D(t, t_0)$ has units:

$$[\$ \cdot \text{res-unit}^{-1} \cdot \text{time-unit}^{-1}] \cdot [\text{dimensionless}] = [\$ \cdot \text{res-unit}^{-1} \cdot \text{time-unit}^{-1}]$$

A finite sum of quantities with identical units preserves those units: $\Sigma_i C_i \cdot D$ has units [$\$ \cdot \text{res-unit}^{-1} \cdot \text{time-unit}^{-1}$]. Integration with respect to t removes the time-unit^{-1} factor, yielding [$\$ \cdot \text{res-unit}^{-1}$]. Dimensional consistency is preserved under the generalization, independent of n. Q.E.D.

A consequence: heterogeneous cost terms (health damage in dollars per ton per year, community-transition cost in dollars per ton per year, ecosystem-cascade cost in dollars per ton per year) can be summed cleanly only because Gate 2 forces each to the same dimensional form. Without the dimensional gate, a mixed sum (e.g. dollars added to ordinal dignity-index points) would be arithmetically coherent but dimensionally meaningless.

§17.5 — Preservation of the Cost Severance Relation

Claim. The cost severance equation CS = RCV − B (§3) holds under the generalized RCV without modification.

Proof. CS is defined as the dollar gap between the true intergenerational cost (RCV) and the accountability bond (B) posted against the extraction. The definition does not depend on the internal structure of RCV — that is, on whether RCV is computed via the two-term form or the generalized sum-of-costs form. CS = RCV − B remains the relation between the computed RCV value and the posted bond, regardless of how RCV is assembled internally. Q.E.D.

Corollary — directional monotonicity. Gate 1 (§7.1) constrains admitted terms to be costs (non-negative in present-value contribution) rather than benefits. Gate 3 (§7.3) constrains them to bounded, integrable functions. Admitting an additional C_{n+1} can only increase RCV, widening the CS gap (or leaving it unchanged if C_{n+1} is identically zero in the current Context). Adding correctly-specified variables cannot falsify CS > 0 for cases where it currently holds; only mischaracterizing existing terms can.

Corollary — published-result preservation. Every CS > 0 finding in the framework's prior worked cases (McDowell coal, Deepwater Horizon, Libby, Exxon Valdez, Norway oil, and the 13 additional cases in §11) remains valid under the generalization. Those computations correspond to the n = 2 special case. Extensions to n > 2 produce larger RCV values and therefore stronger (not weaker) CS > 0 findings.

Corollary — bidirectional preservation. The same preservation extends to the bidirectional form `total CS = (CSD − B₁) + (RCV − B₂)` (per §5.5): the n-term sum-of-costs generalization applies independently to each integrand (backward CSD-side and forward RCV-side), and the directional-monotonicity corollary above operates symmetrically on both. Adding correctly-specified backward variables (legacy-effect C_i under the backward application of the Four Gates) widens the ($CSD - B_1$) gap; adding correctly-specified forward variables widens the ($RCV - B_2$) gap; neither direction's admission can falsify a $CS > 0$ finding established on the other side.

§17.6 — Worked Derivation: The Two-Term Form as Special Case

Starting from the generalized form:

$RCV(R, t_0, Context) = \int_{t_0}^{\infty} \{ \sum_{i=1}^{n} C_i(R, t, Context) \} \cdot D(t, t_0)\, dt$

Specify Context = "standard fossil-fuel extraction with canonical externality profile" and admit only the following two terms:

C_1 — Foreclosure Cost. Under this Context, foreclosure captures the value future users lose by the extracted resource not being available. Functional form:

$C_1(R, t, Context) = [1 - S(t \mid t_0)] \cdot U(R, t, Q(t))$

Gate checks:

• Gate 1 (CIT): under abundance of substitutes at all times, $[1 - S] = 0$; foreclosure vanishes. Pass.

• Gate 2 (dimensional): $[1 - S]$ dimensionless; U has units [$\$ \cdot \text{res-unit}^{-1} \cdot \text{time-unit}^{-1}$]; product has required units. Pass.

• Gate 3 (boundedness): S monotonically increases toward $S_{max} \leq 1$; $[1 - S]$ bounded in $[0, 1]$; U bounded for finite stock Q. Integral converges under Weitzman D. Pass.

• Gate 4 (independence): foreclosure is the first term of the starting set; independence trivially holds. Pass.

C_2 — Externality Tail. Captures ongoing damage flow attributable to the extraction, independent of resource-stock depletion:

$C_2(R, t, Context) = E(R, t)$

Gate checks:

• Gate 1: under abundance of absorption capacity, E-driven damage vanishes. Pass.

• Gate 2: E specified with units [$\$ \cdot \text{res-unit}^{-1} \cdot \text{time-unit}^{-1}$]. Pass.

• Gate 3: E decays with natural sinks (atmospheric CO_2 half-life ~100 years for primary fraction; ~1000 years for residual fraction). Integral converges. Pass.

• Gate 4: E captures damage to absorption systems; C_1 captures resource-stock loss to future users. Cost categories disjoint. Pass.

Substitution. With $n = 2$ and $\{C_1, C_2\}$ as above:

$RCV(R, t_0, Context) = \int_{t_0}^{\infty} \{ [1 - S(t \mid t_0)] \cdot U(R, t, Q(t)) + E(R, t) \} \cdot D(t, t_0)\, dt$

This is the two-term formula from §3.1. The canonical formula is the generalized formula under the specification (Context = standard-fossil-fuel, n = 2, {C_1 = foreclosure, C_2 = externality tail}). Q.E.D.

§17.7 — The Generative Property

The sum-of-costs form supports three modes of variable admission that the two-term form does not.

Extension under the same Context. Adding C_3, C_4, … under the current Context, each passing the four gates (§7.1-§7.4). The worked example in §7.6 demonstrates C_3 (community-transition-reserve) under McDowell-coal Context. Subsequent additions expand the variable set as empirical evidence and methodology development surface them.

Context-specific variable sets. Different Contexts admit different {C_i}. For a fisheries extraction, the foreclosure structure differs from fossil-fuel extraction (substitutability of specific fish species differs from substitutability of fossil energy), and additional terms such as ecosystem-cascade cost may apply that do not apply to fossil fuels. The framework produces Context-specific RCV formulas as the gate discipline reveals what terms belong.

Third-party extension. Future analysts can apply CIT + the four gates to extraction contexts the book does not examine, discovering additional C_i the current author did not identify. The framework does not require revision to admit them; it is structurally open. What the four gates enforce is that the discovery is disciplined, not arbitrary.

Relationship to the empirical variable set. The ten commons categories (Habitability · Spatial · Temporal · Institutional · Kindred · Ecosystem · Political · Cohesion · Epistemic · Autonomy) and the eight-tier decomposition (Direct Health / Environmental Degradation / Community Disruption / Foreclosure / Dynastic Labor / Knowledge-and-Cultural / Political Capture / Temporal Displacement) are organizational scaffolding over variables CIT has surfaced across the case studies this volume examines. They are empirical convergences — variables the discovery method has produced — not the universe of variables the discovery method can produce. New Contexts may surface additional C_i that map onto the existing scaffolding or that require new organizational categories.

§17.8 — Practical Consequence for Published Analyses

Every RCV computation in the book remains valid under the generalized form. The two-term formula applies to fossil-fuel cases (McDowell coal, Deepwater Horizon, Norway oil, Exxon Valdez) and recovers the Ch 6 convergence-table figures. Extensions with additional C_i produce larger RCV values (the §7.6 worked update adds ~\$5–\$10/ton to the McDowell figure); the qualitative direction $CS > 0$ is preserved. Published analyses that cite the two-term form are not invalidated; they are the n = 2 special case of the general form.

A book-internal convention: published analyses that compute RCV from the two-term form will explicitly name the Context and acknowledge the n = 2 restriction in a footnote or methodology paragraph, with a pointer to this Section and to §7.5 (gate-application protocol) for analysts extending the computation. This preserves backward-compatibility with the Ch 6 convergence table and with the 18-case empirical catalog while signaling the framework's structural extensibility — the methodology accommodates additional variables as cases surface them, without requiring methodology revision.

§18. Bibliography

Academic citations referenced inline throughout this Tech Appendix, organized alphabetically by author surname.

- Acemoglu, Daron, and James A. Robinson. *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*. New York: Crown Business, 2012.
- Akerlof, George A. "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism." *Quarterly Journal of Economics* 84, no. 3 (1970): 488–500.
- Anderson, Chris. *The Long Tail: Why the Future of Business Is Selling Less of More*. New York: Hyperion, 2006.
- Anderson, Elizabeth. *Private Government: How Employers Rule Our Lives (and Why We Don't Talk about It)*. Princeton: Princeton University Press, 2017.
- Arrow, Kenneth J., and Anthony C. Fisher. "Environmental Preservation, Uncertainty, and Irreversibility." *Quarterly Journal of Economics* 88, no. 2 (1974): 312–319.
- Atkinson, Anthony B. "On the Measurement of Inequality." *Journal of Economic Theory* 2, no. 3 (1970): 244–263.
- Balboa, Cristina M. "Accountability of Environmental Impact Bonds." 2016.
- Barnosky, Anthony D., et al. "Approaching a State Shift in Earth's Biosphere." *Nature* 486 (2012): 52–58.
- Basel Committee on Banking Supervision. *Basel III: Finalising Post-Crisis Reforms*. Bank for International Settlements, 2017.
- Bassett-Bell, Naa Oyo A., David R. Williams, William A. Darity Jr., et al. "Association Between Racial Wealth Inequities and Racial Disparities in Longevity Among US Adults and Role of Reparations Payments, 1992 to 2018." *JAMA Network Open* 5, no. 11 (2022): e2240519.
- Benjamin, Ruha. *Race After Technology: Abolitionist Tools for the New Jim Code*. Cambridge: Polity Press, 2019.
- Bergson, Abram. "A Reformulation of Certain Aspects of Welfare Economics." *Quarterly Journal of Economics* 52, no. 2 (1938): 310–334.
- Brown Weiss, Edith. *In Fairness to Future Generations: International Law, Common Patrimony, and Intergenerational Equity*. Dobbs Ferry, NY: Transnational Publishers, 1989.
- Brundtland, Gro Harlem (chair), World Commission on Environment and Development. *Our Common Future*. Oxford: Oxford University Press, 1987.
- Bullard, Robert D. *Dumping in Dixie: Race, Class, and Environmental Quality*. Boulder, CO: Westview Press, 1990.
- Coase, Ronald H. "The Problem of Social Cost." *Journal of Law and Economics* 3 (1960): 1–44.
- Coates, Ta-Nehisi. "The Case for Reparations." *The Atlantic*, June 2014.

- Cobb, Charles W., and Paul H. Douglas. "A Theory of Production." *American Economic Review* 18, no. 1 (1928): 139–165.
- Coleman, James S. *Foundations of Social Theory*. Cambridge, MA: Harvard University Press, 1990.
- Costanza, Robert, et al. "The Value of the World's Ecosystem Services and Natural Capital." *Nature* 387 (1997): 253–260.
- Costanza, Robert, et al. "Changes in the Global Value of Ecosystem Services." *Global Environmental Change* 26 (2014): 152–158.
- Coulthard, Glen Sean. *Red Skin, White Masks: Rejecting the Colonial Politics of Recognition*. Minneapolis: University of Minnesota Press, 2014.
- Crawford, Kate. *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. New Haven: Yale University Press, 2021.
- Daily, Gretchen C., ed. *Nature's Services: Societal Dependence on Natural Ecosystems*. Washington, DC: Island Press, 1997.
- Darity, William A., Jr. Interview by Christopher Winn. May 13, 2026. Recording on file with author.
- Darity, William A., Jr., and A. Kirsten Mullen. *From Here to Equality: Reparations for Black Americans in the Twenty-First Century*. Chapel Hill: University of North Carolina Press, 2020.
- Dasgupta, Partha, and Geoffrey M. Heal. *Economic Theory and Exhaustible Resources*. Cambridge: Cambridge University Press, 1979.
- Desmond, Matthew. *Evicted: Poverty and Profit in the American City*. New York: Crown, 2016.
- Dixit, Avinash K., and Robert S. Pindyck. *Investment under Uncertainty*. Princeton: Princeton University Press, 1994.
- Du Bois, W. E. B. *Black Reconstruction in America: 1860–1880*. New York: Harcourt, Brace, 1935.
- Embrechts, Paul, Claudia Klüppelberg, and Thomas Mikosch. *Modelling Extremal Events for Insurance and Finance*. Berlin: Springer, 1997.
- Esping-Andersen, Gøsta. *The Three Worlds of Welfare Capitalism*. Princeton: Princeton University Press, 1990.
- Federici, Silvia. *Caliban and the Witch: Women, the Body, and Primitive Accumulation*. New York: Autonomedia, 2004.
- Folland, Gerald B. *Real Analysis: Modern Techniques and Their Applications*. 2nd ed. New York: John Wiley & Sons, 1999.
- Fricker, Miranda. *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford: Oxford University Press, 2007.
- Habermas, Jürgen. *Theory of Communicative Action*. Translated by Thomas McCarthy. Boston: Beacon Press, 1981.
- Hamilton, James D. "Causes and Consequences of the Oil Shock of 2007–08." *Brookings Papers on Economic Activity* 2009, no. 1 (2009): 215–261.
- Hartwick, John M. "Intergenerational Equity and the Investing of Rents from Exhaustible Resources." *American Economic Review* 67, no. 5 (1977): 972–974.
- Harvey, David. *The New Imperialism*. Oxford: Oxford University Press, 2003.

- Henry, Claude. "Investment Decisions Under Uncertainty: The 'Irreversibility Effect.'" *American Economic Review* 64, no. 6 (1974): 1006–1012.
- Hess, Charlotte, and Elinor Ostrom, eds. *Understanding Knowledge as a Commons: From Theory to Practice*. Cambridge, MA: MIT Press, 2007.
- Hickel, Jason. *Less Is More: How Degrowth Will Save the World*. London: William Heinemann, 2020.
- Hicks, John R. *The Theory of Wages*. London: Macmillan, 1932.
- Hong, Lu, and Scott E. Page. "Groups of Diverse Problem Solvers Can Outperform Groups of High-Ability Problem Solvers." *Proceedings of the National Academy of Sciences* 101, no. 46 (2004): 16385–16389.
- Hotelling, Harold. "The Economics of Exhaustible Resources." *Journal of Political Economy* 39, no. 2 (1931): 137–175.
- Howarth, Richard B., and Richard B. Norgaard. "Intergenerational Resource Rights, Efficiency, and Social Optimality." *American Economic Review* 80, no. 2 (1990): 473–477.
- Howarth, Richard B., and Richard B. Norgaard. "Environmental Valuation under Sustainable Development." *Environmental and Resource Economics* 2, no. 6 (1992): 583–596.
- Hyman, Louis. *Debtor Nation: The History of America in Red Ink*. Princeton: Princeton University Press, 2011.
- Insurance Information Institute. *Replacement Cost vs. Actual Cash Value* (consumer guide).
- Jacobs, Jane. *The Death and Life of Great American Cities*. New York: Random House, 1961.
- Kant, Immanuel. *Groundwork of the Metaphysics of Morals*. 1785. Translated by Mary Gregor and Jens Timmermann. Cambridge: Cambridge University Press, 2012.
- Kilian, Lutz. "Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market." *American Economic Review* 99, no. 3 (2009): 1053–1069.
- Klinenberg, Eric. *Palaces for the People: How Social Infrastructure Can Help Fight Inequality, Polarization, and the Decline of Civic Life*. New York: Crown, 2018.
- Lempert, Robert J., Steven W. Popper, and Steven C. Bankes. *Shaping the Next One Hundred Years: New Methods for Quantitative, Long-Term Policy Analysis*. Santa Monica, CA: RAND, 2003.
- Lenton, Timothy M., et al. "Tipping Elements in the Earth's Climate System." *Proceedings of the National Academy of Sciences* 105, no. 6 (2008): 1786–1793.
- Loomes, Graham, and Robert Sugden. "Regret Theory: An Alternative Theory of Rational Choice under Uncertainty." *Economic Journal* 92, no. 368 (1982): 805–824.
- Mandelbrot, Benoit. "The Variation of Certain Speculative Prices." *Journal of Business* 36, no. 4 (1963): 394–419.
- Mann, Michael. "The Autonomous Power of the State: Its Origins, Mechanisms and Results." *European Journal of Sociology* 25, no. 2 (1984): 185–213.
- Marshall, Alfred. *Principles of Economics*. London: Macmillan, 1890.
- Mazzucato, Mariana. *The Value of Everything: Making and Taking in the Global Economy*. London: Allen Lane, 2018.
- Mian, Atif, and Amir Sufi. *House of Debt: How They (and You) Caused the Great Recession, and How We Can Prevent It from Happening Again*. Chicago: University of Chicago Press, 2014.

- Mill, John Stuart. *On Liberty*. London: John W. Parker and Son, 1859.
- Mosteller, Frederick, and John W. Tukey. *Data Analysis and Regression: A Second Course in Statistics*. Reading, MA: Addison-Wesley, 1977.
- Mussa, Michael, and Sherwin Rosen. "Monopoly and Product Quality." *Journal of Economic Theory* 18, no. 2 (1978): 301–317.
- Nordhaus, William D., and Joseph Boyer. *Warming the World: Economic Models of Global Warming*. Cambridge, MA: MIT Press, 2000.
- North, Douglass C. *Institutions, Institutional Change and Economic Performance*. Cambridge: Cambridge University Press, 1990.
- Nussbaum, Martha C. *Creating Capabilities: The Human Development Approach*. Cambridge, MA: Belknap Press of Harvard University Press, 2011.
- Olson, Mancur. *The Logic of Collective Action: Public Goods and the Theory of Groups*. Cambridge, MA: Harvard University Press, 1965.
- O'Neil, Cathy. *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy*. New York: Crown, 2016.
- Ostrom, Elinor. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press, 1990.
- Parfit, Derek. *Reasons and Persons*. Oxford: Oxford University Press, 1984.
- Patel, Raj, and Jason W. Moore. *A History of the World in Seven Cheap Things: A Guide to Capitalism, Nature, and the Future of the Planet*. Berkeley: University of California Press, 2017.
- Pettit, Philip. *Republicanism: A Theory of Freedom and Government*. Oxford: Oxford University Press, 1997.
- Pigou, Arthur Cecil. *The Economics of Welfare*. London: Macmillan, 1920.
- Pindyck, Robert S. "The Optimal Exploration and Production of Nonrenewable Resources." *Journal of Political Economy* 86, no. 5 (1978): 841–861.
- Pindyck, Robert S. "Volatility and Commodity Price Dynamics." *Energy Journal* 29, no. 1 (2008): 1–26.
- Putnam, Robert D. *Bowling Alone: The Collapse and Revival of American Community*. New York: Simon & Schuster, 2000.
- Reilly, Benjamin. *Democracy and Diversity: Political Engineering in the Asia-Pacific*. Oxford: Oxford University Press, 2001.
- Rio Declaration on Environment and Development. United Nations Conference on Environment and Development, 1992. (Principle 15: Precautionary Approach.)
- Rockström, Johan, et al. "A Safe Operating Space for Humanity." *Nature* 461 (2009): 472–475.
- Rothstein, Richard. *The Color of Law: A Forgotten History of How Our Government Segregated America*. New York: Liveright, 2017.
- Royden, H. L. *Real Analysis*. 3rd ed. New York: Macmillan, 1988.
- Savage, Leonard J. "The Theory of Statistical Decision." *Journal of the American Statistical Association* 46, no. 253 (1951): 55–67.
- Sen, Amartya. *Development as Freedom*. New York: Knopf, 1999.

- Sen, Amartya. *Inequality Reexamined*. Cambridge, MA: Harvard University Press, 1992.
- Slade, Margaret E., and Henry Thille. "Whither Hotelling: Tests of the Theory of Exhaustible Resources." *Annual Review of Resource Economics* 1 (2009): 239–260.
- Solow, Robert M. "A Contribution to the Theory of Economic Growth." *Quarterly Journal of Economics* 70, no. 1 (1956): 65–94.
- Solow, Robert M. "Intergenerational Equity and Exhaustible Resources." *Review of Economic Studies* 41 (Symposium 1974): 29–45.
- Spence, Michael. "Product Differentiation and Welfare." *American Economic Review Papers and Proceedings* 66, no. 2 (1976): 407–414.
- Steffen, Will, et al. "Planetary Boundaries: Guiding Human Development on a Changing Planet." *Science* 347, no. 6223 (2015): 1259855.
- Stern, Nicholas. *The Economics of Climate Change: The Stern Review*. Cambridge: Cambridge University Press, 2007.
- Stigler, George J. "The Theory of Economic Regulation." *Bell Journal of Economics and Management Science* 2, no. 1 (1971): 3–21.
- Stiglitz, Joseph E. *Economics of the Public Sector*. 3rd ed. New York: W. W. Norton, 2000.
- Sugrue, Thomas J. *The Origins of the Urban Crisis: Race and Inequality in Postwar Detroit*. Princeton: Princeton University Press, 1996.
- Taleb, Nassim Nicholas. *The Black Swan: The Impact of the Highly Improbable*. New York: Random House, 2007.
- Taylor, Keeanga-Yamahtta. *Race for Profit: How Banks and the Real Estate Industry Undermined Black Homeownership*. Chapel Hill: University of North Carolina Press, 2019.
- Tilly, Charles. *Coercion, Capital and European States, AD 990–1990*. Cambridge, MA: Blackwell, 1990.
- Tirole, Jean. *The Theory of Industrial Organization*. Cambridge, MA: MIT Press, 1988.
- Tönnies, Ferdinand. *Gemeinschaft und Gesellschaft*. Leipzig: Fues's Verlag, 1887. English translation: *Community and Society*, 1957.
- Tuan, Yi-Fu. *Space and Place: The Perspective of Experience*. Minneapolis: University of Minnesota Press, 1977.
- Tuck, Eve, and K. Wayne Yang. "Decolonization Is Not a Metaphor." *Decolonization: Indigeneity, Education & Society* 1, no. 1 (2012): 1–40.
- Weitzman, Martin L. "Gamma Discounting." *American Economic Review* 91, no. 1 (2001): 260–271.
- Weston, Kath. *Families We Choose: Lesbians, Gays, Kinship*. New York: Columbia University Press, 1991.
- Whyte, Kyle. *Indigenous Environmental Ethics* (collected works on indigenous environmental ethics + climate justice).
- Wolfe, Patrick. *Settler Colonialism and the Transformation of Anthropology: The Politics and Poetics of an Ethnographic Event*. London: Cassell, 1999.